

MAKALAH
“PRAKTIKUM SISTEM OPRASI”

Disusun Untuk Memenuhi Tugas Mata Kuliah
Sistem Oprasi

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BAB I

OPERATING-SYSTEM SERVICES

1.1 Menjelajahi Layanan OS

Lihat Informasi Sistem

```
agilinggara@RagilAnggara: ~  
To run a command as administrator (user "root"), use "sudo <command>".  
See "man sudo_root" for details.  
  
agilinggara@RagilAnggara:~$ uname -a  
Linux RagilAnggara 6.6.87.2-microsoft-standard-WSL2 #1 SMP PREEMPT_DYNAMIC Thu Jun 5 18:30:46 UTC 2025 x86_64 x86_64 x86_64 GNU/Linux  
agilinggara@RagilAnggara:~$ lsb_release -a  
No LSB modules are available.  
Distributor ID: Ubuntu  
Description:    Ubuntu 24.04.3 LTS  
Release:        24.04  
Codename:       noble
```

Lihat Proses yang Berjalan

```
agilinggara@RagilAnggara:~$ ps aux | head -20  
USER      PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND  
root         1  0.1  0.1 21660 11628 ?        Ss   18:56   0:00 /sbin/init  
root         2  0.0  0.0   3072  1920 ?        SL   18:56   0:00 /init  
root         7  0.0  0.0   3072  1792 ?        SL   18:56   0:00 plan9 --control-socket 7 --log-level 4 --server-fd 8 --pipe-fd 10 --log-truncate  
root        40  0.0  0.2 66832 18628 ?        S<s  18:56   0:00 /usr/lib/systemd/systemd-journald  
root        90  0.0  0.0  25132  6144 ?        Ss   18:56   0:00 /usr/lib/systemd/systemd-udevd  
systemd+   112  0.0  0.1 21456 12672 ?        Ss   18:56   0:00 /usr/lib/systemd/systemd-resolved  
systemd+   114  0.0  0.0  91024  7424 ?        Ssl  18:56   0:00 /usr/lib/systemd/systemd-timesyncd  
root       177  0.0  0.0   4236  2560 ?        Ss   18:56   0:00 /usr/sbin/cron -f -P  
message+   178  0.0  0.0   9596  4480 ?        Ss   18:56   0:00 @dbus-daemon --system --address=systemd: --nofork --nopidfile --systemd-activation --sysl  
og-only  
root       191  0.0  0.1 17964  8320 ?        Ss   18:56   0:00 /usr/lib/systemd/systemd-logind  
root       198  0.0  0.1 1756096 12160 ?       Ssl  18:56   0:00 /usr/libexec/wsl-pro-service -vv  
root       207  0.0  0.0   3160  1920 hvc0     Ss+  18:56   0:00 /sbin/agetty -o -p -- \u --noclear --keep-baud - 115200,38400,9600 vt220  
syslog     226  0.0  0.0 222508  5248 ?        Ssl  18:56   0:00 /usr/sbin/rsyslogd -n -iNONE  
root       235  0.0  0.0   3116  1792 tty1     Ss+  18:56   0:00 /sbin/agetty -o -p -- \u --noclear - linux  
root       255  0.0  0.2 107032 22268 ?        Ssl  18:56   0:00 /usr/bin/python3 /usr/share/unattended-upgrades/unattended-upgrade-shutdown --wait-for-si  
gnal  
root       362  0.0  0.0   6660  4352 pts/1    Ss   18:56   0:00 /bin/login -f  
agilingg+  408  0.0  0.1 20312 10880 ?        Ss   18:56   0:00 /usr/lib/systemd/systemd --user  
agilingg+  409  0.0  0.0  21152  3516 ?        S    18:56   0:00 (sd-pam)  
agilingg+  464  0.0  0.0   6056  5248 pts/1    S+   18:56   0:00 -bash
```

Lihat Pengguna Memori

```
agilinggara@RagilAnggara:~$ free -h  
              total        used        free      shared  buff/cache   available  
Mem:          7.5Gi        535Mi        6.3Gi        3.5Mi        798Mi        6.9Gi  
Swap:         2.0Gi          0B        2.0Gi
```

Lihat informasi CPU

```
agilinggara@RagilAnggara:~$ lscpu  
Architecture: x86_64  
CPU op-mode(s): 32-bit, 64-bit  
Address sizes: 48 bits physical, 48 bits virtual  
Byte Order: Little Endian  
CPU(s): 16
```

```
agilinggara@RagilAnggara: ~  
CPU(s): 16  
On-line CPU(s) list: 0-15  
Vendor ID: AuthenticAMD  
Model name: AMD Ryzen 9 5900HX with Radeon Graphics  
CPU family: 25  
Model: 80  
Thread(s) per core: 2  
Core(s) per socket: 8  
Socket(s): 1  
Stepping: 0  
BogoMIPS: 6587.45  
Flags: fpu vme de pse tsc msr pae mce cx8 apic sep mtrr pge mca cmov pat pse36 clflush mmx fxsr sse sse2 ht syscall nx mmxext fxsr_opt pd  
pe1gb rdtscp lm constant_tsc rep_good nopl tsc_reliable nonstop_tsc cpuid extd_apicid tsc_known_freq pni pclmulqdq ssse3 fma cx16  
sse4_1 sse4_2 movbe popcnt aes xsave avx f16c rdrand hypervisor lahf_lm cmp_legacy svm cr8_legacy abm sse4a misalignsse 3dnowprefe  
tch osvw topoext perfctr_core ssbd ibrs ibpb stibp vmmcall fsgsbase bmi1 avx2 smep bmi2 erms invpcid rdseed adx smap clflushopt cl  
wb sha_ni xsaveopt xsavec xgetbv1 xsaves clzero xsaveerptr arat npt nrip_save tsc_scale vmcb_clean flushbyasid decodeassists pause  
filter pfthreshold v_vmsave_vmload umip vaes vpclmulqdq rdpid fsrm  
Virtualization features:  
Virtualization: AMD-V  
Hypervisor vendor: Microsoft  
Virtualization type: full  
Caches (sum of all):  
L1d: 256 KiB (8 instances)  
L1i: 256 KiB (8 instances)  
L2: 4 MiB (8 instances)  
L3: 16 MiB (1 instance)  
NUMA:  
NUMA node(s): 1  
NUMA node0 CPU(s): 0-15  
Vulnerabilities:  
Gather data sampling: Not affected  
Itlb multihit: Not affected  
L1tf: Not affected  
Mds: Not affected  
Meltdown: Not affected  
Mmio stale data: Not affected  
Reg file data sampling: Not affected  
Retbleed: Not affected  
Spec rstack overflow: Vulnerable: Safe RET, no microcode  
Spec store bypass: Mitigation; Speculative Store Bypass disabled via prctl  
Spectre v1: Mitigation; usercopy/swapgs barriers and __user pointer sanitization  
Spectre v2: Mitigation; Retpolines; IBPB conditional; IBRS_FW; STIBP always-on; RSB filling; PBRSE-eIBRS Not affected; BHI Not affected  
Srbds: Not affected  
Tsx async abort: Not affected
```

Lihat perangkat yang terpasang

```
agilinggara@RagilAnggara:~$ lsblk  
NAME  
sda 8:0 0 388.4M 1 disk  
sdb 8:16 0 186M 1 disk  
sdc 8:32 0 2G 0 disk [SWAP]  
sdd 8:48 0 1T 0 disk /mnt/wslg/distro  
agilinggara@RagilAnggara:~$
```

1.2 Program C- Menggunakan Layanan OS

Buat file os_services.c

```
agilinggara@RagilAnggara: ~  
agilinggara@RagilAnggara:~$ mkdir praktikum1  
agilinggara@RagilAnggara:~$ cd praktikum1  
agilinggara@RagilAnggara:~/praktikum1$ pwd  
/home/agilinggara/praktikum1  
agilinggara@RagilAnggara:~/praktikum1$ ls  
agilinggara@RagilAnggara:~/praktikum1$ nano os_services.c
```

```
GNU nano 7.2 os_services.c
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/types.h>
#include <sys/stat.h>
#include <time.h>

int main() {
    // 1. Program Execution Service
    printf("=== Program Execution Service ===\n");
    printf("Process ID (PID): %d\n", getpid());
    printf("Parent Process ID (PPID): %d\n", getppid());

    // 2. I/O Operations Service
    printf("\n=== I/O Operations Service ===\n");
    FILE *fp = fopen("test_file.txt", "w");
    if (fp != NULL) {
        fprintf(fp, "Testing I/O operations\n");
        fclose(fp);
        printf("File created and written successfully\n");
    }

    // 3. File System Manipulation
    printf("\n=== File System Manipulation ===\n");
    struct stat file_stat;
    if (stat("test_file.txt", &file_stat) == 0) {
        printf("File size: %ld bytes\n", file_stat.st_size);
        printf("File permissions: %o\n", file_stat.st_mode & 0777);
    }

    // 4. Communications
    printf("\n=== Communications ===\n");
    printf("Hostname: %s\n", getenv("HOSTNAME"));

    // 5. Error Detection
    printf("\n=== Error Detection ===\n");
    FILE *fp2 = fopen("nonexistent.txt", "r");
    if (fp2 == NULL) {
        perror("Error opening file");
    }
}

^G Help      ^O Write Out  ^W Where Is   ^K Cut        ^T Execute    ^C Location   ^U Undo       ^-A Set Mark  ^-] To Bracket
^X Exit      ^R Read File  ^_ Replace    ^U Paste      ^J Justify    ^/_ Go To Line ^-E Redo      ^-G Copy      ^_ Where Was
```

```
// 6. Resource Allocation
printf("\n=== Resource Allocation ===\n");
int *ptr = (int*)malloc(100 * sizeof(int));
if (ptr != NULL) {
    printf("Memory allocated successfully\n");
    free(ptr);
}

return 0;
}
```

```
^G Help      ^O Write Out  ^W Where Is   ^K Cut        ^T Execute    ^C Location   ^U Undo       ^-A Set Mark  ^-] To Bracket
^X Exit      ^R Read File  ^_ Replace    ^U Paste      ^J Justify    ^/_ Go To Line ^-E Redo      ^-G Copy      ^_ Where Was
```

Compile dan Jalankan

```
agilnggara@RagilAnggara:~/praktikum1$ ls
os_services.c
agilnggara@RagilAnggara:~/praktikum1$ gcc -o os_services os_services.c
Command 'gcc' not found, but can be installed with:
sudo apt install gcc
```

```
agilnggara@RagilAnggara:~/praktikum1$ gcc -o os_services os_services.c
./os_services
=== Program Execution Service ===
Process ID (PID): 3564
Parent Process ID (PPID): 549

=== I/O Operations Service ===
File created and written successfully

=== File System Manipulation ===
File size: 23 bytes
File permissions: 644

=== Communications ===
Hostname: (null)

=== Error Detection ===
Error opening file: No such file or directory

=== Resource Allocation ===
Memory allocated successfully
```

BAB II

USER AND OPERATING-SYSTEM INTERFACE

2.1 Command Line Interface (CLI)

Navigasi Dasar

```
agilnggara@RagilAnggara: / ~$ pwd
/home/agilnggara
agilnggara@RagilAnggara:~$ ls -la
total 36
drwxr-xr-x  5 agilnggara agilnggara 4096 Oct 12 19:23 .
drwxr-xr-x  3 root      root      4096 Oct  8 08:12 ..
-rw-r--r--  1 agilnggara agilnggara 307 Oct 12 19:58 .bash_history
-rw-r--r--  1 agilnggara agilnggara 220 Oct  8 08:12 .bash_logout
-rw-r--r--  1 agilnggara agilnggara 3771 Oct  8 08:12 .bashrc
drwx----- 2 agilnggara agilnggara 4096 Oct  8 08:13 .cache
drwxr-xr-x  3 agilnggara agilnggara 4096 Oct 12 19:16 .local
-rw-rw-r--  1 agilnggara agilnggara  0 Oct 12 18:56 .motd_shown
-rw-r--r--  1 agilnggara agilnggara 807 Oct  8 08:12 .profile
-rw-r--r--  1 agilnggara agilnggara  0 Oct 12 19:23 .sudo_as_admin_successful
drwxr-xr-x  2 agilnggara agilnggara 4096 Oct 12 19:30 praktikum1
agilnggara@RagilAnggara:~$ cd /tmp
agilnggara@RagilAnggara:/tmp$ mkdir test_dir
agilnggara@RagilAnggara:/tmp$ cd test_dir
```

Pipelining

```
agilnggara@RagilAnggara: ~$ ls -l | grep "^d"
drwxr-xr-x  2 agilnggara agilnggara 4096 Oct 12 19:30 praktikum1
agilnggara@RagilAnggara:~$ ps aux | grep bash | wc -l
3
```

Redirection

```
agilnggara@RagilAnggara:~$ echo "Hello World" > output.txt
agilnggara@RagilAnggara:~$ cat output.txt
Hello World
agilnggara@RagilAnggara:~$ cat output.txt >> output.txt
cat: output.txt: input file is output file
agilnggara@RagilAnggara:~$ cat < output.txt
Hello World
```

Background Proseses

```
agilnggara@RagilAnggara:~$ sleep 30 &
[1] 488
agilnggara@RagilAnggara:~$ jobs
[1]+  Running                  sleep 30 &
agilnggara@RagilAnggara:~$ fg %1
sleep 30
```

2.2 Program C - Shell Sederhana

Buat file simple_shell.c:

```
agilnggara@RagilAnggara:~$ cd ~/praktikum1
agilnggara@RagilAnggara:~/praktikum1$ pwd
/home/agilnggara/praktikum1
agilnggara@RagilAnggara:~/praktikum1$ mkdir shell
agilnggara@RagilAnggara:~/praktikum1$ cd shell
agilnggara@RagilAnggara:~/praktikum1/shell$ nano simple_shell.c
```

```
GNU nano 7.2 simple_shell.c *
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <unistd.h>
#include <sys/wait.h>

#define MAX_INPUT 1024
#define MAX_ARGS 64

int main() {
    char input[MAX_INPUT];
    char *args[MAX_ARGS];

    while (1) {
        printf("myshell> ");
        fflush(stdout);

        // Baca input user
        if (fgets(input, MAX_INPUT, stdin) == NULL) {
            break;
        }

        // Hapus newline di akhir input
        input[strcspn(input, "\n")] = 0;

        // Jika user ketik "exit", keluar dari shell
        if (strcmp(input, "exit") == 0) {
            break;
        }

        // Pisahkan input jadi argumen (contoh: "ls -l" + ["ls", "-l"])
        int argc = 0;
        char *token = strtok(input, " ");
        while (token != NULL && argc < MAX_ARGS - 1) {
            args[argc++] = token;
            token = strtok(NULL, " ");
        }
        args[argc] = NULL; // penanda akhir argumen

        if (argc == 0) continue; // jika input kosong, lanjut ke loop berikutnya
    }
}
```

```
// Buat proses baru (fork)
pid_t pid = fork();

if (pid == 0) {
    // === CHILD PROCESS ===
    execvp(args[0], args); // Jalankan perintah
    perror("execvp failed"); // kalau gagal
    exit(1);
} else if (pid > 0) {
    // === PARENT PROCESS ===
    wait(NULL); // tunggu anak selesai
} else {
    perror("fork failed");
}

return 0;
}
```

Compile dan Jalankan

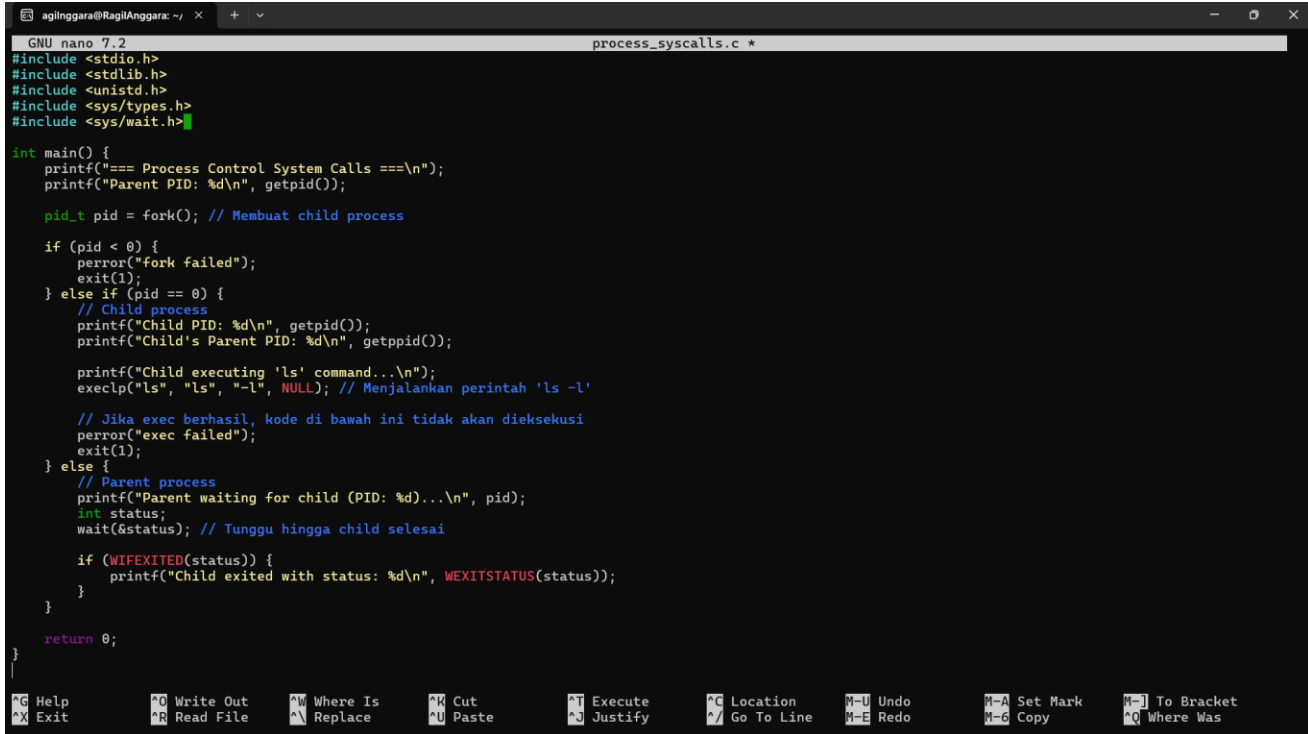
```
agilnggara@RagilAnggara:~/praktikum1/shell$ gcc -o simple_shell simple_shell.c
agilnggara@RagilAnggara:~/praktikum1/shell$ ./simple_shell
myshell> ls
simple_shell  simple_shell.c
myshell> pwd
/home/agilnggara/praktikum1/shell
myshell> echo Hello World
Hello World
```

BAB III SYSTEM CALLS

3.1 Process Control System Calls

Buat file `process_syscalls.c`

```
agilnggara@RagilAnggara:~$ cd ~/praktikum1
agilnggara@RagilAnggara:~/praktikum1$ pwd
/home/agilnggara/praktikum1
agilnggara@RagilAnggara:~/praktikum1$ mkdir syscalls
agilnggara@RagilAnggara:~/praktikum1$ nano process_syscalls.c
```



```
GNU nano 7.2 process_syscalls.c
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/types.h>
#include <sys/wait.h>

int main() {
    printf("=== Process Control System Calls ===\n");
    printf("Parent PID: %d\n", getpid());

    pid_t pid = fork(); // Membuat child process

    if (pid < 0) {
        perror("fork failed");
        exit(1);
    } else if (pid == 0) {
        // Child process
        printf("Child PID: %d\n", getpid());
        printf("Child's Parent PID: %d\n", getppid());

        printf("Child executing 'ls' command...\n");
        execvp("ls", "ls", "-l", NULL); // Menjalankan perintah 'ls -l'

        // Jika exec berhasil, kode di bawah ini tidak akan dieksekusi
        perror("exec failed");
        exit(1);
    } else {
        // Parent process
        printf("Parent waiting for child (PID: %d)...\n", pid);
        int status;
        wait(&status); // Tunggu hingga child selesai

        if (WIFEXITED(status)) {
            printf("Child exited with status: %d\n", WEXITSTATUS(status));
        }
    }

    return 0;
}
```

Compile dan Jalankan

```
agilnggara@RagilAnggara:~/praktikum1$ gcc -o process_syscalls process_syscalls.c
agilnggara@RagilAnggara:~/praktikum1$ ./process_syscalls
=== Process Control System Calls ===
Parent PID: 681
Parent waiting for child (PID: 682)...
Child PID: 682
Child's Parent PID: 681
Child executing 'ls' command...
total 56
-rwxr-xr-x 1 agilnggara agilnggara 16488 Oct 12 19:26 os_services
-rw-r--r-- 1 agilnggara agilnggara 1456 Oct 12 19:19 os_services.c
-rwxr-xr-x 1 agilnggara agilnggara 16360 Oct 13 00:33 process_syscalls
-rw-r--r-- 1 agilnggara agilnggara 1050 Oct 13 00:33 process_syscalls.c
drwxr-xr-x 2 agilnggara agilnggara 4096 Oct 12 23:09 shell
drwxr-xr-x 2 agilnggara agilnggara 4096 Oct 13 00:31 syscalls
-rw-r--r-- 1 agilnggara agilnggara 23 Oct 12 19:26 test_file.txt
Child exited with status: 0
```

3.2 File Management System Calls

Buat file `file_syscalls.c`

```
agilnggara@RagilAnggara:~$ cd ~/praktikum1
agilnggara@RagilAnggara:~/praktikum1$ cd syscalls
agilnggara@RagilAnggara:~/praktikum1/syscalls$ nano file_syscalls.c
```

```
GNU nano 7.2 file_syscalls.c *
#include <stdio.h>
#include <stdlib.h>
#include <fcntl.h>
#include <unistd.h>
#include <string.h>

int main() {
printf("=== File Management System Calls ===\n");
// open() system call
int fd = open("syscall_test.txt", O_CREAT | O_WRONLY | O_TRUNC, 0644);
if (fd < 0) {
perror("open failed");
exit(1);
}
printf("File descriptor: %d\n", fd);
// write() system call
const char *text = "Hello from system calls!\n";
ssize_t bytes_written = write(fd, text, strlen(text));
printf("Bytes written: %d\n", bytes_written);
// close() system call
close(fd);
printf("File closed\n");
// read() system call
fd = open("syscall_test.txt", O_RDONLY);
if (fd < 0) {
perror("open for reading failed");
exit(1);
}
char buffer[100];
ssize_t bytes_read = read(fd, buffer, sizeof(buffer) - 1);
buffer[bytes_read] = '\0';
printf("Read from file: %s", buffer);
close(fd);
// unlink() system call (delete file)
// unlink("syscall_test.txt");
return 0;
}

^G Help      ^O Write Out  ^W Where Is   ^K Cut        ^T Execute    ^C Location   ^U Undo       ^A Set Mark   ^- To Bracket
^X Exit      ^R Read File  ^N Replace    ^U Paste      ^J Justify    ^_ Go To Line ^E Redo       ^M Copy       ^Q Where Was
```

Compile dan Jalankan

```
agilnggara@RagilAnggara:~/praktikum1/syscalls$ gcc -o file_syscalls file_syscalls.c
agilnggara@RagilAnggara:~/praktikum1/syscalls$ ./file_syscalls
=== File Management System Calls ===
File descriptor: 3
Bytes written: 25
File closed
Read from file: Hello from system calls!
```

3.3 Information Maintenance System Calls

Buat file info_syscalls.c:

```
agilnggara@RagilAnggara:~$ cd ~/praktikum1
agilnggara@RagilAnggara:~/praktikum1$ cd syscalls
agilnggara@RagilAnggara:~/praktikum1/syscalls$ nano info_syscalls.c
```

```
GNU nano 7.2 info_syscalls.c *
#include <unistd.h>
#include <sys/types.h>
#include <time.h>
#include <sys/times.h>

int main() {
printf("=== Information Maintenance System Calls ===\n");
// getpid() and getppid()
printf("Process ID: %d\n", getpid());
printf("Parent Process ID: %d\n", getppid());
// getuid() and getgid()
printf("User ID: %d\n", getuid());
printf("Group ID: %d\n", getgid());
// time()
time_t current_time = time(NULL);
printf("Current time: %s", ctime(&current_time));
// times()
struct tms time_buf;
clock_t elapsed = times(&time_buf);
printf("Clock ticks since boot: %ld\n", elapsed);/ sleep()
printf("Sleeping for 2 seconds...\n");
sleep(2);
printf("Awake!\n");
return 0;
}
```

Compile dan Jalankan

```
agilnggara@RagilAnggara:~/praktikum1/syscalls$ gcc -o info_syscalls info_syscalls.c
agilnggara@RagilAnggara:~/praktikum1/syscalls$ ./info_syscalls
=== Information Maintenance System Calls ===

Process ID: 920
Parent Process ID: 892
User ID: 1000
Group ID: 1000
Current time: Mon Oct 13 09:45:22 2025
Clock ticks since boot: 1718116487
Sleeping for 2 seconds...
Awake!
```

BAB IV SYSTEM SERVICES

4.1 Menggunakan Systemd

Lihat Semua Service

```
agilnggara@RagilAnggara: ~$ systemctl list-units --type=service
UNIT                                LOAD    ACTIVE SUB    DESCRIPTION
console-getty.service              loaded active running Console Getty
console-setup.service              loaded active exited Set console font and keymap
cron.service                       loaded active running Regular background program processing daemon
dbus.service                       loaded active running D-Bus System Message Bus
getty@tty1.service                 loaded active running Getty on tty1
keyboard-setup.service             loaded active exited Set the console keyboard layout
kmod-static-nodes.service          loaded active exited Create List of Static Device Nodes
polkit.service                     loaded active running Authorization Manager
rsyslog.service                    loaded active running System Logging Service
setvtrgb.service                   loaded active exited Set console scheme
snapd.seeded.service              loaded active exited Wait until snapd is fully seeded
systemd-binfmt.service             loaded active exited Set Up Additional Binary Formats
systemd-journal-flush.service       loaded active exited Flush Journal to Persistent Storage
systemd-journald.service           loaded active running Journal Service
systemd-logind.service             loaded active running User Login Management
systemd-modules-load.service        loaded active exited Load Kernel Modules
systemd-remount-fs.service          loaded active exited Remount Root and Kernel File Systems
systemd-resolved.service            loaded active running Network Name Resolution
systemd-sysctl.service             loaded active exited Apply Kernel Variables
systemd-timesyncd.service           loaded active running Network Time Synchronization
systemd-tmpfiles-setup-dev-early.service loaded active exited Create Static Device Nodes in /dev gracefully
systemd-tmpfiles-setup-dev.service loaded active exited Create Static Device Nodes in /dev
systemd-tmpfiles-setup.service      loaded active exited Create Volatile Files and Directories
systemd-udev-trigger.service        loaded active exited Coldplug All udev Devices
systemd-udevd.service              loaded active running Rule-based Manager for Device Events and Files
systemd-update-utmp.service         loaded active exited Record System Boot/Shutdown in UTMP
systemd-user-sessions.service       loaded active exited Permit User Sessions
unattended-upgrades.service         loaded active running Unattended Upgrades Shutdown
lines 1-29... skipping...
UNIT                                LOAD    ACTIVE SUB    DESCRIPTION
console-getty.service              loaded active running Console Getty
console-setup.service              loaded active exited Set console font and keymap
cron.service                       loaded active running Regular background program processing daemon
dbus.service                       loaded active running D-Bus System Message Bus
getty@tty1.service                 loaded active running Getty on tty1
keyboard-setup.service             loaded active exited Set the console keyboard layout
kmod-static-nodes.service          loaded active exited Create List of Static Device Nodes
polkit.service                     loaded active running Authorization Manager
rsyslog.service                    loaded active running System Logging Service
setvtrgb.service                   loaded active exited Set console scheme
snapd.seeded.service              loaded active exited Wait until snapd is fully seeded
systemd-binfmt.service             loaded active exited Set Up Additional Binary Formats
systemd-journal-flush.service       loaded active exited Flush Journal to Persistent Storage
systemd-journald.service           loaded active running Journal Service
systemd-logind.service             loaded active running User Login Management
systemd-modules-load.service        loaded active exited Load Kernel Modules
systemd-remount-fs.service          loaded active exited Remount Root and Kernel File Systems
systemd-resolved.service            loaded active running Network Name Resolution
systemd-sysctl.service             loaded active exited Apply Kernel Variables
systemd-timesyncd.service           loaded active running Network Time Synchronization
systemd-tmpfiles-setup-dev-early.service loaded active exited Create Static Device Nodes in /dev gracefully
systemd-tmpfiles-setup-dev.service loaded active exited Create Static Device Nodes in /dev
systemd-tmpfiles-setup.service      loaded active exited Create Volatile Files and Directories
systemd-udev-trigger.service        loaded active exited Coldplug All udev Devices
systemd-udevd.service              loaded active running Rule-based Manager for Device Events and Files
systemd-update-utmp.service         loaded active exited Record System Boot/Shutdown in UTMP
systemd-user-sessions.service       loaded active exited Permit User Sessions
unattended-upgrades.service         loaded active running Unattended Upgrades Shutdown
user-runtime-dir@1000.service       loaded active exited User Runtime Directory /run/user/1000
user@1000.service                  loaded active running User Manager for UID 1000
wsl-pro.service                    loaded active running Bridge to Ubuntu Pro agent on Windows

Legend: LOAD    → Reflects whether the unit definition was properly loaded.
          ACTIVE → The high-level unit activation state, i.e. generalization of SUB.
          SUB    → The low-level unit activation state, values depend on unit type.

31 loaded units listed. Pass --all to see loaded but inactive units, too.
To show all installed unit files use 'systemctl list-unit-files'.
~
~
~
lines 1-39/39 (END)
```

Status Service

```
agilnggara@RagilAnggara: ~  
agilnggara@RagilAnggara:~$ systemctl status ssh  
○ ssh.service - OpenBSD Secure Shell server  
   Loaded: loaded (/usr/lib/systemd/system/ssh.service; disabled; preset:  
   Active: inactive (dead)  
TriggeredBy: ● ssh.socket  
   Docs: man:sshd(8)  
        man:sshd_config(5)  
[lines 1-6/6 (END)]...skipping...  
○ ssh.service - OpenBSD Secure Shell server  
   Loaded: loaded (/usr/lib/systemd/system/ssh.service; disabled; preset: enabled)  
   Active: inactive (dead)  
TriggeredBy: ● ssh.socket  
   Docs: man:sshd(8)  
        man:sshd_config(5)  
~  
~  
~  
~
```

Lihat Log Service Dan Lihat Service Yang Berjalan

```
agilnggara@RagilAnggara: ~  
agilnggara@RagilAnggara:~$ journalctl -u ssh -n 20  
-- No entries --  
agilnggara@RagilAnggara:~$ systemctl list-unit-files --type=service --state=enabled  
UNIT FILE                                STATE PRESET  
apparmor.service                        enabled enabled  
apport.service                         enabled enabled  
cloud-config.service                  enabled enabled  
cloud-final.service                   enabled enabled  
cloud-init-local.service              enabled enabled  
cloud-init.service                   enabled enabled  
console-setup.service                 enabled enabled  
cron.service                          enabled enabled  
dmesg.service                         enabled enabled  
e2scrub_reap.service                 enabled enabled  
getty@.service                       enabled enabled  
keyboard-setup.service               enabled enabled  
landscape-client.service             enabled enabled  
networkd-dispatcher.service          enabled enabled  
rsyslog.service                      enabled enabled  
setvtrgb.service                    enabled enabled  
snapd.apparmor.service              enabled enabled  
snapd.autoimport.service            enabled enabled  
snapd.core-fixup.service            enabled enabled  
snapd.recovery-chooser-trigger.service enabled enabled  
snapd.seeded.service               enabled enabled  
snapd.service                       enabled enabled  
snapd.system-shutdown.service        enabled enabled  
systemd-pstore.service              enabled enabled  
systemd-resolved.service            enabled enabled  
systemd-timesyncd.service           enabled enabled  
ua-reboot-cmds.service              enabled enabled  
ubuntu-advantage.service            enabled enabled  
unattended-upgrades.service         enabled enabled  
wsl-pro.service                     enabled enabled  
30 unit files listed.
```

4.2 Membuat Custom System Service

Buat program sederhana my_daemon.c

```
agilnggara@RagilAnggara: ~  
agilnggara@RagilAnggara:~$ cd ~/praktikum1  
agilnggara@RagilAnggara:~/praktikum1$ pwd  
/home/agilnggara/praktikum1  
agilnggara@RagilAnggara:~/praktikum1$ mkdir daemon  
agilnggara@RagilAnggara:~/praktikum1$ nano my_daemon.c
```

```
GNU nano 7.2 my_daemon.c *
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <syslog.h>
#include <signal.h>
#include <sys/stat.h>

void signal_handler(int sig) {
    if (sig == SIGTERM) {
        syslog(LOG_INFO, "My daemon stopped");
        closelog();
        exit(0);
    }
}

int main() {
    pid_t pid = fork();
    if (pid < 0) exit(1);
    if (pid > 0) exit(0); // Parent keluar, child jadi daemon

    if (setsid() < 0) exit(1);

    signal(SIGTERM, signal_handler);

    // Tutup stdin, stdout, stderr
    close(STDIN_FILENO);
    close(STDOUT_FILENO);
    close(STDERR_FILENO);

    // Tulis log ke syslog
    openlog("my_daemon", LOG_PID, LOG_DAEMON);
    syslog(LOG_INFO, "My daemon started");

    // Loop utama daemon
    while (1) {
        syslog(LOG_INFO, "Daemon is running...");
        sleep(30);
    }

    return 0;
}

^G Help      ^O Write Out  ^W Where Is   ^K Cut        ^T Execute    ^C Location   ^M-U Undo     ^M-A Set Mark ^M-] To Bracket
^X Exit      ^R Read File  ^\ Replace    ^U Paste      ^J Justify    ^/_ Go To Line ^M-E Redo     ^M-6 Copy     ^_Q Where Was
```

Compile dan Buat service file /etc/systemd/system/my_daemon.service

```
agilnggara@RagilAnggara:~/praktikum1$ gcc -o my_daemon my_daemon.c
agilnggara@RagilAnggara:~/praktikum1$ ls
daemon my_daemon my_daemon.c os_services os_services.c process_syscalls process_syscalls.c shell syscalls test_file.txt
agilnggara@RagilAnggara:~/praktikum1$ sudo nano /etc/systemd/system/my_daemon.service
[sudo] password for agilnggara:
```

```
GNU nano 7.2 /etc/systemd/system/my_daemon.service *
[Unit]
Description=My Custom Daemon
After=network.target

[Service]
Type=forking
ExecStart=/home/agilnggara/praktikum1/my_daemon
Restart=on-failure

[Install]
WantedBy=multi-user.target
```

Kelola Service

```
agilnggara@RagilAnggara:~/praktikum1$ sudo systemctl daemon-reload
agilnggara@RagilAnggara:~/praktikum1$ sudo systemctl start my_daemon
agilnggara@RagilAnggara:~/praktikum1$ sudo systemctl status my_daemon
● my_daemon.service - My Custom Daemon
   Loaded: loaded (/etc/systemd/system/my_daemon.service; disabled; preset: enabled)
   Active: active (running) since Mon 2025-10-13 13:03:25 WIB; 25s ago
     Process: 2088 ExecStart=/home/agilnggara/praktikum1/my_daemon (code=exited, status=0/SUCCESS)
    Main PID: 2089 (my_daemon)
       Tasks: 1 (limit: 9154)
      Memory: 436.0K (peak: 548.0K)
         CPU: 2ms
        CGroup: /system.slice/my_daemon.service
                └─2089 /home/agilnggara/praktikum1/my_daemon

Oct 13 13:03:25 RagilAnggara systemd[1]: Starting my_daemon.service - My Custom Daemon...
Oct 13 13:03:25 RagilAnggara my_daemon[2089]: My daemon started
Oct 13 13:03:25 RagilAnggara my_daemon[2089]: Daemon is running...
Oct 13 13:03:25 RagilAnggara systemd[1]: Started my_daemon.service - My Custom Daemon.
agilnggara@RagilAnggara:~/praktikum1$
```

```
agilnggara@RagilAnggara:~/praktikum1$ sudo journalctl -u my_daemon -n 20
Oct 13 12:57:31 RagilAnggara systemd[1]: Starting my_daemon.service - My Custom Daemon...
Oct 13 12:57:31 RagilAnggara systemd[1]: my_daemon.service: Control process exited, code=exited, status=203/EXEC
Oct 13 12:57:31 RagilAnggara systemd[1]: my_daemon.service: Failed with result 'exit-code'.
Oct 13 12:57:31 RagilAnggara systemd[1]: Failed to start my_daemon.service - My Custom Daemon.
Oct 13 12:57:31 RagilAnggara systemd[1]: my_daemon.service: Scheduled restart job, restart counter is at 5.
Oct 13 12:57:31 RagilAnggara systemd[1]: my_daemon.service: Start request repeated too quickly.
Oct 13 12:57:31 RagilAnggara systemd[1]: my_daemon.service: Failed with result 'exit-code'.
Oct 13 12:57:31 RagilAnggara systemd[1]: Failed to start my_daemon.service - My Custom Daemon.
Oct 13 13:03:25 RagilAnggara systemd[1]: Starting my_daemon.service - My Custom Daemon...
Oct 13 13:03:25 RagilAnggara my_daemon[2089]: My daemon started
Oct 13 13:03:25 RagilAnggara my_daemon[2089]: Daemon is running...
Oct 13 13:03:25 RagilAnggara systemd[1]: Started my_daemon.service - My Custom Daemon.
Oct 13 13:03:58 RagilAnggara my_daemon[2089]: Daemon is running...
Oct 13 13:04:30 RagilAnggara my_daemon[2089]: Daemon is running...
Oct 13 13:05:03 RagilAnggara my_daemon[2089]: Daemon is running...
Oct 13 13:05:37 RagilAnggara my_daemon[2089]: Daemon is running...
Oct 13 13:06:07 RagilAnggara my_daemon[2089]: Daemon is running...
Oct 13 13:06:40 RagilAnggara my_daemon[2089]: Daemon is running...
Oct 13 13:07:12 RagilAnggara my_daemon[2089]: Daemon is running...
Oct 13 13:07:45 RagilAnggara my_daemon[2089]: Daemon is running...
agilnggara@RagilAnggara:~/praktikum1$ sudo systemctl stop my_daemon
agilnggara@RagilAnggara:~/praktikum1$ sudo systemctl disable my_daemon
agilnggara@RagilAnggara:~/praktikum1$ sudo rm /etc/systemd/system/my_daemon.service
agilnggara@RagilAnggara:~/praktikum1$ sudo systemctl daemon-reload
agilnggara@RagilAnggara:~/praktikum1$
```

BAB V LINKERS AND LOADERS

5.1 Proses Linking

Buat 3 File;

```
agilnggara@RagilAnggara: ~$ cd ~/praktikum1
agilnggara@RagilAnggara:~/praktikum1$ pwd
/home/agilnggara/praktikum1
agilnggara@RagilAnggara:~/praktikum1$ mkdir linking
agilnggara@RagilAnggara:~/praktikum1$ cd linking
agilnggara@RagilAnggara:~/praktikum1/linking$ nano math_ops.h
```

```
GNU nano 7.2 math_ops.h *
#ifndef MATH_OPS_H
#define MATH_OPS_H

int add(int a, int b);
int multiply(int a, int b);

#endif
```

```
agilnggara@RagilAnggara:~/praktikum1/linking$ nano math_ops.c
```

```
GNU nano 7.2 math_ops.c *
#include "math_ops.h"

int add(int a, int b) {
    return a + b;
}

int multiply(int a, int b) {
    return a * b;
}
```

```
agilnggara@RagilAnggara:~/praktikum1/linking$ nano main.c
```

```
GNU nano 7.2 main.c *
#include <stdio.h>
#include "math_ops.h"

int main() {
    printf("5 + 3 = %d\n", add(5, 3));
    printf("5 * 3 = %d\n", multiply(5, 3));
    return 0;
}
```

Proses Kompilasi Bertahap

```
agilnggara@RagilAnggara:~/praktikum1/linking$ gcc -E main.c -o main.i
agilnggara@RagilAnggara:~/praktikum1/linking$ gcc -E math_ops.c -o math_ops.i
agilnggara@RagilAnggara:~/praktikum1/linking$ less main.i
```

```

agilnggara@RagilAnggara: ~/ + v
# 0 "<built-in>"
# 0 "<command-line>"
# 1 "/usr/include/stdc-predef.h" 1 3 4
# 0 "<command-line>" 2
# 1 "main.c"
# 1 "/usr/include/stdio.h" 1 3 4
# 28 "/usr/include/stdio.h" 3 4
# 1 "/usr/include/x86_64-linux-gnu/bits/libc-header-start.h" 1 3 4
# 33 "/usr/include/x86_64-linux-gnu/bits/libc-header-start.h" 3 4
# 1 "/usr/include/features.h" 1 3 4
# 394 "/usr/include/features.h" 3 4
# 1 "/usr/include/features-time64.h" 1 3 4
# 20 "/usr/include/features-time64.h" 3 4
# 1 "/usr/include/x86_64-linux-gnu/bits/wordsize.h" 1 3 4
# 21 "/usr/include/features-time64.h" 2 3 4
# 1 "/usr/include/x86_64-linux-gnu/bits/timesize.h" 1 3 4
# 19 "/usr/include/x86_64-linux-gnu/bits/timesize.h" 3 4
# 1 "/usr/include/x86_64-linux-gnu/bits/wordsize.h" 1 3 4
# 20 "/usr/include/x86_64-linux-gnu/bits/timesize.h" 2 3 4
# 22 "/usr/include/features-time64.h" 2 3 4
# 395 "/usr/include/features.h" 2 3 4
# 502 "/usr/include/features.h" 3 4
# 1 "/usr/include/x86_64-linux-gnu/sys/cdefs.h" 1 3 4
# 576 "/usr/include/x86_64-linux-gnu/sys/cdefs.h" 3 4
# 1 "/usr/include/x86_64-linux-gnu/bits/wordsize.h" 1 3 4
# 577 "/usr/include/x86_64-linux-gnu/sys/cdefs.h" 2 3 4
# 1 "/usr/include/x86_64-linux-gnu/bits/long-double.h" 1 3 4
# 578 "/usr/include/x86_64-linux-gnu/sys/cdefs.h" 2 3 4
# 503 "/usr/include/features.h" 2 3 4
# 526 "/usr/include/features.h" 3 4
# 1 "/usr/include/x86_64-linux-gnu/gnu/stubs.h" 1 3 4
# 10 "/usr/include/x86_64-linux-gnu/gnu/stubs.h" 3 4
# 1 "/usr/include/x86_64-linux-gnu/gnu/stubs-64.h" 1 3 4
# 11 "/usr/include/x86_64-linux-gnu/gnu/stubs.h" 2 3 4
# 527 "/usr/include/features.h" 2 3 4
# 34 "/usr/include/x86_64-linux-gnu/bits/libc-header-start.h" 2 3 4
# 29 "/usr/include/stdio.h" 2 3 4

```

```

agilnggara@RagilAnggara:~/praktikum1/linking$ gcc -S main.c -o main.s
agilnggara@RagilAnggara:~/praktikum1/linking$ gcc -S math_ops.c -o math_ops.s
agilnggara@RagilAnggara:~/praktikum1/linking$ cat main.s | head -20
.file "main.c"
.text
.section .rodata
.LC0:
.string "5 + 3 = %d\n"
.LC1:
.string "5 * 3 = %d\n"
.text
.globl main
.type main, @function
main:
.LFB0:
.cfi_startproc
endbr64
pushq %rbp
.cfi_def_cfa_offset 16
.cfi_offset 6, -16
movq %rsp, %rbp
.cfi_def_cfa_register 6
movl $3, %esi
agilnggara@RagilAnggara:~/praktikum1/linking$ |

```

```

agilnggara@RagilAnggara:~/praktikum1/linking$ gcc -c main.c -o main.o
agilnggara@RagilAnggara:~/praktikum1/linking$ gcc -c math_ops.c -o math_ops.o

```

```

agilnggara@RagilAnggara:~/praktikum1/linking$ gcc main.o math_ops.o -o program
agilnggara@RagilAnggara:~/praktikum1/linking$ ./program
5 + 3 = 8
5 * 3 = 15

```

```
agilnggara@RagilAnggara: ~/j + v
agilnggara@RagilAnggara:~/praktikum1/linking$ nm main.o
U add
0000000000000000 T main
U multiply
U printf
agilnggara@RagilAnggara:~/praktikum1/linking$ nm math_ops.o
0000000000000000 T add
0000000000000018 T multiply
agilnggara@RagilAnggara:~/praktikum1/linking$ nm program
0000000000003dc8 d _DYNAMIC
0000000000003fb8 d _GLOBAL_OFFSET_TABLE_
0000000000002000 R _IO_stdin_used
w _ITM_deregisterTMCloneTable
w _ITM_registerTMCloneTable
0000000000002148 r __FRAME_END__
000000000000201c r __GNU_EH_FRAME_HDR
0000000000004010 D __TMC_END__
000000000000038c r __abi_tag
0000000000004010 B __bss_start
w __cxa_finalize@GLIBC_2.2.5
D __data_start
0000000000001100 t __do_global_dtors_aux
0000000000003dc0 d __do_global_dtors_aux_fini_array_entry
0000000000004008 D __dso_handle
0000000000003db8 d __frame_dummy_init_array_entry
w __gmon_start__
U __libc_start_main@GLIBC_2.34
0000000000004010 D _edata
0000000000004018 B _end
00000000000011d4 T _fini
0000000000001000 T _init
0000000000001060 T _start
00000000000011a2 T add
0000000000004010 b completed.0
0000000000004000 W data_start
0000000000001090 t deregister_tm_clones
0000000000001140 t frame_dummy
0000000000001149 T main
00000000000011ba T multiply
U printf@GLIBC_2.2.5
00000000000010c0 t register_tm_clones

agilnggara@RagilAnggara:~/praktikum1/linking$ ldd program
linux-vdso.so.1 (0x00007ffdcdd27000)
libc.so.6 => /lib/x86_64-linux-gnu/libc.so.6 (0x000078c127e00000)
/lib64/ld-linux-x86-64.so.2 (0x000078c128031000)
agilnggara@RagilAnggara:~/praktikum1/linking$
```

5.2 Static vs Dynamic Linking

Static Linking

```
agilnggara@RagilAnggara:~/praktikum1/linking$ gcc main.c math_ops.c -o program_static -static
agilnggara@RagilAnggara:~/praktikum1/linking$ ls -lh program_static
-rwxr-xr-x 1 agilnggara agilnggara 767K Oct 13 16:51 program_static
```

Dynamic linking (default)

```
agilnggara@RagilAnggara:~/praktikum1/linking$ gcc main.c math_ops.c -o program_dynamic
agilnggara@RagilAnggara:~/praktikum1/linking$ ls -lh program_dynamic
-rwxr-xr-x 1 agilnggara agilnggara 16K Oct 13 16:53 program_dynamic
```

Bandingkan ukuran

```
agilnggara@RagilAnggara:~/praktikum1/linking$ ls -lh program_static program_dynamic
-rwxr-xr-x 1 agilnggara agilnggara 16K Oct 13 16:53 program_dynamic
-rwxr-xr-x 1 agilnggara agilnggara 767K Oct 13 16:51 program_static
```

Lihat dependencies

```
agilnggara@RagilAnggara:~/praktikum1/linking$ ldd program_static
not a dynamic executable
agilnggara@RagilAnggara:~/praktikum1/linking$ ldd program_dynamic
linux-vdso.so.1 (0x00007ffcc64c5000)
libc.so.6 => /lib/x86_64-linux-gnu/libc.so.6 (0x00007329aee00000)
/lib64/ld-linux-x86-64.so.2 (0x00007329af0c4000)
agilnggara@RagilAnggara:~/praktikum1/linking$
```

5.3 Membuat dan Menggunakan Shared Library

Compile sebagai shared library

```

agilnggara@RagilAnggara:~/praktikum1/linking$ gcc -c -fPIC math_ops.c -o math_ops.o
agilnggara@RagilAnggara:~/praktikum1/linking$ gcc -shared -o libmathops.so math_ops.o
agilnggara@RagilAnggara:~/praktikum1/linking$ ls
libmathops.so  main.c  main.i  main.o  main.s  math_ops.c  math_ops.h  math_ops.i  math_ops.o  math_ops.s  program  program_dynamic  program_static
agilnggara@RagilAnggara:~/praktikum1/linking$ |

```

Compile program yang menggunakan library & Jalankan dengan LD_LIBRARY_PATH

```

agilnggara@RagilAnggara:~/praktikum1/linking$ gcc main.c -L. -lmathops -o program_shared
agilnggara@RagilAnggara:~/praktikum1/linking$ LD_LIBRARY_PATH=. ./program_shared
5 + 3 = 8
5 * 3 = 15
agilnggara@RagilAnggara:~/praktikum1/linking$ |

```

Lihat informasi loader

```

agilnggara@RagilAnggara:~/praktikum1/linking$ readelf -d program_shared
Dynamic section at offset 0x2da8 contains 28 entries:
  Tag               Type              Name/Value
0x0000000000000001 (NEEDED)          Shared library: [libmathops.so]
0x0000000000000001 (NEEDED)          Shared library: [libc.so.6]
0x000000000000000c (INIT)            0x1000
0x000000000000000d (FINI)            0x11e4
0x0000000000000019 (INIT_ARRAY)      0x3d98
0x000000000000001b (INIT_ARRAYSZ)      8 (bytes)
0x000000000000001a (FINI_ARRAY)      0x3da0
0x000000000000001c (FINI_ARRAYSZ)      8 (bytes)
0x000000006ffffef5 (GNU_HASH)          0x3b0
0x0000000000000005 (STRTAB)            0x4b0
0x0000000000000006 (SYMTAB)            0x3d8
0x000000000000000a (STRSZ)              170 (bytes)
0x000000000000000b (SYMENT)             24 (bytes)
0x0000000000000015 (DEBUG)              0x0
0x0000000000000003 (PLTGOT)             0x3fa8
0x0000000000000002 (PLTRELSZ)           72 (bytes)
0x0000000000000014 (PLTREL)             RELA
0x0000000000000017 (JMPREL)             0x660
0x0000000000000007 (RELA)               0x5a0
0x0000000000000008 (RELASZ)             192 (bytes)
0x0000000000000009 (RELAENT)            24 (bytes)
0x000000000000001e (FLAGS)              BIND_NOW
0x000000006fffffbb (FLAGS_1)            Flags: NOW PIE
0x000000006ffffffe (VERNEED)            0x570
0x000000006fffffff (VERNEEDNUM)         1
0x000000006ffffff0 (VERSYM)             0x55a
0x000000006ffffff9 (RELACOUNT)          3
0x0000000000000000 (NULL)               0x0
agilnggara@RagilAnggara:~/praktikum1/linking$ |

```

BAB VI

WHY APPLICATIONS ARE OPERATING-SYSTEM SPECIFIC

6.1 System Call Differences

Buat file `os_specific.c`

```
agilnggara@RagilAnggara: ~/ +
agilnggara@RagilAnggara:~$ cd ~/praktikum1
agilnggara@RagilAnggara:~/praktikum1$ mkdir os_specific
agilnggara@RagilAnggara:~/praktikum1$ cd os_specific
agilnggara@RagilAnggara:~/praktikum1/os_specific$ nano os_specific.c

GNU nano 7.2 os_specific.c *
#include <stdio.h>

#ifdef __linux__
#include <unistd.h>
#include <sys/utsname.h>

void print_os_info() {
    struct utsname info;
    uname(&info);
    printf("OS: Linux\n");
    printf("System: %s\n", info.sysname);
    printf("Release: %s\n", info.release);
    printf("Version: %s\n", info.version);
}

#elif _WIN32
#include <windows.h>

void print_os_info() {
    printf("OS: Windows\n");
    // Windows-specific code
}

#else
void print_os_info() {
    printf("Unknown OS\n");
}
#endif

int main() {
    printf("=== OS-Specific Application ===\n");
    print_os_info();

#ifdef __linux__
    printf("\nLinux-specific: Process ID = %d\n", getpid());
#endif

    return 0;
}

Help Write Out Where Is Cut Execute Location M-U Undo M-A Set Mark M-] To Bracket
Exit Read File Replace AU Paste AO Justify G Go To Line M-E Redo M-G Copy M-Q Where Was
```

Compile dan Jalankan

```
agilnggara@RagilAnggara:~/praktikum1/os_specific$ gcc -o os_specific os_specific.c
agilnggara@RagilAnggara:~/praktikum1/os_specific$ ./os_specific
=== OS-Specific Application ===
OS: Linux
System: Linux
Release: 6.6.87.2-microsoft-standard-WSL2
Version: #1 SMP PREEMPT_DYNAMIC Thu Jun  5 18:30:46 UTC 2025
Linux-specific: Process ID = 609
agilnggara@RagilAnggara:~/praktikum1/os_specific$
```

6.2 Binary Format Analysis

Lihat format executable

```
agilnggara@RagilAnggara:~/praktikum1/os_specific$ file os_specific
os_specific: ELF 64-bit LSB pie executable, x86-64, version 1 (SYSV), dynamically linked, interpreter /lib64/ld-linux-x86-64.so.2, BuildID[sha1]=334d4d381bc
b2c27c8d33b5f83d6d9031f808181, for GNU/Linux 3.2.0, not stripped
agilnggara@RagilAnggara:~/praktikum1/os_specific$
```

Lihat header ELF

```
agilnggara@RagilAnggara:~/praktikum1/os_specific$ readelf -h os_specific
ELF Header:
  Magic:   7f 45 4c 46 02 01 01 00 00 00 00 00 00 00 00 00
  Class:       ELF64
  Data:       2's complement, little endian
  Version:    1 (current)
  OS/ABI:     UNIX - System V
  ABI Version: 0
  Type:       DYN (Position-Independent Executable file)
  Machine:    Advanced Micro Devices X86-64
  Version:    0x1
  Entry point address: 0x10e0
  Start of program headers: 64 (bytes into file)
  Start of section headers: 14200 (bytes into file)
  Flags:      0x0
  Size of this header: 64 (bytes)
  Size of program headers: 56 (bytes)
  Number of program headers: 13
  Size of section headers: 64 (bytes)
  Number of section headers: 31
  Section header string table index: 30
agilnggara@RagilAnggara:~/praktikum1/os_specific$ |
```

Lihat section headers

```
agilnggara@RagilAnggara:~/praktikum1/os_specific$ readelf -s os_specific

Symbol table '.dynsym' contains 11 entries:
Num:  Value              Size Type Bind Vis Ndx Name
 0: 0000000000000000      0 NOTYPE LOCAL DEFAULT UND
 1: 0000000000000000      0 FUNC GLOBAL DEFAULT UND _[...]@GLIBC_2.34 (2)
 2: 0000000000000000      0 NOTYPE WEAK DEFAULT UND _ITM_deregisterT[...]
 3: 0000000000000000      0 FUNC GLOBAL DEFAULT UND puts@GLIBC_2.2.5 (3)
 4: 0000000000000000      0 FUNC GLOBAL DEFAULT UND [...]@GLIBC_2.2.5 (3)
 5: 0000000000000000      0 FUNC GLOBAL DEFAULT UND [...]@GLIBC_2.4 (4)
 6: 0000000000000000      0 FUNC GLOBAL DEFAULT UND [...]@GLIBC_2.2.5 (3)
 7: 0000000000000000      0 FUNC GLOBAL DEFAULT UND uname@GLIBC_2.2.5 (3)
 8: 0000000000000000      0 NOTYPE WEAK DEFAULT UND __gmon_start__
 9: 0000000000000000      0 NOTYPE WEAK DEFAULT UND _ITM_registerTMC[...]
10: 0000000000000000      0 FUNC WEAK DEFAULT UND [...]@GLIBC_2.2.5 (3)

Symbol table '.symtab' contains 41 entries:
Num:  Value              Size Type Bind Vis Ndx Name
 0: 0000000000000000      0 NOTYPE LOCAL DEFAULT UND
 1: 0000000000000000      0 FILE LOCAL DEFAULT ABS Sctrl.o
 2: 0000000000000038c     32 OBJECT LOCAL DEFAULT 4 __abi_tag
 3: 0000000000000000      0 FILE LOCAL DEFAULT ABS crtstuff.c
 4: 00000000000001110      0 FUNC LOCAL DEFAULT 16 deregister_tm_clones
 5: 00000000000001140      0 FUNC LOCAL DEFAULT 16 register_tm_clones
 6: 00000000000001180      0 FUNC LOCAL DEFAULT 16 __do_global_ctors_aux
 7: 00000000000004010      1 OBJECT LOCAL DEFAULT 26 completed.0
 8: 00000000000003da0      0 OBJECT LOCAL DEFAULT 22 __do_global_dtor[...]
 9: 000000000000011c0      0 FUNC LOCAL DEFAULT 16 frame_dummy
10: 00000000000003498      0 OBJECT LOCAL DEFAULT 21 __frame_dummy_in[...]
11: 0000000000000000      0 FILE LOCAL DEFAULT ABS os_specific.c
12: 0000000000000000      0 FILE LOCAL DEFAULT ABS crtstuff.c
13: 00000000000002180      0 OBJECT LOCAL DEFAULT 20 __FRAME_END__
14: 0000000000000000      0 FILE LOCAL DEFAULT ABS
15: 00000000000003da8      0 OBJECT LOCAL DEFAULT 23 __DYNAMIC
16: 0000000000000207c      0 NOTYPE LOCAL DEFAULT 19 __GNU_EH_FRAME_HDR
17: 00000000000003f98      0 OBJECT LOCAL DEFAULT 24 __GLOBAL_OFFSET_TABLE_
18: 0000000000000000      0 FUNC GLOBAL DEFAULT UND __libc_start_mai[...]
19: 0000000000000000      0 NOTYPE WEAK DEFAULT UND _ITM_deregisterT[...]
20: 00000000000004000      0 NOTYPE WEAK DEFAULT 25 data_start
21: 0000000000000000      0 FUNC GLOBAL DEFAULT UND puts@GLIBC_2.2.5
22: 0000000000000000      0 FUNC GLOBAL DEFAULT UND getpid@GLIBC_2.2.5
23: 00000000000004010      0 NOTYPE GLOBAL DEFAULT 25 _edata
24: 00000000000012c8      0 FUNC GLOBAL HIDDEN 17 _fini
25: 0000000000000000      0 FUNC GLOBAL DEFAULT UND __stack_chk_fail[...]

26: 0000000000000000      0 FUNC GLOBAL DEFAULT UND printf@GLIBC_2.2.5
27: 0000000000000000      0 FUNC GLOBAL DEFAULT UND uname@GLIBC_2.2.5
28: 00000000000004000      0 NOTYPE GLOBAL DEFAULT 25 __data_start
29: 0000000000000000      0 NOTYPE WEAK DEFAULT UND __gmon_start__
30: 00000000000004008      0 OBJECT GLOBAL HIDDEN 25 __dso_handle
31: 0000000000002000      4 OBJECT GLOBAL DEFAULT 18 _IO_stdin_used
32: 00000000000004018      0 NOTYPE GLOBAL DEFAULT 26 __end
33: 00000000000010e0      38 FUNC GLOBAL DEFAULT 16 __start
34: 00000000000004010      0 NOTYPE GLOBAL DEFAULT 26 __bss_start
35: 0000000000001282      67 FUNC GLOBAL DEFAULT 16 main
36: 00000000000011c9      185 FUNC GLOBAL DEFAULT 16 print_os_info
37: 00000000000004010      0 OBJECT GLOBAL HIDDEN 25 __TMC_END__
38: 0000000000000000      0 NOTYPE WEAK DEFAULT UND _ITM_registerTMC[...]
39: 0000000000000000      0 FUNC WEAK DEFAULT UND __cxa_finalize@G[...]
40: 0000000000001000      0 FUNC GLOBAL HIDDEN 12 __init

agilnggara@RagilAnggara:~/praktikum1/os_specific$ |
```

Lihat program headers

```
agilnggara@RagilAnggara: ~/X + v
agilnggara@RagilAnggara:~/praktikum1/os_specific$ readelf -l os_specific

ELF file type is DYN (Position-Independent Executable file)
Entry point 0x10e0
There are 13 program headers, starting at offset 64

Program Headers:
Type           Offset             VirtAddr           PhysAddr
               FileSiz            MemSiz             Flags  Align
PHDR           0x0000000000000040 0x0000000000000040 0x0000000000000040
               0x0000000000002d8 0x0000000000002d8  R      0x8
INTERP         0x000000000000318 0x000000000000318 0x000000000000318
               0x00000000000001c 0x00000000000001c  R      0x1
LOAD [Requesting program interpreter: /lib64/ld-linux-x86-64.so.2]
               0x000000000000000 0x000000000000000 0x000000000000000
               0x000000000000730 0x000000000000730  R      0x1000
LOAD           0x000000000001000 0x000000000001000 0x0000000000001000
               0x0000000000002d5 0x0000000000002d5  R E    0x1000
LOAD           0x000000000002000 0x000000000002000 0x0000000000002000
               0x000000000000184 0x000000000000184  R      0x1000
LOAD           0x000000000003d98 0x000000000003d98 0x0000000000003d98
               0x000000000000278 0x000000000000278  RW     0x1000
DYNAMIC        0x000000000002da8 0x000000000002da8 0x0000000000003da8
               0x0000000000001f0 0x0000000000001f0  RW     0x8
NOTE           0x000000000000338 0x000000000000338 0x000000000000338
               0x000000000000030 0x000000000000030  R      0x8
NOTE           0x000000000000368 0x000000000000368 0x000000000000368
               0x000000000000044 0x000000000000044  R      0x4
GNU_PROPERTY   0x000000000000338 0x000000000000338 0x000000000000338
               0x000000000000030 0x000000000000030  R      0x8
GNU_EH_FRAME   0x000000000000207c 0x000000000000207c 0x000000000000207c
               0x00000000000003c 0x00000000000003c  R      0x4
GNU_STACK      0x0000000000000000 0x0000000000000000 0x0000000000000000
               0x000000000000000 0x000000000000000  RW     0x10
GNU_RELRO      0x0000000000003d98 0x0000000000003d98 0x0000000000003d98
               0x000000000000268 0x000000000000268  R      0x1

Section to Segment mapping:
Segment Sections...
00
01
02 .interp
03 .init .plt .got .plt.sec .text .fini
04 .rodata .eh_frame_hdr .eh_frame
05 .init_array .fini_array .dynamic .got .data .bss
06 .dynamic
07 .note.gnu.property
08 .note.gnu.build-id .note.ABI-tag
09 .note.gnu.property
10 .eh_frame_hdr
11
12 .init_array .fini_array .dynamic .got

agilnggara@RagilAnggara:~/praktikum1/os_specific$ |
```

Disassemble

```
agilnggara@RagilAnggara: ~/X + v
agilnggara@RagilAnggara:~/praktikum1/os_specific$ objdump -d os_specific | head -50

os_specific:      file format elf64-x86-64

Disassembly of section .init:

000000000001000 <.init>:
1000: f3 0f 1e fa          endbr64
1004: 48 83 ec 08          sub    $0x8,%rsp
1008: 48 8b 05 d9 2f 00 00 mov    0x2fd9(%rip),%rax      # 3fe8 <__gmon_start__@Base>
100f: 48 85 c0             test   %rax,%rax
1012: 74 02              je     1016 <_init+0x16>
1014: ff d0              call   %rax
1016: 48 83 c4 08          add    $0x8,%rsp
101a: c3                 ret

Disassembly of section .plt:

000000000001020 <.plt>:
1020: ff 35 7a 2f 00 00    push   0x2f7a(%rip)          # 3fa0 <_GLOBAL_OFFSET_TABLE_+0x8>
1026: ff 25 7c 2f 00 00    jmp    *0x2f7c(%rip)        # 3fa8 <_GLOBAL_OFFSET_TABLE_+0x10>
102c: 0f 1f 40 00          nopl   0x0(%rax)
1030: f3 0f 1e fa          endbr64
1034: 68 00 00 00 00      push   $0x0
1039: e9 c2 ff ff ff      jmp    1020 <_init+0x20>
103e: 66 90              xchg   %ax,%ax
1040: f3 0f 1e fa          endbr64
1044: 68 01 00 00 00      push   $0x1
1049: e9 d2 ff ff ff      jmp    1020 <_init+0x20>
104e: 66 90              xchg   %ax,%ax
1050: f3 0f 1e fa          endbr64
1054: 68 02 00 00 00      push   $0x2
1059: e9 c2 ff ff ff      jmp    1020 <_init+0x20>
105e: 66 90              xchg   %ax,%ax
1060: f3 0f 1e fa          endbr64
1064: 68 03 00 00 00      push   $0x3
1069: e9 b2 ff ff ff      jmp    1020 <_init+0x20>
106e: 66 90              xchg   %ax,%ax
1070: f3 0f 1e fa          endbr64
1074: 68 04 00 00 00      push   $0x4
1079: e9 a2 ff ff ff      jmp    1020 <_init+0x20>
107e: 66 90              xchg   %ax,%ax

Disassembly of section .plt.got:

000000000001080 <__cxa_finalize@plt>:
1080: f3 0f 1e fa          endbr64
1084: ff 25 6e 2f 00 00    jmp    *0x2f6e(%rip)        # 3ff8 <__cxa_finalize@GLIBC_2.2.5>
108a: 66 0f 1f 44 00 00    nopw   0x0(%rax,%rax,1)

agilnggara@RagilAnggara:~/praktikum1/os_specific$ |
```

BAB VII

OS Design and Implementation

7.1 Kernel Module Sederhana

Buat file `hello_module.c`;

```
agilnggara@RagilAnggara: ~/ x + v
agilnggara@RagilAnggara:~$ cd ~/praktikum1
agilnggara@RagilAnggara:~/praktikum1$ mkdir kernel_module
agilnggara@RagilAnggara:~/praktikum1$ cd kernel_module
agilnggara@RagilAnggara:~/praktikum1/kernel_module$ nano hello_module.c
```

```
GNU nano 7.2 hello_module.c *
#include <linux/init.h>
#include <linux/module.h>
#include <linux/kernel.h>

MODULE_LICENSE("GPL");
MODULE_AUTHOR("Mahasiswa Sistem Operasi");
MODULE_DESCRIPTION("A simple Linux kernel module");
MODULE_VERSION("1.0");

// Fungsi yang dijalankan saat modul dimuat
static int __init hello_init(void) {
    printk(KERN_INFO "Hello Module: Loaded successfully!\n");
    return 0;
}

// Fungsi yang dijalankan saat modul dihapus
static void __exit hello_exit(void) {
    printk(KERN_INFO "Hello Module: Unloaded successfully!\n");
}

// Registrasikan fungsi init dan exit
module_init(hello_init);
module_exit(hello_exit);

^G Help      ^O Write Out  ^W Where Is   ^K Cut        ^T Execute
^X Exit      ^R Read File  ^\ Replace    ^U Paste      ^J Justify
^_ Location  ^_/ Go To Line ^M-U Undo     ^M-A Set Mark ^M-] To Bracket
^M-G Copy   ^M-Q Where Was
```

Makefile

```
agilnggara@RagilAnggara:~/praktikum1/kernel_module$ nano makefile
```

```
GNU nano 7.2 makefile
obj-m += hello_module.o

all:
    make -C /lib/modules/$(shell uname -r)/build M=$(PWD) modules

clean:
    make -C /lib/modules/$(shell uname -r)/build M=$(PWD) clean
```

Compile and Load

```
agilnggara@RagilAnggara:~/praktikum1/karnel_module$ make
make -C /lib/modules/6.6.87.2-microsoft-standard-WSL2/build M=/home/agilnggara/praktikum1/karnel_module modules
make[1]: *** /lib/modules/6.6.87.2-microsoft-standard-WSL2/build: No such file or directory. Stop.
make: *** [makefile:4: all] Error 2
agilnggara@RagilAnggara:~/praktikum1/karnel_module$ sudo insmod hello_module.ko
[sudo] password for agilnggara:
insmod: ERROR: could not load module hello_module.ko: No such file or directory
agilnggara@RagilAnggara:~/praktikum1/karnel_module$ lsmod | grep hello
agilnggara@RagilAnggara:~/praktikum1/karnel_module$ dmesg | tail
[ 4.530618] ACPI: battery: Slot [BAT1] (battery present)
[ 4.912486] kvm_amd: TSC scaling supported
[ 4.912495] kvm_amd: Nested Virtualization enabled
[ 4.912498] kvm_amd: Nested Paging enabled
[ 4.912504] kvm_amd: kvm_amd: Hyper-V enlightened NPT TLB flush enabled
[ 4.912506] kvm_amd: kvm_amd: Hyper-V Direct TLB Flush enabled
[ 4.912521] kvm_amd: Virtual VMLOAD VMSAVE supported
[ 6.735666] WSL (267) ERROR: CheckConnection: getaddrinfo() failed: -5
[ 7.767320] systemd-journald[44]: /var/log/journal/5e0b2178f2a64652ac34e2eddec893ce/user-1000.journal: Journal file uses a different sequence number ID,
rotating.
[ 48.707499] hv_balloon: Max. dynamic memory size: 7886 MB
agilnggara@RagilAnggara:~/praktikum1/karnel_module$ sudo rmmod hello_module
rmmod: ERROR: Module hello_module is not currently loaded
agilnggara@RagilAnggara:~/praktikum1/karnel_module$ dmesg | tail
[ 4.530618] ACPI: battery: Slot [BAT1] (battery present)
[ 4.912486] kvm_amd: TSC scaling supported
[ 4.912495] kvm_amd: Nested Virtualization enabled
[ 4.912498] kvm_amd: Nested Paging enabled
[ 4.912504] kvm_amd: kvm_amd: Hyper-V enlightened NPT TLB flush enabled
[ 4.912506] kvm_amd: kvm_amd: Hyper-V Direct TLB Flush enabled
[ 4.912521] kvm_amd: Virtual VMLOAD VMSAVE supported
[ 6.735666] WSL (267) ERROR: CheckConnection: getaddrinfo() failed: -5
[ 7.767320] systemd-journald[44]: /var/log/journal/5e0b2178f2a64652ac34e2eddec893ce/user-1000.journal: Journal file uses a different sequence number ID,
rotating.
[ 48.707499] hv_balloon: Max. dynamic memory size: 7886 MB
agilnggara@RagilAnggara:~/praktikum1/karnel_module$ |
```

7.2 Monitoring Kernel

Lihat informasi kernel

```
agilnggara@RagilAnggara:~/praktikum1/karnel_module$ uname -r
6.6.87.2-microsoft-standard-WSL2
agilnggara@RagilAnggara:~/praktikum1/karnel_module$ cat /proc/version
Linux version 6.6.87.2-microsoft-standard-WSL2 (root@439a258ad544) (gcc (GCC) 11.2.0, GNU ld (GNU Binutils) 2.37) #1 SMP PREEMPT_DYNAMIC Thu Jun  5 18:30:46
UTC 2025
agilnggara@RagilAnggara:~/praktikum1/karnel_module$ |
```

Lihat parameter kernel

```
agilnggara@RagilAnggara:~/praktikum1/karnel_module$ sysctl -a | head -20
sysctl: abi.vsyscall32 = 1
permission denied on key 'kernel.cad_pid'debug.exception-trace = 1

debug.kprobes-optimization = 1
dev.cdrom.autoclose = 1
dev.cdrom.autoeject = 0
dev.cdrom.check_media = 0
dev.cdrom.debug = 0
dev.cdrom.info = CD-ROM information, Id: cdrom.c 3.20 2003/12/17
dev.cdrom.info =
dev.cdrom.info = drive name:
dev.cdrom.info = drive speed:
dev.cdrom.info = drive # of slots:
dev.cdrom.info = Can close tray:
dev.cdrom.info = Can open tray:
dev.cdrom.info = Can lock tray:
dev.cdrom.info = Can change speed:
dev.cdrom.info = Can select disk:
dev.cdrom.info = Can read multisession:
dev.cdrom.info = Can read MCN:
dev.cdrom.info = Reports media changed:
```

Lihat kernel ring buffer

```
agilnggara@RagilAnggara:~/praktikum1/kernel_module$ dmesg | tail -30
[ 1.900487] sd 0:0:0:3: [sdd] Mode Sense: 0f 00 00 00
[ 1.902154] sd 0:0:0:3: [sdd] Write cache: enabled, read cache: enabled, doesn't support DPO or FUA
[ 1.907488] sd 0:0:0:3: [sdd] Attached SCSI disk
[ 2.433322] EXT4-fs (sdd): recovery complete
[ 2.435965] EXT4-fs (sdd): mounted filesystem 77bd2e57-3ee6-49e5-99a7-b00ce1b47a2f r/w with ordered data mode. Quota mode: none.
[ 3.438484] pulseaudio[298]: memfd_create() called without MFD_EXEC or MFD_NOEXEC_SEAL set
[ 3.628343] misc dxg: dxgk: dxgkio_is_feature_enabled: Ioctl failed: -22
[ 3.639099] misc dxg: dxgk: dxgkio_query_adapter_info: Ioctl failed: -22
[ 3.640519] misc dxg: dxgk: dxgkio_query_adapter_info: Ioctl failed: -22
[ 3.641839] misc dxg: dxgk: dxgkio_query_adapter_info: Ioctl failed: -22
[ 3.643404] misc dxg: dxgk: dxgkio_query_adapter_info: Ioctl failed: -2
[ 3.649825] misc dxg: dxgk: dxgkio_query_adapter_info: Ioctl failed: -22
[ 3.651154] misc dxg: dxgk: dxgkio_query_adapter_info: Ioctl failed: -22
[ 3.652611] misc dxg: dxgk: dxgkio_query_adapter_info: Ioctl failed: -22
[ 3.654175] misc dxg: dxgk: dxgkio_query_adapter_info: Ioctl failed: -2
[ 3.789353] Failed to connect to bus: No such file or directory
[ 4.050486] systemd-journald[44]: Collecting audit messages is disabled.
[ 4.155532] systemd-journald[44]: Received client request to flush runtime journal.
[ 4.157044] systemd-journald[44]: File /var/log/journal/5e0b2178f2a64652ac34e2eddec893ce/system.journal corrupted or uncleanly shut down, renaming and replacing.
[ 4.500100] ACPI: AC: AC Adapter [AC1] (off-line)
[ 4.530618] ACPI: battery: Slot [BAT1] (battery present)
[ 4.912486] kvm_amd: TSC scaling supported
[ 4.912495] kvm_amd: Nested Virtualization enabled
[ 4.912498] kvm_amd: Nested Paging enabled
[ 4.912504] kvm_amd: kvm_amd: Hyper-V enlightened NPT TLB flush enabled
[ 4.912506] kvm_amd: kvm_amd: Hyper-V Direct TLB Flush enabled
[ 4.912521] kvm_amd: Virtual VMLOAD VMSAVE supported
[ 6.735666] WSL (267) ERROR: CheckConnection: getaddrinfo() failed: -5
[ 7.767320] systemd-journald[44]: /var/log/journal/5e0b2178f2a64652ac34e2eddec893ce/user-1000.journal: Journal file uses a different sequence number ID, rotating.
[ 48.707499] hv_balloon: Max. dynamic memory size: 7886 MB
```

Lihat kernel modules

```
agilnggara@RagilAnggara:~/praktikum1/kernel_module$ lsmod
Module                  Size  Used by
intel_rapl_msr          16384   0
intel_rapl_common       32768   1 intel_rapl_msr
kvm_amd                 126976   0
kvm                     970752   1 kvm_amd
irqbypass              12288   1 kvm
crc32c_intel            16384   0
battery                 20480   0
ac                      16384   0
sch_fq_codel            16384   1
configfs                53248   1
autofs4                 45056   0
br_netfilter            28672   0
bridge                  282624   1 br_netfilter
stp                     12288   1 bridge
llc                     12288   2 bridge,stp
ip_tables                28672   0
tun                     53248   0
```

Informasi modul tertentu

```
agilnggara@RagilAnggara:~/praktikum1/kernel_module$ modinfo ext4
name: ext4
filename: (builtin)
softdep: pre: crc32c
license: GPL
file: fs/ext4/ext4
description: Fourth Extended Filesystem
author: Remy Card, Stephen Tweedie, Andrew Morton, Andreas Dilger, Theodore Ts'o and others
alias: fs-ext4
alias: ext3
alias: fs-ext3
```

BAB VIII

Operating-System Structure

8.1 Eksplorasi Struktur Layering

Hardware

```
agilnggara@RagilAnggara: ~  
agilnggara@RagilAnggara:~$ lspci  
Command 'lspci' not found, did you mean:  
  command 'lscpi' from deb nilfs-tools (2.2.9-1)  
  command 'lscpu' from deb util-linux (2.39.3-9ubuntu6.3)  
  command 'lspci' from deb pciutils (1:3.10.0-2)  
Try: sudo apt install <deb name>  
agilnggara@RagilAnggara:~$ lsusb  
agilnggara@RagilAnggara:~$ cat /proc/cpuinfo  
processor       : 0  
vendor_id      : AuthenticAMD  
cpu family     : 25  
model          : 80  
model name     : AMD Ryzen 9 5900HX with Radeon Graphics  
stepping       : 0  
microcode      : 0xffffffff  
cpu MHz        : 3293.728  
cache size     : 512 KB  
physical id    : 0  
siblings       : 16  
core id        : 0  
cpu cores      : 8  
apicid         : 0  
initial apicid : 0  
fpu            : yes  
fpu_exception  : yes  
cpuid level    : 13  
wp             : yes  
flags          : fpu vme de pse tsc msr pae mce cx8 apic sep mtrr pge mca cmov pat pse36 clflush mmx fxsr sse sse2 ht syscall nx mmxext fxsr_op  
t pdpe1gb rdtscp lm constant_tsc rep_good nopl tsc_reliable nonstop_tsc cpuid extd_apicid tsc_known_freq pni pclmulqdq ssse3 fma cx16 sse4_1 sse  
4_2 movbe popcnt aes xsave avx f16c rdrand hypervisor lahf_lm cmp_legacy svm cr8_legacy abm sse4a misalignsse 3dnowprefetch osvw topoext perfctr  
_core ssbd ibrs ibpb stibp vmmcall fsgsbase bmi1 avx2 smep bmi2 erms invpcid rdseed adx smap clflushopt clwb sha_ni xsaveopt xsavec xgetbv1 xsav  
es clzero xsaveerptr arat npt nrip_save tsc_scale vmcb_clean flushbyasid decodeassists pausefilter pfthreshold v_vmsave_vmload umip vaes vpclmul  
qdq rdpid fsrm  
bugs           : sysret_ss_attrs null_seg spectre_v1 spectre_v2 spec_store_bypass srs0 ibpb_no_ret  
bogomips       : 6587.45  
TLB size       : 2560 4K pages  
clflush size   : 64  
cache_alignmen : 64  
address sizes  : 48 bits physical, 48 bits virtual  
power managemt:
```

Kernel

```
agilnggara@RagilAnggara:~$ uname -a  
Linux RagilAnggara 6.6.87.2-microsoft-standard-WSL2 #1 SMP PREEMPT_DYNAMIC Thu Jun  5 18:30:46 UTC 2025 x86_64 x86_64 x86_64 GNU/Linux  
agilnggara@RagilAnggara:~$ cat /proc/sys/kernel/ostype  
Linux  
agilnggara@RagilAnggara:~$ |
```

System Calls

[illegible]

```

agilinggara@RagilAnggara: ~
operat(AT_FDCWD, "/proc/mounts", 0_RDONLY|O_CLOEXEC) = 3
fstab(3, {st_mode=S_IFREG|0644, st_size=0, ...}) = 0
read(3, "none /usr/lib/modules/6.6.87-2-wr... 1024) = 1024
read(3, "\devpts /dev/pts devpts rw,nosuid,... 1024) = 1024
read(3, "\lnt,size=65536,trans=fd,rfd=6,... 1024) = 745
read(3, "... 1024)
close(3)
access("/etc/selinux/config", F_OK) = -1 ENOENT (No such file or directory)
operat(AT_FDCWD, "/usr/lib/locale/locale-archive", 0_RDONLY|O_CLOEXEC) = -1 ENOENT (No such file or directory)
operat(AT_FDCWD, "/usr/share/locale/locale.alias", 0_RDONLY|O_CLOEXEC) = 3
fstab(3, {st_mode=S_IFREG|0644, st_size=2996, ...}) = 0
read(3, "\ Locale name alias data base.\n#... 4096) = 2996
read(3, "... 4096)
close(3)
operat(AT_FDCWD, "/usr/lib/locale/C.UTF-8/LC_IDENTIFICATION", 0_RDONLY|O_CLOEXEC) = -1 ENOENT (No such file or directory)
operat(AT_FDCWD, "/usr/lib/locale/C.UTF-8/LC_IDENTIFICATION", 0_RDONLY|O_CLOEXEC) = 3
fstab(3, {st_mode=S_IFREG|0644, st_size=258, ...}) = 0
mmap(NULL, 258, PROT_READ, MAP_PRIVATE, 3, 0) = 0x7869819e2000
close(3)
operat(AT_FDCWD, "/usr/lib/x86_64-linux-gnu/gconv/gconv-modules.cache", 0_RDONLY|O_CLOEXEC) = 3
fstab(3, {st_mode=S_IFREG|0644, st_size=27028, ...}) = 0
mmap(NULL, 27028, PROT_READ, MAP_SHARED, 3, 0) = 0x78698190d000
close(3)
futex(0x7869818b702c, FUTEX_WAKE_PRIVATE, 2147483647) = 0
operat(AT_FDCWD, "/usr/lib/locale/C.UTF-8/LC_MEASUREMENT", 0_RDONLY|O_CLOEXEC) = -1 ENOENT (No such file or directory)
operat(AT_FDCWD, "/usr/lib/locale/C.UTF-8/LC_MEASUREMENT", 0_RDONLY|O_CLOEXEC) = 3
fstab(3, {st_mode=S_IFREG|0644, st_size=23, ...}) = 0
mmap(NULL, 23, PROT_READ, MAP_PRIVATE, 3, 0) = 0x7869819e1000
close(3)
operat(AT_FDCWD, "/usr/lib/locale/C.UTF-8/LC_TELEPHONE", 0_RDONLY|O_CLOEXEC) = -1 ENOENT (No such file or directory)
operat(AT_FDCWD, "/usr/lib/locale/C.UTF-8/LC_TELEPHONE", 0_RDONLY|O_CLOEXEC) = 3
fstab(3, {st_mode=S_IFREG|0644, st_size=47, ...}) = 0
mmap(NULL, 47, PROT_READ, MAP_PRIVATE, 3, 0) = 0x7869819e0000
close(3)
operat(AT_FDCWD, "/usr/lib/locale/C.UTF-8/LC_ADDRESS", 0_RDONLY|O_CLOEXEC) = -1 ENOENT (No such file or directory)
operat(AT_FDCWD, "/usr/lib/locale/C.UTF-8/LC_ADDRESS", 0_RDONLY|O_CLOEXEC) = 3
fstab(3, {st_mode=S_IFREG|0644, st_size=127, ...}) = 0
mmap(NULL, 127, PROT_READ, MAP_PRIVATE, 3, 0) = 0x7869819df000
close(3)
operat(AT_FDCWD, "/usr/lib/locale/C.UTF-8/LC_NAME", 0_RDONLY|O_CLOEXEC) = -1 ENOENT (No such file or directory)
operat(AT_FDCWD, "/usr/lib/locale/C.UTF-8/LC_NAME", 0_RDONLY|O_CLOEXEC) = 3
fstab(3, {st_mode=S_IFREG|0644, st_size=62, ...}) = 0
mmap(NULL, 62, PROT_READ, MAP_PRIVATE, 3, 0) = 0x7869819de000
close(3)
operat(AT_FDCWD, "/usr/lib/locale/C.UTF-8/LC_PAPER", 0_RDONLY|O_CLOEXEC) = -1 ENOENT (No such file or directory)
operat(AT_FDCWD, "/usr/lib/locale/C.UTF-8/LC_PAPER", 0_RDONLY|O_CLOEXEC) = 3
fstab(3, {st_mode=S_IFREG|0644, st_size=34, ...}) = 0
mmap(NULL, 34, PROT_READ, MAP_PRIVATE, 3, 0) = 0x78698199c000
close(3)
operat(AT_FDCWD, "/usr/lib/locale/C.UTF-8/LC_MESSAGES", 0_RDONLY|O_CLOEXEC) = -1 ENOENT (No such file or directory)
operat(AT_FDCWD, "/usr/lib/locale/C.UTF-8/LC_MESSAGES", 0_RDONLY|O_CLOEXEC) = 3
fstab(3, {st_mode=S_IFDIR|0755, st_size=4096, ...}) = 0
close(3)
operat(AT_FDCWD, "/usr/lib/locale/C.UTF-8/LC_MESSAGES/SYS_LC_MESSAGES", 0_RDONLY|O_CLOEXEC) = 3
fstab(3, {st_mode=S_IFREG|0644, st_size=48, ...}) = 0
mmap(NULL, 48, PROT_READ, MAP_PRIVATE, 3, 0) = 0x78698190b000
close(3)
operat(AT_FDCWD, "/usr/lib/locale/C.UTF-8/LC_MONETARY", 0_RDONLY|O_CLOEXEC) = -1 ENOENT (No such file or directory)
operat(AT_FDCWD, "/usr/lib/locale/C.UTF-8/LC_MONETARY", 0_RDONLY|O_CLOEXEC) = 3
fstab(3, {st_mode=S_IFREG|0644, st_size=270, ...}) = 0
mmap(NULL, 270, PROT_READ, MAP_PRIVATE, 3, 0) = 0x78698190a000
close(3)
operat(AT_FDCWD, "/usr/lib/locale/C.UTF-8/LC_COLLATE", 0_RDONLY|O_CLOEXEC) = -1 ENOENT (No such file or directory)
operat(AT_FDCWD, "/usr/lib/locale/C.UTF-8/LC_COLLATE", 0_RDONLY|O_CLOEXEC) = 3
fstab(3, {st_mode=S_IFREG|0644, st_size=1406, ...}) = 0
mmap(NULL, 1406, PROT_READ, MAP_PRIVATE, 3, 0) = 0x786981909000
close(3)

```

```

close(1)
openat(AT_FDCWD, "/usr/lib/locale/C.UTF-8/LC_TIME", O_RDONLY|O_CLOEXEC) = -1 ENOENT (No such file or directory)
openat(AT_FDCWD, "/usr/lib/locale/C.utf8/LC_TIME", O_RDONLY|O_CLOEXEC) = 3
fstat(3, {st_mode=S_IFREG|0644, st_size=3360, ...}) = 0
mmap(NULL, 3360, PROT_READ, MAP_PRIVATE, 3, 0) = 0x786981980000
close(3)
openat(AT_FDCWD, "/usr/lib/locale/C.UTF-8/LC_NUMERIC", O_RDONLY|O_CLOEXEC) = -1 ENOENT (No such file or directory)
openat(AT_FDCWD, "/usr/lib/locale/C.utf8/LC_NUMERIC", O_RDONLY|O_CLOEXEC) = 3
fstat(3, {st_mode=S_IFREG|0644, st_size=50, ...}) = 0
mmap(NULL, 50, PROT_READ, MAP_PRIVATE, 3, 0) = 0x786981980700
close(3)
openat(AT_FDCWD, "/usr/lib/locale/C.UTF-8/LC_CTYPE", O_RDONLY|O_CLOEXEC) = -1 ENOENT (No such file or directory)
openat(AT_FDCWD, "/usr/lib/locale/C.utf8/LC_CTYPE", O_RDONLY|O_CLOEXEC) = 3
fstat(3, {st_mode=S_IFREG|0644, st_size=36460, ...}) = 0
mmap(NULL, 36460, PROT_READ, MAP_PRIVATE, 3, 0) = 0x78698198ae000
close(3)
ioctl(1, TCGETS, {c_iflag=ICRNL|IXON, c_oflag=NLCR|CR0|TAB0|B50|VTF0|PPOST|ONLCR, c_lflag=ISIG|ICANON|ECHO|ECHOK|IEXTEN|ECHOCTL|ECHONL, ...}) = 0
ioctl(1, TIOCGWINSZ, {ws_row=40, ws_col=140, ws_xpixel=0, ws_ypixel=0}) = 0
openat(AT_FDCWD, "...", O_RDONLY|O_NONBLOCK|O_CLOEXEC|O_DIRECTORY) = 3
fstat(3, {st_mode=S_IFDIR|0750, st_size=4096, ...}) = 0
getdents64(3, 0x568f256868a0 /* 13 entries */, 32768) = 424
getdents64(3, 0x568f256868a0 /* 0 entries */, 32768) = 0
close(3)
fstat(1, {st_mode=S_IFCHR|0620, st_rdev=makedev(0x88, 0), ...}) = 0
write(1, "output.txt praktikum1\n", 23output.txt praktikum1
) = 23
close(1)
close(2)
exit_group(0)
+++ exited with 0 +++
agilnggara@RagilAnggara:~$ |

```

System Programs

```

agilnggara@RagilAnggara:~$ ps aux
USER      PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND
root         1  0.1  0.1 21824 12132 ?        Ss   19:49   0:00 /sbin/init
root         2  0.0  0.0  3072  1792 ?        Sl   19:49   0:00 /init
root         7  0.0  0.0  3072  1792 ?        Sl   19:49   0:00 plan9 --control-socket 7 --log-level 4 --server-fd 8 --pipe-fd 10 --log-truncate
root        41  0.0  0.1 50428 15488 ?        S<s  19:49   0:00 /usr/lib/systemd/systemd-journald
root        91  0.0  0.0 25052  6144 ?        Ss   19:49   0:00 /usr/lib/systemd/systemd-udev
systemd+  112  0.0  0.1 21456 12160 ?        Ss   19:49   0:00 /usr/lib/systemd/systemd-resolved
systemd+  113  0.0  0.0  91024  7552 ?        Ssl  19:49   0:00 /usr/lib/systemd/systemd-timesyncd
root       180  0.0  0.0  4236  2560 ?        Ss   19:49   0:00 /usr/sbin/cron -f -p
message+  181  0.0  0.0  9596  4864 ?        Ss   19:49   0:00 @dbus-daemon --system --address=systemd: --nofork --nopidfile --systemd-activation --syslog-only
root       188  0.0  0.1 17960  8192 ?        Ss   19:49   0:00 /usr/lib/systemd/systemd-logind
root       190  0.0  0.1 1756096 12160 ?       Ssl  19:49   0:00 /usr/libexec/wsl-pro-service -vv
syslog    196  0.0  0.0 222508  5504 ?        Ssl  19:49   0:00 /usr/sbin/rsyslogd -n -iNONE
root      204  0.0  0.0  3160  1920 hvc0     Ss+  19:49   0:00 /sbin/agetty -o -p -- \u --noclear --keep-baud - 115200,38400,9600 vt220
root      212  0.0  0.0  3116  1792 tty1     Ss+  19:49   0:00 /sbin/agetty -o -p -- \u --noclear - linux
root      218  0.0  0.2 107032 22528 ?       Ssl  19:49   0:00 /usr/bin/python3 /usr/share/unattended-upgrades/unattended-upgrade-shutdown --wait-for-signal
root      304  0.0  0.0  3076   896 ?        Ss   19:49   0:00 /init
root      305  0.0  0.0  3092  1024 ?        S   19:49   0:00 /init
agilngg+  306  0.0  0.0  6072  4992 pts/0     Ss   19:49   0:00 -bash
root      308  0.0  0.0  6660  4480 pts/1     Ss   19:49   0:00 /bin/login -f
agilngg+  354  0.0  0.1 20308 11264 ?       Ss   19:49   0:00 /usr/lib/systemd/systemd --user
agilngg+  355  0.0  0.0 21156  3520 ?        S   19:49   0:00 (sd-pam)
agilngg+  379  0.0  0.0  6072  4992 pts/1     S+   19:49   0:00 -bash
root      559  0.0  0.2 370084 20352 ?       Ssl  19:52   0:00 /usr/libexec/packagekitd
polkitd   564  0.0  0.0 308164  7680 ?        Ssl  19:52   0:00 /usr/lib/polkit-1/polkitd --no-debug
agilngg+  593  0.0  0.0  8200  4096 pts/0     R+   19:57   0:00 ps aux
agilnggara@RagilAnggara:~$ |

```

```

agilnggara@RagilAnggara:~$ top
top - 19:58:30 up 9 min, 1 user, load average: 0.00, 0.00, 0.00
Tasks: 24 total, 1 running, 23 sleeping, 0 stopped, 0 zombie
%Cpu(s):  0.0 us,  0.0 sy,  0.0 ni,100.0 id,  0.0 wa,  0.0 hi,  0.0 si,  0.0 st
MiB Mem : 7638.2 total, 6670.8 free,  553.9 used, 566.1 buff/cache
MiB Swap: 2048.0 total, 2048.0 free,  0.0 used, 7084.3 avail Mem

```

	PID	USER	PR	NI	VIRT	RES	SHR	S	%CPU	%MEM	TIME+	COMMAND
	595	agilngg+	20	0	9308	5504	3328	R	0.3	0.1	0:00.02	top
	1	root	20	0	21824	12132	9188	S	0.0	0.2	0:00.53	systemd
	2	root	20	0	3072	1792	1792	S	0.0	0.0	0:00.00	init-systemd(Ub
	7	root	20	0	3072	1792	1792	S	0.0	0.0	0:00.00	init
	41	root	19	-1	50428	15488	14720	S	0.0	0.2	0:00.16	systemd-journal
	91	root	20	0	25052	6144	4864	S	0.0	0.1	0:00.14	systemd-udev
	112	systemd+	20	0	21456	12160	10240	S	0.0	0.2	0:00.11	systemd-resolve
	113	systemd+	20	0	91024	7552	6656	S	0.0	0.1	0:00.08	systemd-timesyn
	180	root	20	0	4236	2560	2432	S	0.0	0.0	0:00.00	cron
	181	message+	20	0	9596	4864	4352	S	0.0	0.1	0:00.05	dbus-daemon
	188	root	20	0	17960	8192	7296	S	0.0	0.1	0:00.06	systemd-logind
	190	root	20	0	1756096	12160	10368	S	0.0	0.2	0:00.07	wsl-pro-service
	196	syslog	20	0	222508	5504	4480	S	0.0	0.1	0:00.04	rsyslogd
	204	root	20	0	3160	1920	1792	S	0.0	0.0	0:00.00	agetty
	212	root	20	0	3116	1792	1664	S	0.0	0.0	0:00.00	agetty
	218	root	20	0	107032	22528	13184	S	0.0	0.3	0:00.08	unattended-upgr
	304	root	20	0	3076	896	896	S	0.0	0.0	0:00.00	SessionLeader
	305	root	20	0	3092	1024	896	S	0.0	0.0	0:00.02	Relay(306)
	306	agilngg+	20	0	6072	4992	3456	S	0.0	0.1	0:00.04	bash
	308	root	20	0	6660	4480	3840	S	0.0	0.1	0:00.01	login
	354	agilngg+	20	0	20308	11264	9216	S	0.0	0.1	0:00.09	systemd
	355	agilngg+	20	0	21156	3520	1792	S	0.0	0.0	0:00.00	(sd-pam)
	379	agilngg+	20	0	6072	4992	3456	S	0.0	0.1	0:00.02	bash
	564	polkitd	20	0	308164	7680	6912	S	0.0	0.1	0:00.05	polkitd

User Application

```

agilnggara@RagilAnggara:~$ which bash
/usr/bin/bash
agilnggara@RagilAnggara:~$ which gcc
/usr/bin/gcc
agilnggara@RagilAnggara:~$ |

```

8.2 Microkernel vs Monolithic

Lihat ukuran kernel (Monolithic)

```

agilnggara@RagilAnggara:~$ ls -lh /boot/vmlinuz-$(uname -r)
ls: cannot access '/boot/vmlinuz-6.87.2-microsoft-standard-WSL2': No such file or directory

```

“Kernel WSL2 tidak memiliki direktori /boot karena kernel di-manage langsung oleh Windows, bukan oleh Linux guest system.”

Lihat built-in vs module

```
agilnggara@RagilAnggara:~$ cat /boot/config-$(uname -r) | grep "=y" | wc -l
cat: /boot/config-6.6.87.2-microsoft-standard-WSL2: No such file or directory
0
agilnggara@RagilAnggara:~$ cat /boot/config-$(uname -r) | grep "=m" | wc -l
cat: /boot/config-6.6.87.2-microsoft-standard-WSL2: No such file or directory
0
```

Lihat loaded modules

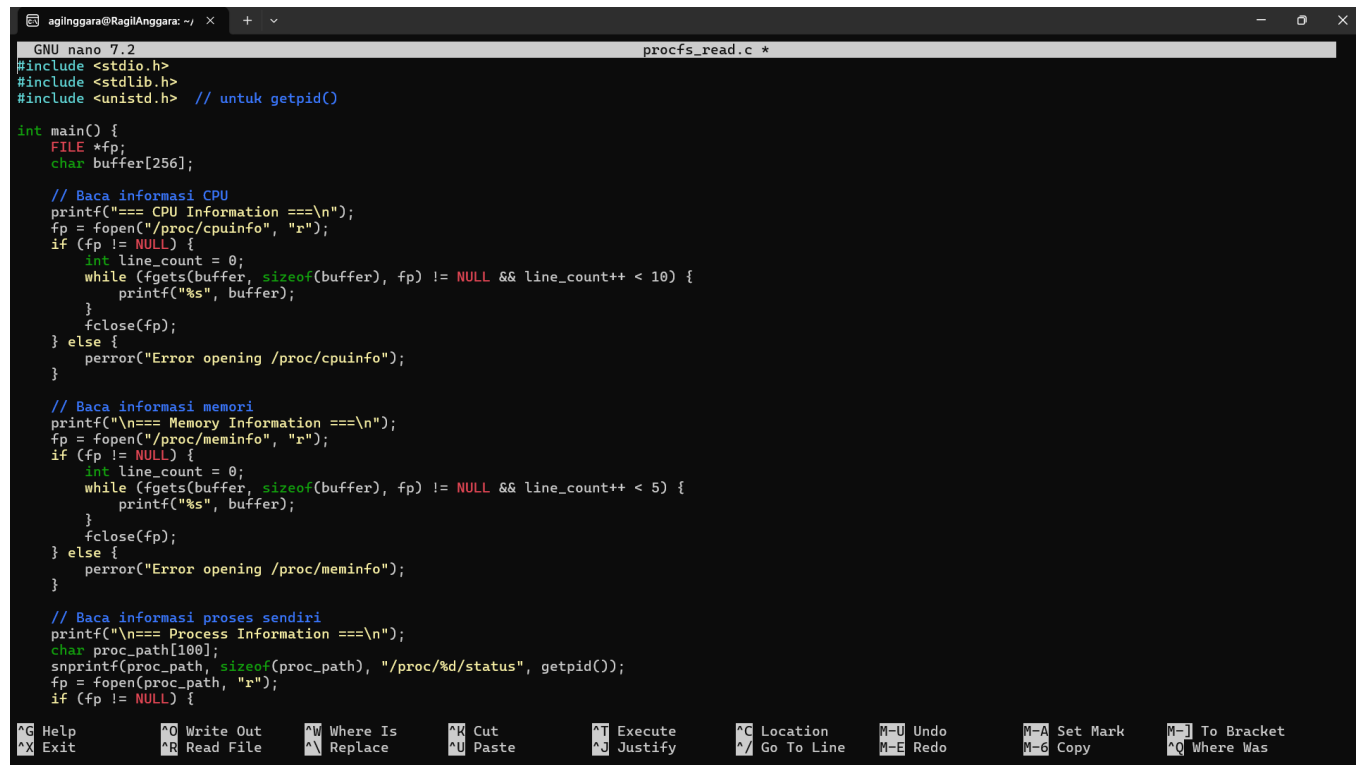
```
agilnggara@RagilAnggara:~$ lsmod | wc -l
18
```

8.3 Virtual File System

Buat file procfs_read.c

```
agilnggara@RagilAnggara:~$ cd ~/praktikum1
agilnggara@RagilAnggara:~/praktikum1$ mkdir procfs_read
```

```
agilnggara@RagilAnggara:~/praktikum1$ cd procfs_read
agilnggara@RagilAnggara:~/praktikum1/procfs_read$ nano procfs_read.c
```



```
GNU nano 7.2 procfs_read.c *
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h> // untuk getpid()

int main() {
    FILE *fp;
    char buffer[256];

    // Baca informasi CPU
    printf("=== CPU Information ===\n");
    fp = fopen("/proc/cpuinfo", "r");
    if (fp != NULL) {
        int line_count = 0;
        while (fgets(buffer, sizeof(buffer), fp) != NULL && line_count++ < 10) {
            printf("%s", buffer);
        }
        fclose(fp);
    } else {
        perror("Error opening /proc/cpuinfo");
    }

    // Baca informasi memori
    printf("\n=== Memory Information ===\n");
    fp = fopen("/proc/meminfo", "r");
    if (fp != NULL) {
        int line_count = 0;
        while (fgets(buffer, sizeof(buffer), fp) != NULL && line_count++ < 5) {
            printf("%s", buffer);
        }
        fclose(fp);
    } else {
        perror("Error opening /proc/meminfo");
    }

    // Baca informasi proses sendiri
    printf("\n=== Process Information ===\n");
    char proc_path[100];
    snprintf(proc_path, sizeof(proc_path), "/proc/%d/status", getpid());
    fp = fopen(proc_path, "r");
    if (fp != NULL) {
        // ... (rest of the code) ...
    }
}
```

```

if (fp != NULL) {
    int line_count = 0;
    while (fgets(buffer, sizeof(buffer), fp) != NULL && line_count++ < 10) {
        printf("%s", buffer);
    }
    fclose(fp);
} else {
    perror("Error opening process status");
}

return 0;
}

```

^G Help ^O Write Out ^W Where Is ^K Cut ^T Execute ^C Location M-U Undo M-A Set Mark M-] To Bracket
 ^X Exit ^R Read File ^\ Replace ^U Paste ^J Justify ^/ Go To Line M-E Redo M-G Copy ^Q Where Was

```

agilnggara@RagilAnggara:~/praktikum1/procfs_read$ gcc -o procfs_read procfs_read.c
agilnggara@RagilAnggara:~/praktikum1/procfs_read$ ./procfs_read

```

```

=== CPU Information ===

```

```

processor       : 0
vendor_id      : AuthenticAMD
cpu family     : 25
model          : 80
model name     : AMD Ryzen 9 5900HX with Radeon Graphics
stepping       : 0
microcode      : 0xffffffff
cpu MHz        : 3293.728
cache size     : 512 KB
physical id    : 0

```

```

=== Memory Information ===

```

```

MemTotal:      7821500 kB
MemFree:       7255740 kB
MemAvailable:  7264692 kB
Buffers:       2484 kB
Cached:        138984 kB

```

```

=== Process Information ===

```

```

Name:   procfs_read
Umask:  0022
State:  R (running)
Tgid:   460
Ngid:   0
Pid:    460
PPid:   431
TracerPid: 0
Uid:    1000 1000 1000 1000
Gid:    1000 1000 1000 1000

```

```

agilnggara@RagilAnggara:~/praktikum1/procfs_read$

```

BAB IX

BUILDING AND BOOTING AN OPERATING SYSTEM

9.1 Analisis Boot Process

Lihat Boot Messages

```
agilnggara@RagilAnggara: ~  
agilnggara@RagilAnggara:~$ dmesg | less  
  
[ 0.000000] Linux version 6.6.87.2-microsoft-standard-WSL2 (root@439a258ad544) (gcc (GCC) 11.2.0, GNU ld (GNU Binutils) 2.37) #1 SMP PREEMPT_DYNAMIC Thu  
Jun 5 18:30:46 UTC 2025  
[ 0.000000] Command line: initrd=\initrd.img WSL_ROOT_INIT=1 panic=-1 nr_cpus=16 hv_utils.timesync_implicit=1 console=hvc0 debug ptty.legacy_count=0 WSL_E  
NABLE_CRASH_DUMP=1  
[ 0.000000] KERNEL supported cpus:  
[ 0.000000] Intel GenuineIntel  
[ 0.000000] AMD AuthenticAMD  
[ 0.000000] BIOS-provided physical RAM map:  
[ 0.000000] BIOS-e820: [mem 0x0000000000000000-0x000000000009ffff] usable  
[ 0.000000] BIOS-e820: [mem 0x00000000000e0000-0x00000000000e0fff] reserved  
[ 0.000000] BIOS-e820: [mem 0x0000000001000000-0x00000000001fffff] ACPI data  
[ 0.000000] BIOS-e820: [mem 0x0000000002000000-0x0000000007ffffff] usable  
[ 0.000000] BIOS-e820: [mem 0x0000000100000000-0x00000001f4dfffff] usable  
[ 0.000000] NX (Execute Disable) protection: active  
[ 0.000000] APIC: Static calls initialized  
[ 0.000000] DMI not present or invalid.  
[ 0.000000] Hypervisor detected: Microsoft Hyper-V  
[ 0.000000] Hyper-V: privilege flags low 0xae7f, high 0x3b8030, hints 0x900c2c, misc 0xe0bed7b6  
[ 0.000000] Hyper-V: Nested features: 0x4a0000  
[ 0.000000] Hyper-V: LAPIC Timer Frequency: 0xc3500  
[ 0.000000] Hyper-V: Using hypercall for remote TLB flush  
[ 0.000000] clocksource: hyperv_clocksource_tsc_page: mask: 0xffffffffffffff max_cycles: 0x24e6a1710, max_idle_ns: 440795202120 ns  
[ 0.000000] clocksource: hyperv_clocksource_msr: mask: 0xffffffffffffff max_cycles: 0x24e6a1710, max_idle_ns: 440795202120 ns  
[ 0.000000] tsc: Detected 3293.728 MHz processor  
[ 0.000038] e820: update [mem 0x00000000-0x000000fff] usable ==> reserved  
[ 0.000041] e820: remove [mem 0x000a0000-0x0000ffff] usable  
[ 0.000043] last_pfn = 0x1f4e00 max_arch_pfn = 0x400000000  
[ 0.000067] MTRR map: 5 entries (4 fixed + 1 variable; max 20), built from 8 variable MTRRs  
[ 0.000070] x86/PAT: Configuration [0-7]: WB WC UC- UC WB WP UC- WT  
[ 0.000111] last_pfn = 0xf8000 max_arch_pfn = 0x400000000  
[ 0.000122] Using GB pages for direct mapping  
[ 0.000218] RAMDISK: [mem 0x04be4000-0x04e7ffff]  
[ 0.000220] ACPI: Early table checksum verification disabled  
[ 0.000223] ACPI: RSDP 0x000000000000E000 000024 (v02 VIRTUAL)  
[ 0.000227] ACPI: XSDT 0x0000000000100000 000044 (v01 VIRTUAL MICROSOFT 00000001 MSFT 00000001)  
[ 0.000231] ACPI: FACP 0x0000000000101000 000114 (v06 VIRTUAL MICROSOFT 00000001 MSFT 00000001)  
[ 0.000234] ACPI: DSDT 0x00000000001011B8 01E11C (v02 MSFTVM DSDT01 00000001 INTL 20230628)  
[ 0.000236] ACPI: FACS 0x0000000000101114 000040  
[ 0.000238] ACPI: OEM0 0x0000000000101154 000064 (v01 VIRTUAL MICROSOFT 00000001 MSFT 00000001)  
[ 0.000240] ACPI: SRAT 0x000000000011F2D4 0003B0 (v02 VIRTUAL MICROSOFT 00000001 MSFT 00000001)  
[ 0.000242] ACPI: APIC 0x000000000011F684 0000C8 (v04 VIRTUAL MICROSOFT 00000001 MSFT 00000001)  
[ 0.000243] ACPI: Reserving FACP table memory at [mem 0x101000-0x101113]  
[ 0.000244] ACPI: Reserving DSDT table memory at [mem 0x1011B8-0x11f2d3]
```

Lihat systemd boot analysis

```
agilnggara@RagilAnggara: ~  
agilnggara@RagilAnggara:~$ systemd-analyze  
Startup finished in 1.446s (userspace).  
graphical.target reached after 1.407s in userspace.  
agilnggara@RagilAnggara:~$ systemd-analyze blame  
in 3.489s: motd-news.service  
752ms landscape-client.service  
333ms snapd.seeded.service  
321ms dev-sdd.device  
263ms wsl-pro.service  
230ms snapd.service  
166ms systemd-resolved.service  
162ms systemd-udev-trigger.service  
155ms user@1000.service  
115ms rsyslog.service  
103ms systemd-timesyncd.service  
83ms snapd.socket  
83ms systemd-logind.service  
60ms systemd-journal-flush.service  
52ms systemd-journald.service  
52ms keyboard-setup.service  
50ms systemd-udev.service  
49ms systemd-tmpfiles-clean.service  
40ms a2scrub_reap.service  
38ms dbus.service  
34ms user-runtime-dir@1000.service  
31ms systemd-tmpfiles-setup.service  
24ms dev-hugepages.mount  
24ms dev-mqueue.mount  
23ms sys-kernel-debug.mount  
22ms systemd-sysctl.service  
22ms systemd-user-sessions.service  
22ms sys-kernel-tracing.mount  
22ms systemd-tmpfiles-setup-dev-early.service  
21ms systemd-binfmt.service  
20ms systemd-update-utmp.service  
17ms modprobe@dm_mod.service  
17ms kmod-static-nodes.service  
17ms modprobe@drm.service  
17ms modprobe@efi-ptstore.service  
17ms modprobe@fuse.service  
17ms modprobe@configfs.service  
16ms modprobe@loop.service  
15ms systemd-modules-load.service  
14ms systemd-tmpfiles-setup-dev.service  
14ms systemd-remount-fs.service  
12ms setvtrgb.service  
11ms console-setup.service
```

```

agilnggara@RagilAnggara:~$ systemd-analyze critical-chain
The time when unit became active or started is printed after the "@" character.
The time the unit took to start is printed after the "+" character.

graphical.target @1.407s
├─multi-user.target @1.407s
│   └─snapd.seeded.service @661ms +333ms
│       └─basic.target @642ms
│           └─sockets.target @642ms
│               └─snapd.socket @559ms +83ms
│                   └─sysinit.target @549ms
│                       └─systemd-resolved.service @382ms +166ms
│                           └─systemd-tmpfiles-setup.service @345ms +31ms
│                               └─systemd-journal-flush.service @281ms +60ms
│                                   └─systemd-journald.service @212ms +52ms
│                                       └─systemd-journald.socket @203ms
│                                           └─mount @165ms
│                                               └─.slice @164ms
agilnggara@RagilAnggara:~$

```

Lihat GRUB configuration

```

agilnggara@RagilAnggara:~$ cat /boot/grub/grub.cfg | head -50
cat: /boot/grub/grub.cfg: No such file or directory

```

Kernel WSL2 tidak memiliki direktori /boot karena kernel di-manage langsung oleh Windows, bukan oleh Linux guest system. Akibatnya, file konfigurasi GRUB (/boot/grub/grub.cfg) tidak tersedia, dan analisis konfigurasi GRUB tidak dapat dilakukan di lingkungan WSL2.

Sebagai pengganti, analisis boot tetap dapat dilakukan melalui perintah lain:

```

agilnggara@RagilAnggara:~$ dmesg | head -30
[    0.000000] Linux version 6.6.87.2-microsoft-standard-WSL2 (root@439a258ad544) (gcc (GCC) 11.2.0, GNU ld (GNU Binutils) 2.37) #1 SMP PREEMPT_DYNAMIC Thu
Jun  5 18:30:46 UTC 2025
[    0.000000] Command line: initrd=\initrd.img WSL_ROOT_INIT=1 panic=-1 nr_cpus=16 hv_utils.timesync_implicit=1 console=hvc0 debug pty.legacy_count=0 WSL_E
NABLE_CRASH_DUMP=1
[    0.000000] KERNEL supported cpus:
[    0.000000]   Intel GenuineIntel
[    0.000000]   AMD AuthenticAMD
[    0.000000] BIOS-provided physical RAM map:
[    0.000000] BIOS-e820: [mem 0x0000000000000000-0x000000000009ffff] usable
[    0.000000] BIOS-e820: [mem 0x00000000000a0000-0x00000000000e0fff] reserved
[    0.000000] BIOS-e820: [mem 0x0000000000100000-0x000000000001ffff] ACPI data
[    0.000000] BIOS-e820: [mem 0x0000000000200000-0x0000000000f7ffff] usable
[    0.000000] BIOS-e820: [mem 0x0000000010000000-0x000000001f4dffff] usable
[    0.000000] NX (Execute Disable) protection: active
[    0.000000] APIC: Static calls initialized
[    0.000000] DMI not present or invalid.
[    0.000000] Hypervisor detected: Microsoft Hyper-V
[    0.000000] Hyper-V: privilege flags low 0xae7f, high 0x3b8030, hints 0x900c2c, misc 0xe0bed7b6
[    0.000000] Hyper-V: Nested Features: 0x4a0000
[    0.000000] Hyper-V: LAPIC Timer Frequency: 0xc3500
[    0.000000] Hyper-V: Using hypercall fox remote TLB flush
[    0.000000] clocksource: hyperv_clocksource_tsc_page: mask: 0xffffffffffffffff max_cycles: 0x24e6a1710, max_idle_ns: 440795202120 ns
[    0.000000] clocksource: hyperv_clocksource_msr: mask: 0xffffffffffffffff max_cycles: 0x24e6a1710, max_idle_ns: 440795202120 ns
[    0.000000] tsc: Detected 3293.728 MHz processor
[    0.000038] e820: update [mem 0x00000000-0x000000ffff] usable ==> reserved
[    0.000041] e820: remove [mem 0x000a0000-0x000fffff] usable
[    0.000043] last_pfn = 0x1f4e00 max_arch_pfn = 0x400000000
[    0.000067] MTRR map: 5 entries (4 fixed + 1 variable; max 20), built from 8 variable MTRRs
[    0.000070] x86/PAT: Configuration [0-7]: WB WC UC- UC WB WP UC- WT
[    0.000111] last_pfn = 0xf8000 max_arch_pfn = 0x400000000
[    0.000122] Using GB pages for direct mapping
[    0.000218] RAMDISK: [mem 0x04be4000-0x04e7ffff]
agilnggara@RagilAnggara:~$

```

```

agilnggara@RagilAnggara:~$ uname -a
Linux RagilAnggara 6.6.87.2-microsoft-standard-WSL2 #1 SMP PREEMPT_DYNAMIC Thu Jun  5 18:30:46 UTC 2025 x86_64 x86_64 x86_64 GNU/Linux
agilnggara@RagilAnggara:~$ cat /proc/version
Linux version 6.6.87.2-microsoft-standard-WSL2 (root@439a258ad544) (gcc (GCC) 11.2.0, GNU ld (GNU Binutils) 2.37) #1 SMP PREEMPT_DYNAMIC Thu Jun  5 18:30:46
UTC 2025
agilnggara@RagilAnggara:~$

```

Lihat initramfs

```

agilnggara@RagilAnggara:~$ lsinitramfs /boot/initrd.img-$(uname -r) | head -20
/usr/bin/unmkinitramfs: 64: cannot open /boot/initrd.img-6.6.87.2-microsoft-standard-WSL2: No such file
/usr/bin/unmkinitramfs: 57: cannot open /boot/initrd.img-6.6.87.2-microsoft-standard-WSL2: No such file
/usr/bin/unmkinitramfs: 38: cannot open /boot/initrd.img-6.6.87.2-microsoft-standard-WSL2: No such file
cpio: premature end of archive

```

Kernel WSL2 tidak memiliki file initrd.img karena sistem tidak menggunakan mekanisme *initramfs* seperti Linux native. Kernel dimuat langsung oleh Windows host tanpa melalui GRUB atau tahapan boot tradisional, sehingga perintah *lsinitramfs* tidak dapat dijalankan di WSL2.

9.2 Kernel Parameters

Lihat kernel parameters saat boot

```
agilnggara@RagilAnggara:~$ cat /proc/cmdline
initrd=\initrd.img WSL_ROOT_INIT=1 panic=-1 nr_cpus=16 hv_utils.timesync_implicit=1 console=hvc0 debug ptty.legacy_count=0 WSL_ENABLE_CRASH_DUMP=1
```

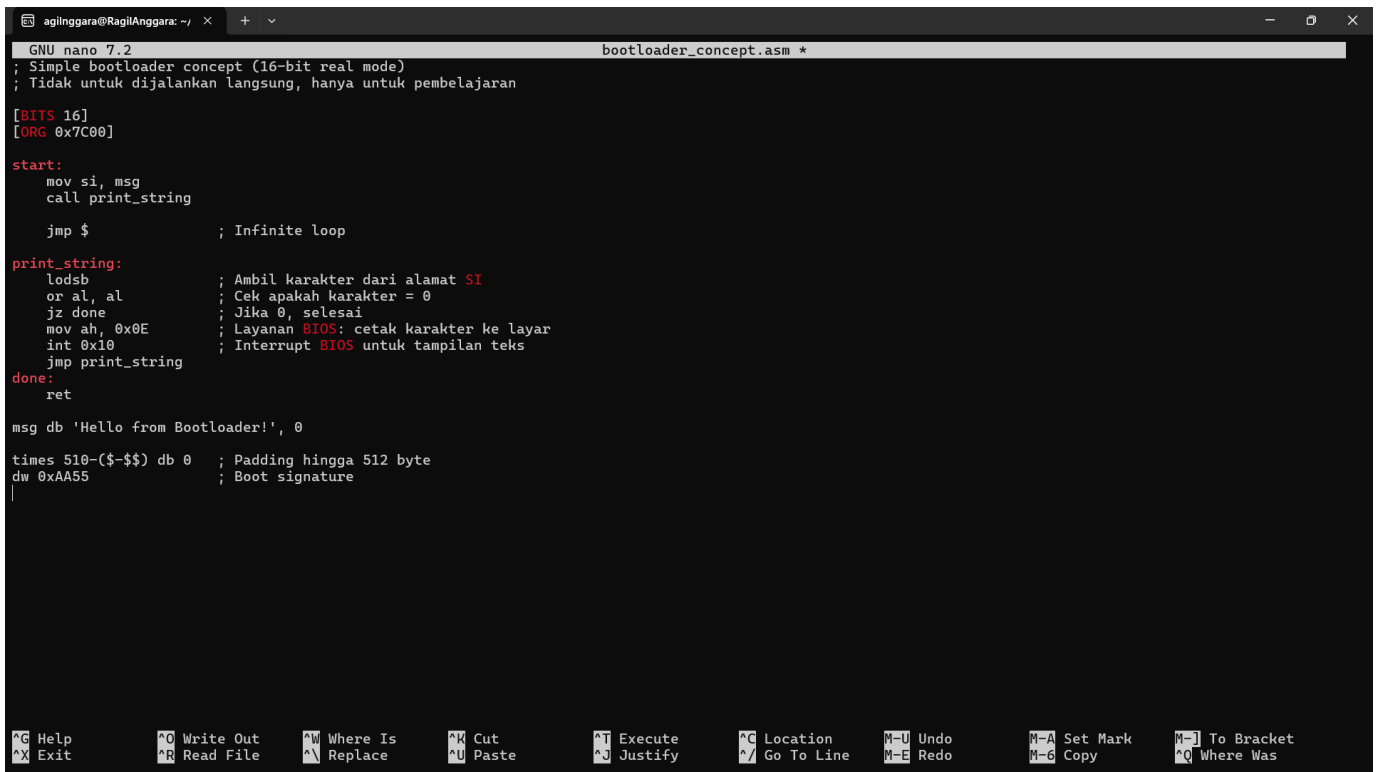
```
agilnggara@RagilAnggara:~/praktikum1/kernel_module$ cd ~/praktikum1
agilnggara@RagilAnggara:~/praktikum1$ sudo nano /etc/default/grub
```



9.3 Simple Bootloader (Konsep)

Buat file bootloader_concept.asm (untuk pembelajaran)

```
agilnggara@RagilAnggara:~/praktikum1$ mkdir bootloader
agilnggara@RagilAnggara:~/praktikum1$ cd bootloader
agilnggara@RagilAnggara:~/praktikum1/bootloader$ nano bootloader_concept.asm
```



```
agilnggara@RagilAnggara:~/praktikum1/bootloader$ nasm -f bin bootloader_concept.asm -o bootloader_concept.bin
agilnggara@RagilAnggara:~/praktikum1/bootloader$ ls -lh bootloader_concept.bin
-rw-r--r-- 1 agilnggara agilnggara 512 Oct 14 09:32 bootloader_concept.bin
```

BAB X

10.1 Using strace

Trace system calls

[illegible]

Trace dengan detail

[illegible]

Hitung waktu system calls

```

agilinggara@RagilAnggara:~$ strace -c ls
output.txt praktikum1
% time      seconds  usecs/call     calls    errors syscall
-----
33.62      0.000886         25        35        13 openat
13.13      0.000246         14        24         0 close
12.14      0.000329         10        30         0 mmap
11.69      0.000308         30        10         0 read
11.57      0.000305         13        23         0 fstat
 4.97      0.000131         26         5         0 mprotect
 1.75      0.000046         23         2         0 getdents64
 1.63      0.000043         21         2         1 statfs
 1.48      0.000039         19         2         0 ioctl
 1.44      0.000038         38         1         0 write
 1.14      0.000030         10         3         0 brk
 1.02      0.000027         27         1         0 munmap
 0.72      0.000019         19         1         0 futex
 0.68      0.000018         9         2         2 access
 0.53      0.000014         14         1         0 getrandom
 0.53      0.000014         14         1         0 rseq
 0.49      0.000013         13         1         0 arch_prctl
 0.49      0.000013         13         1         0 set_tid_address
 0.49      0.000013         13         1         0 prlimit64
 0.46      0.000012         12         1         0 set_robust_list
 0.00      0.000000         0         2         0 pread64
 0.00      0.000000         0         1         0 execve
-----
100.00      0.002635         17       150         16 total
agilinggara@RagilAnggara:~$ |

```

Trace dan simpan ke file

```
agilingara@RagilAnggara:~$ strace -o trace.txt ls  
output.txt praktikum1 trace.txt  
agilingara@RagilAnggara:~$ cat trace.txt  
execve("/usr/bin/lis", ["lis"], 0x7fffd23c9680 /* 25 vars */) = 0  
brk(NULL)                                = 0x58f79fd2800  
mmap(NULL, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_ANONYMOUS, -1, 0) = 0x700ce3908000  
access("/etc/ld.so.preload", R_OK)  
= -1 ENOENT (No such file or directory)  
openat(AT_FDCWD, "/etc/ld.so.cache", O_RDONLY|O_CLOEXEC) = 3  
statfs(3, {fst_mode=S_IFREG|0644, st_size=26935, ...}) = 0  
mmap(NULL, 20935, PROT_READ, MAP_PRIVATE, 3, 0) = 0x700ce3903000  
close(3)                                 = 0  
openat(AT_FDCWD, "/lib/x86_64-linux-gnu/libselinux.so.1", O_RDONLY|O_CLOEXEC) = 3  
read(3, "\177ELF\211\1\0\0\0\0\0\0\0\3\0e-\01\0\0\0\0\0\0\220\243\2\0\0\0\0...", 832) = 832  
fstat(3, {st_mode=S_IFREG|0644, st_size=174472, ...}) = 0  
mmap(NULL, 181960, PROT_READ, MAP_PRIVATE|MAP_DENYWRITE, 3, 0) = 0x700ce38d6000  
mmap(0x700ce38dc000, 118784, PROT_READ|PROT_EXEC, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x0000) = 0x700ce38dc000  
mmap(0x700ce38fe000, 24576, PROT_READ, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x23000) = 0x700ce38fe000  
mmap(0x700ce38ff000, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x29000) = 0x700ce38ff000  
mmap(0x700ce3901000, 5832, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_ANONYMOUS, -1, 0) = 0x700ce3901000  
close(3)                                 = 0  
openat(AT_FDCWD, "/lib/x86_64-linux-gnu/libc.so.6", O_RDONLY|O_CLOEXEC) = 3  
read(3, "\177ELF\211\1\0\0\0\0\0\0\0\3\0e-\01\0\0\0\0\0\0\220\243\2\0\0\0\0...", 832) = 832  
pread64(3, "\6\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0... ", 784, 64) = 784  
fstat(3, {st_mode=S_IFREG|0755, st_size=2125328, ...}) = 0  
pread64(3, "\6\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0... ", 784, 64) = 784  
mmap(NULL, 2178256, PROT_READ, MAP_PRIVATE|MAP_DENYWRITE, 3, 0) = 0x700ce3600000  
mmap(0x700ce3620000, 1695632, PROT_READ|PROT_EXEC, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x20000) = 0x700ce3620000  
mmap(0x700ce37b0000, 323584, PROT_READ, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x1db000) = 0x700ce37b0000  
mmap(0x700ce37bf000, 24576, PROT_READ, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x1fee00) = 0x700ce37bff000  
mmap(0x700ce3850000, 52624, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_ANONYMOUS, -1, 0) = 0x700ce3850000  
close(3)                                 = 0  
openat(AT_FDCWD, "/lib/x86_64-linux-gnu/libcrypt-2.8.so.0", O_RDONLY|O_CLOEXEC) = 3  
read(3, "\177ELF\211\1\0\0\0\0\0\0\0\3\0e-\01\0\0\0\0\0\0\220\243\2\0\0\0\0...", 832) = 832  
fstat(3, {st_mode=S_IFREG|0644, st_size=625344, ...}) = 0  
mmap(NULL, 627472, PROT_READ, MAP_PRIVATE|MAP_DENYWRITE, 3, 0) = 0x700ce383c000  
mmap(0x700ce383e000, 450560, PROT_READ|PROT_EXEC, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x20000) = 0x700ce383e000  
mmap(0x700ce38ac000, 163840, PROT_READ, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x70000) = 0x700ce38ac000  
mmap(0x700ce38d0000, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x7f800) = 0x700ce38d0000  
close(3)                                 = 0  
mmap(NULL, 12288, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_ANONYMOUS, -1, 0) = 0x700ce3839000  
arch_prctl(&ARCH_SET_FS, 0x700ce3839000) = 0  
set_tid_address(0x700ce3839ad0)         = 624  
set_robust_list(0x700ce3839ae0, 24)     = 0  
rseq(0x700ce383al20, 0x20, 0, 0x53653053) = 0  
mprotect(0x700ce37fff00, 16384, PROT_READ) = 0  
mprotect(0x700ce38d0000, 4096, PROT_READ) = 0  
mprotect(0x700ce38ff000, 4096, PROT_READ) = 0  
mprotect(0x58f775620000, 8192, PROT_READ) = 0  
mprotect(0x700ce3940000, 8192, PROT_READ) = 0  
prlimit64(0, RLIMIT_STACK, NULL, ({rlim_cur=8192*1024, rlim_max=RLIM64_INFINITY})) = 0  
munmap(0x700ce3903000, 20935)           = 0  
statfs("/sys/fs/selinux", {ftype=SVFS_MAGIC, f_bsize=4096, f_blocks=0, f_bfree=0, f_avail=0, f_files=0, f_ffree=0, f_fsid=[val={0x36118bcf, 0xddcd9ed4}], f_namelen=255, f_rsize=4096, f_flags=ST_VALID|ST_N  
OSUID|ST_NODEV|ST_NOEXEC|ST_NOWRIE)) = 0  
statfs("/selinux", {ftype=VFAT_VS360} = -1 ENOENT (No such file or directory)  
getrandom("x\xdc\xac\x40\x23\x85\xfd\x20\x11", 8, GRND_NONBLOCK) = 8  
brk(NULL)                                = 0x58f79fd2800  
brk(0x58f79fd2800)                       = 0x58f79fd2800  
openat(AT_FDCWD, "/proc/filesystems", O_RDONLY|O_CLOEXEC) = 3  
fstat(3, {st_mode=S_IFREG|0644, st_size=0, ...}) = 0
```

```
agilinggara@RagilAnggara: ~  
read(3, "nodev\tsysfs\nnnodev\ttmpfs\nnnodev\tbtd"... , 1024) = 442  
close(3) = 0  
openat(AT_FDCWD, "/proc/mounts", O_RDONLY|O_CLOEXEC) = 3  
fstat(3, {st_mode=S_IFREG|0444, st_size=0, ...}) = 0  
read(3, "none /usr/lib/modules/6.8.87.2-m"... , 1024) = 1024  
read(3, "\ndepts /dev/pts depts rw,nosui"... , 1024) = 1024  
read(3, "lient,msize=65536,trans=f,d,rfdr"... , 1024) = 745  
read(3, "", 1024) = 0  
close(3) = 0  
access("/etc/selinux/config", F_OK) = -1 ENOENT (No such file or directory)  
openat(AT_FDCWD, "/usr/lib/locale/locale-archive", O_RDONLY|O_CLOEXEC) = -1 ENOENT (No such file or directory)  
openat(AT_FDCWD, "/usr/share/locale/locale.alias", O_RDONLY|O_CLOEXEC) = 3  
fstat(3, {st_mode=S_IFREG|0644, st_size=2996, ...}) = 0  
read(3, "# locale name alias data base.\n"... , 4096) = 2996  
read(3, "", 4096) = 0  
close(3) = 0  
openat(AT_FDCWD, "/usr/lib/locale/C.UTF-8/LC_IDENTIFICATION", O_RDONLY|O_CLOEXEC) = -1 ENOENT (No such file or directory)  
openat(AT_FDCWD, "/usr/lib/locale/C.utf8/LC_IDENTIFICATION", O_RDONLY|O_CLOEXEC) = 3  
fstat(3, {st_mode=S_IFREG|0644, st_size=258, ...}) = 0  
mmap(NULL, 258, PROT_READ, MAP_PRIVATE, 3, 0) = 0x700ce3907000  
close(3) = 0  
openat(AT_FDCWD, "/usr/lib/x86_64-linux-gnu/gconv/gconv-modules.cache", O_RDONLY|O_CLOEXEC) = 3  
fstat(3, {st_mode=S_IFREG|0644, st_size=27028, ...}) = 0  
mmap(NULL, 27028, PROT_READ, MAP_SHARED, 3, 0) = 0x700ce3832000  
close(3) = 0  
futext(0x700ce388072c, FUTEX_NAME_PRIVATE, 2147483647) = 0  
openat(AT_FDCWD, "/usr/lib/locale/C.UTF-8/LC_MEASUREMENT", O_RDONLY|O_CLOEXEC) = -1 ENOENT (No such file or directory)  
openat(AT_FDCWD, "/usr/lib/locale/C.utf8/LC_MEASUREMENT", O_RDONLY|O_CLOEXEC) = 3  
fstat(3, {st_mode=S_IFREG|0644, st_size=23, ...}) = 0  
mmap(NULL, 23, PROT_READ, MAP_PRIVATE, 3, 0) = 0x700ce3906000  
close(3) = 0  
openat(AT_FDCWD, "/usr/lib/locale/C.UTF-8/LC_TELEPHONE", O_RDONLY|O_CLOEXEC) = -1 ENOENT (No such file or directory)  
openat(AT_FDCWD, "/usr/lib/locale/C.utf8/LC_TELEPHONE", O_RDONLY|O_CLOEXEC) = 3  
fstat(3, {st_mode=S_IFREG|0644, st_size=47, ...}) = 0  
mmap(NULL, 47, PROT_READ, MAP_PRIVATE, 3, 0) = 0x700ce3905000  
close(3) = 0  
openat(AT_FDCWD, "/usr/lib/locale/C.UTF-8/LC_ADDRESS", O_RDONLY|O_CLOEXEC) = -1 ENOENT (No such file or directory)  
openat(AT_FDCWD, "/usr/lib/locale/C.utf8/LC_ADDRESS", O_RDONLY|O_CLOEXEC) = 3  
fstat(3, {st_mode=S_IFREG|0644, st_size=127, ...}) = 0  
mmap(NULL, 127, PROT_READ, MAP_PRIVATE, 3, 0) = 0x700ce3904000  
close(3) = 0  
openat(AT_FDCWD, "/usr/lib/locale/C.UTF-8/LC_NAME", O_RDONLY|O_CLOEXEC) = -1 ENOENT (No such file or directory)  
openat(AT_FDCWD, "/usr/lib/locale/C.utf8/LC_NAME", O_RDONLY|O_CLOEXEC) = 3  
fstat(3, {st_mode=S_IFREG|0644, st_size=62, ...}) = 0  
mmap(NULL, 62, PROT_READ, MAP_PRIVATE, 3, 0) = 0x700ce3903000  
close(3) = 0  
openat(AT_FDCWD, "/usr/lib/locale/C.UTF-8/LC_PAPER", O_RDONLY|O_CLOEXEC) = -1 ENOENT (No such file or directory)  
openat(AT_FDCWD, "/usr/lib/locale/C.utf8/LC_PAPER", O_RDONLY|O_CLOEXEC) = 3  
fstat(3, {st_mode=S_IFREG|0644, st_size=34, ...}) = 0  
mmap(NULL, 34, PROT_READ, MAP_PRIVATE, 3, 0) = 0x700ce3831000  
close(3) = 0  
openat(AT_FDCWD, "/usr/lib/locale/C.UTF-8/LC_MESSAGES", O_RDONLY|O_CLOEXEC) = -1 ENOENT (No such file or directory)  
openat(AT_FDCWD, "/usr/lib/locale/C.utf8/LC_MESSAGES", O_RDONLY|O_CLOEXEC) = 3  
fstat(3, {st_mode=S_IFDIR|0755, st_size=4096, ...}) = 0  
close(3) = 0  
openat(AT_FDCWD, "/usr/lib/locale/C.utf8/LC_MESSAGES/SYS_LC_MESSAGES", O_RDONLY|O_CLOEXEC) = 3  
fstat(3, {st_mode=S_IFREG|0644, st_size=48, ...}) = 0  
mmap(NULL, 48, PROT_READ, MAP_PRIVATE, 3, 0) = 0x700ce3830000  
close(3) = 0
```

```
agilinggara@RagilAnggara: ~  
mmap(NULL, 62, PROT_READ, MAP_PRIVATE, 3, 0) = 0x700ce3903000  
close(3) = 0  
openat(AT_FDCWD, "/usr/lib/locale/C.UTF-8/LC_PAPER", O_RDONLY|O_CLOEXEC) = -1 ENOENT (No such file or directory)  
openat(AT_FDCWD, "/usr/lib/locale/C.utf8/LC_PAPER", O_RDONLY|O_CLOEXEC) = 3  
fstat(3, {st_mode=S_IFREG|0644, st_size=34, ...}) = 0  
mmap(NULL, 34, PROT_READ, MAP_PRIVATE, 3, 0) = 0x700ce3831000  
close(3) = 0  
openat(AT_FDCWD, "/usr/lib/locale/C.UTF-8/LC_MESSAGES", O_RDONLY|O_CLOEXEC) = -1 ENOENT (No such file or directory)  
openat(AT_FDCWD, "/usr/lib/locale/C.utf8/LC_MESSAGES", O_RDONLY|O_CLOEXEC) = 3  
fstat(3, {st_mode=S_IFDIR|0755, st_size=4096, ...}) = 0  
close(3) = 0  
openat(AT_FDCWD, "/usr/lib/locale/C.utf8/LC_MESSAGES/SYS_LC_MESSAGES", O_RDONLY|O_CLOEXEC) = 3  
fstat(3, {st_mode=S_IFREG|0644, st_size=48, ...}) = 0  
mmap(NULL, 48, PROT_READ, MAP_PRIVATE, 3, 0) = 0x700ce3830000  
close(3) = 0  
openat(AT_FDCWD, "/usr/lib/locale/C.UTF-8/LC_MONETARY", O_RDONLY|O_CLOEXEC) = -1 ENOENT (No such file or directory)  
openat(AT_FDCWD, "/usr/lib/locale/C.utf8/LC_MONETARY", O_RDONLY|O_CLOEXEC) = 3  
fstat(3, {st_mode=S_IFREG|0644, st_size=270, ...}) = 0  
mmap(NULL, 270, PROT_READ, MAP_PRIVATE, 3, 0) = 0x700ce382f000  
close(3) = 0  
openat(AT_FDCWD, "/usr/lib/locale/C.UTF-8/LC_COLLATE", O_RDONLY|O_CLOEXEC) = -1 ENOENT (No such file or directory)  
openat(AT_FDCWD, "/usr/lib/locale/C.utf8/LC_COLLATE", O_RDONLY|O_CLOEXEC) = 3  
fstat(3, {st_mode=S_IFREG|0644, st_size=1406, ...}) = 0  
mmap(NULL, 1406, PROT_READ, MAP_PRIVATE, 3, 0) = 0x700ce382e000  
close(3) = 0  
openat(AT_FDCWD, "/usr/lib/locale/C.UTF-8/LC_TIME", O_RDONLY|O_CLOEXEC) = -1 ENOENT (No such file or directory)  
openat(AT_FDCWD, "/usr/lib/locale/C.utf8/LC_TIME", O_RDONLY|O_CLOEXEC) = 3  
fstat(3, {st_mode=S_IFREG|0644, st_size=3360, ...}) = 0  
mmap(NULL, 3360, PROT_READ, MAP_PRIVATE, 3, 0) = 0x700ce382d000  
close(3) = 0  
openat(AT_FDCWD, "/usr/lib/locale/C.UTF-8/LC_NUMERIC", O_RDONLY|O_CLOEXEC) = -1 ENOENT (No such file or directory)  
openat(AT_FDCWD, "/usr/lib/locale/C.utf8/LC_NUMERIC", O_RDONLY|O_CLOEXEC) = 3  
fstat(3, {st_mode=S_IFREG|0644, st_size=50, ...}) = 0  
mmap(NULL, 50, PROT_READ, MAP_PRIVATE, 3, 0) = 0x700ce382c000  
close(3) = 0  
openat(AT_FDCWD, "/usr/lib/locale/C.UTF-8/LC_CTYPE", O_RDONLY|O_CLOEXEC) = -1 ENOENT (No such file or directory)  
openat(AT_FDCWD, "/usr/lib/locale/C.utf8/LC_CTYPE", O_RDONLY|O_CLOEXEC) = 3  
fstat(3, {st_mode=S_IFREG|0644, st_size=360460, ...}) = 0  
mmap(NULL, 360460, PROT_READ, MAP_PRIVATE, 3, 0) = 0x700ce35a7000  
close(3) = 0  
ioctl(1, TCGETS, {c_iflag=ICRNL|IXON, c_oflag=NL0|CR0|TAB0|BS0|VT0|FF0|OPOST|ONLCR, c_cflag=B38400|CS8|CREAD, c_lflag=ISIG|ICANON|ECHO|ECHOE|ECHOK|IEXTEN|ECHOCTL|ECHOKE, ...}) = 0  
ioctl(1, TIOCGWINSZ, {ws_row=30, ws_col=120, ws_xpixel=0, ws_ypixel=0}) = 0  
openat(AT_FDCWD, ".*", O_RDONLY|O_NONBLOCK|O_CLOEXEC|O_DIRECTORY) = 3  
fstat(3, {st_mode=S_IFDIR|0750, st_size=4096, ...}) = 0  
getdents64(3, 0x50f79fd5b8a0 /* 16 entries */, 32768) = 520  
getdents64(3, 0x50f79fd5b8a0 /* 0 entries */, 32768) = 0  
close(3) = 0  
fstat(1, {st_mode=S_IFCHR|0620, st_rdev=makedev(0x88, 0), ...}) = 0  
write(1, "output.txt praktikum1\ttrace.txt"... , 33) = 33  
close(1) = 0  
close(2) = 0  
exit-group(0)  
+++ exited with 0 +++
```

10.2 Using Ltrace

Trace library calls

```
agilnggara@RagilAnggara: ~  
agilnggara@RagilAnggara:~$ ltrace ls  
strcmp("ls", "/") = nil  
setlocale(LC_ALL, "") = "C.UTF-8"  
bindtextdomain("coreutils", "/usr/share/locale") = "/usr/share/locale"  
textdomain("coreutils") = "coreutils"  
__cxa_atexit(0x5802e1b3080, 0, 0x5802e1b3080, 0) = 0  
getopt_long(1, 0x7fffd34815f48, "abcdghiklenopqrstuvw:ABCFGHI:...", 0x5802e1b2340, -1) = -1  
getenv("LS_BLOCK_SIZE") = nil  
getenv("BLOCK_SIZE") = nil  
getenv("BLOCKSIZE") = nil  
getenv("POSIXLY_CORRECT") = nil  
getenv("BLOCK_SIZE") = nil  
isatty(1) = 1  
ioctl(1, 21523, 0x7fffd34815d60) = 0  
getenv("TABSIZE") = nil  
getenv("QUOTING_STYLE") = nil  
__errno_location() = 0x7e421fa866a0  
__errno_location() = 0x7e421fa866a0  
getenv("TZ") = nil  
reallocarray(0, 100, 200, 0x58030a8516c0) = 0x58030a8516d0  
strlen("") = 1  
__memcpy_chk(0x58030a856850, 0x5802e1b9aedc, 2, 2) = 0x58030a856850  
__errno_location() = 0x7e421fa866a0  
opendir(".") = 0x58030a856870  
readdir(0x58030a856870) = 0x58030a8568a0  
__errno_location() = 0x7e421fa866a0  
__ctype_get_mb_cur_max() = 6  
strlen("praktikum1") = 10  
strlen("praktikum1") = 10  
__memcpy_chk(0x58030a85e8b0, 0x58030a8568b3, 11, 11) = 0x58030a85e8b0  
readdir(0x58030a856870) = 0x58030a8568c0  
readdir(0x58030a856870) = 0x58030a8568d0  
readdir(0x58030a856870) = 0x58030a856900  
readdir(0x58030a856870) = 0x58030a856930  
readdir(0x58030a856870) = 0x58030a856950  
readdir(0x58030a856870) = 0x58030a856960  
readdir(0x58030a856870) = 0x58030a856980  
readdir(0x58030a856870) = 0x58030a8569b0  
readdir(0x58030a856870) = 0x58030a8569d0  
readdir(0x58030a856870) = 0x58030a8569f0  
__errno_location() = 0x7e421fa866a0  
__ctype_get_mb_cur_max() = 6  
strlen("output.txt") = 10  
strlen("output.txt") = 10  
__memcpy_chk(0x58030a85e8d0, 0x58030a856a03, 11, 11) = 0x58030a85e8d0  
readdir(0x58030a856870) = 0x58030a856a10  
__errno_location() = 0x7e421fa866a0  
__ctype_get_mb_cur_max() = 6  
strlen("trace.txt") = 9  
strlen("trace.txt") = 9  
__memcpy_chk(0x58030a85e8f0, 0x58030a856a23, 10, 10) = 0x58030a85e8f0  
readdir(0x58030a856870) = 0x58030a856a30  
readdir(0x58030a856870) = 0x58030a856a50  
readdir(0x58030a856870) = 0x58030a856a70  
readdir(0x58030a856870) = 0x58030a856a90  
readdir(0x58030a856870) = 0  
closedir(0x58030a856870) = 0  
reallocarray(0, 3, 24, 0x7e421f91c63b) = 0x58030a856870  
strlen("praktikum1") = 10
```

```
__ctype_get_mb_cur_max() = 6  
__ctype_get_mb_cur_max() = 6  
strlen("output.txt") = 10  
__ctype_get_mb_cur_max() = 6  
__ctype_get_mb_cur_max() = 6  
strlen("trace.txt") = 9  
__ctype_get_mb_cur_max() = 6  
__ctype_get_mb_cur_max() = 6  
setjmp(0x5802e1b3380, 0x7fffd3481394f, 0, 0) = 0  
__errno_location() = 0x7e421fa866a0  
strcoll("output.txt", "trace.txt") = -5  
__errno_location() = 0x7e421fa866a0  
strcoll("praktikum1", "output.txt") = 1  
__errno_location() = 0x7e421fa866a0  
strcoll("praktikum1", "trace.txt") = -4  
reallocarray(0, 3, 48, 116) = 0x58030a8568c0  
reallocarray(0, 21, 8, 1) = 0x58030a856960  
strlen("output.txt") = 10  
fwrite_unlocked("output.txt", 1, 10, 0x7e421fa845c0) = 10  
__overflow(0x7e421fa845c0, 32, 3, 8) = 32  
__overflow(0x7e421fa845c0, 32, 4, 8) = 32  
strlen("praktikum1") = 10  
fwrite_unlocked("praktikum1", 1, 10, 0x7e421fa845c0) = 10  
__overflow(0x7e421fa845c0, 9, 7, 8) = 9  
strlen("trace.txt") = 9  
fwrite_unlocked("trace.txt", 1, 9, 0x7e421fa845c0) = 9  
__overflow(0x7e421fa845c0, 10, 10, 10, output.txt praktikum1 trace.txt) = 10  
__fpending(0x7e421fa845c0, 0, 1, 1) = 0  
fileno(0x7e421fa845c0) = 1  
__freanding(0x7e421fa845c0, 0, 1, 1) = 0  
__freanding(0x7e421fa845c0, 0, 0, 1) = 0  
fflush(0x7e421fa845c0) = 0  
fclose(0x7e421fa845c0) = 0  
__fpending(0x7e421fa844e0, 0, 0x7e421fa01e00, 1) = 0  
fileno(0x7e421fa844e0) = 2  
__freanding(0x7e421fa844e0, 0, 0x7e421fa01e00, 1) = 0  
__freanding(0x7e421fa844e0, 0, 0, 1) = 0  
fflush(0x7e421fa844e0) = 0  
fclose(0x7e421fa844e0) = 0  
+++ exited (status 0) +++  
agilnggara@RagilAnggara:~$ |
```

10.3 Using GDB

Buat program dengan bug buggy.c

```
agilnggara@RagilAnggara: ~  
agilnggara@RagilAnggara:~$ cd ~/praktikum1  
agilnggara@RagilAnggara:~/praktikum1$ mkdir buggy  
agilnggara@RagilAnggara:~/praktikum1$ cd buggy  
agilnggara@RagilAnggara:~/praktikum1/buggy$ nano buggy.c
```

```
GNU nano 7.2                                buggy.c *
#include <stdio.h>
#include <stdlib.h>

void buggy_function(int n) {
    int *array = (int*)malloc(n * sizeof(int));
    for (int i = 0; i <= n; i++) { // Bug: should be i < n
        array[i] = i * 2;
    }
    for (int i = 0; i < n; i++) {
        printf("array[%d] = %d\n", i, array[i]);
    }
    free(array);
}

int main(){
    buggy_function(5);
    return 0;
}
```

Debug dengan GDB

```
agilnggara@RagilAnggara:~/praktikum1/buggy$ gcc -g -o buggy buggy.c\
> wxit
/usr/bin/ld: cannot find buggy.cwxit: No such file or directory
collect2: error: ld returned 1 exit status
agilnggara@RagilAnggara:~/praktikum1/buggy$ gcc -g -o buggy buggy.c
agilnggara@RagilAnggara:~/praktikum1/buggy$ gdb ./buggy
GNU gdb (Ubuntu 15.0.50.20240403-0ubuntu1) 15.0.50.20240403-git
Copyright (C) 2024 Free Software Foundation, Inc.
License GPLv3+: GNU GPL version 3 or later <http://gnu.org/licenses/gpl.html>
This is free software: you are free to change and redistribute it.
There is NO WARRANTY, to the extent permitted by law.
Type "show copying" and "show warranty" for details.
This GDB was configured as "x86_64-linux-gnu".
Type "show configuration" for configuration details.
For bug reporting instructions, please see:
<https://www.gnu.org/software/gdb/bugs/>.
Find the GDB manual and other documentation resources online at:
<http://www.gnu.org/software/gdb/documentation/>.

For help, type "help".
Type "apropos word" to search for commands related to "word"...
Reading symbols from ./buggy...
(gdb) |
```

```
(gdb) break main
Breakpoint 1 at 0x1238: file buggy.c, line 15.
(gdb) run
Starting program: /home/agilnggara/praktikum1/buggy/buggy
[Thread debugging using libthread_db enabled]
Using host libthread_db library "/lib/x86_64-linux-gnu/libthread_db.so.1".

Breakpoint 1, main () at buggy.c:15
15     buggy_function(5);
(gdb) next
array[0] = 0
array[1] = 2
array[2] = 4
array[3] = 6
array[4] = 8
16     return 0;
(gdb) step
17 }
(gdb) print n
No symbol "n" in current context.
(gdb) continue
Continuing.
[Inferior 1 (process 715) exited normally]
(gdb) backtrace
No stack.
(gdb) print n
No symbol "n" in current context.
(gdb) quit|
```

10.4 Kernel Debugging dengan dmesg

Lihat kernel Log

```
agilinggara@RagilAnggara: ~/praktikum1/buggy$ dmesg
0.000000 Linux version 6.6.87.2-microsoft-standard-WSL2 (root@439a258ad544) (gcc (GCC) 11.2.0, GNU ld (GNU Binutils) 2.37) #1 SMP PREEMPT_DYNAMIC Thu Jun  5 18:30:46 UTC 2025
0.000000 Command line: initrd=initrd.img WSL_ROOT_INIT=1 panic=-1 nr_cpus=16 hv_utils.timesync_implicit=1 console=hvc0 debug ptty_legacy_count=0 WSL_ENABLE_CRASH_DUMP=1
0.000000 KERNEL supported cpus:
0.000000 Intel GenuineIntel
0.000000 AMD AuthenticAMD
0.000000 BIOS-provided physical RAM map:
0.000000 BIOS-e820: [mem 0x0000000000000000-0x0000000000009ffff] usable
0.000000 BIOS-e820: [mem 0x000000000000e000-0x000000000000e0ffff] reserved
0.000000 BIOS-e820: [mem 0x000000000000100000-0x0000000000001fffff] ACPI data
0.000000 BIOS-e820: [mem 0x000000000000200000-0x0000000000002fffff] usable
0.000000 BIOS-e820: [mem 0x000000000000300000-0x0000000000003fffff] usable
0.000000 NX (Execute Disable) protection: active
0.000000 APIC: Static calls initialized
0.000000 DMI not present or invalid.
0.000000 Hypervisor detected: Microsoft Hyper-V
0.000000 Hyper-V: privilege flags low 0xae7f, high 0x3b8030, hints 0x900c2c, misc 0xe0bed7b6
0.000000 Hyper-V: Nested features: 0x4a0000
0.000000 Hyper-V: LAPIC Timer Frequency: 0xc3500
0.000000 Hyper-V: Using hypercall for remote TLB flush
0.000000 clocksource: hyperv_clocksource_tsc_page: mask: 0xffffffffffffffff max_cycles: 0x24e6a1710, max_idle_ns: 440795202120 ns
0.000000 clocksource: hyperv_clocksource_msr: mask: 0xffffffffffffffff max_cycles: 0x24e6a1710, max_idle_ns: 440795202120 ns
0.000000 tsc: Detected 3293.728 MHz processor
0.000114 e820: update [mem 0x00000000-0x000000fff] usable ==> reserved
0.000121 e820: remove [mem 0x00000000-0x000000fff] usable
0.000128 last_pfn = 0x1f4e00 max_arch_pfn = 0x400000000
0.000193 MTRR map: 5 entries (4 fixed + 1 variable; max 20), built from 8 variable MTRRs
0.000200 x86/PAT: Configuration [0-7]: WB WC UC- UC WB WP UC- WT
0.000310 last_pfn = 0xf8000 max_arch_pfn = 0x400000000
0.000333 Using GB pages for direct mapping
0.000567 RAMDISK: [mem 0x04be4000-0x04e7ffff]
0.000572 ACPI: Early table checksum verification disabled
0.000577 ACPI: RSDP 0x0000000000000000 000024 (v02 VIRTUAL)
0.000585 ACPI: XSDT 0x0000000000100000 000044 (v01 VIRTUAL MICROSOFT 00000001 MSFT 00000001)
0.000594 ACPI: FACP 0x0000000000010000 000114 (v06 VIRTUAL MICROSOFT 00000001 MSFT 00000001)
0.000604 ACPI: DSDT 0x0000000000010100 01E11C (v02 MSTMVM DSDT01 00000001 INTL 20230628)
0.000610 ACPI: FACS 0x0000000000010114 000040
0.000616 ACPI: OEM0 0x0000000000010114 000064 (v01 VIRTUAL MICROSOFT 00000001 MSFT 00000001)
0.000622 ACPI: SRAT 0x0000000000011F2D 0003B0 (v02 VIRTUAL MICROSOFT 00000001 MSFT 00000001)
0.000628 ACPI: APIC 0x0000000000011F68 0000C8 (v04 VIRTUAL MICROSOFT 00000001 MSFT 00000001)
0.000634 ACPI: Reserving FACP table memory at [mem 0x101000-0x101113]
0.000636 ACPI: Reserving DSDT table memory at [mem 0x1011B8-0x11f2D3]
0.000638 ACPI: Reserving FACS table memory at [mem 0x101114-0x101153]
0.000640 ACPI: Reserving OEM0 table memory at [mem 0x101154-0x1011B7]
0.000642 ACPI: Reserving SRAT table memory at [mem 0x11f2D4-0x11f683]
0.000644 ACPI: Reserving APIC table memory at [mem 0x11f684-0x11f74b]
0.000699 SRAT: PXM 0 -> APIC 0x00 -> Node 0
0.000702 SRAT: PXM 0 -> APIC 0x01 -> Node 0
0.000704 SRAT: PXM 0 -> APIC 0x02 -> Node 0
0.000705 SRAT: PXM 0 -> APIC 0x03 -> Node 0
0.000707 SRAT: PXM 0 -> APIC 0x04 -> Node 0
0.000708 SRAT: PXM 0 -> APIC 0x05 -> Node 0
0.000710 SRAT: PXM 0 -> APIC 0x06 -> Node 0
```

```
0.000711 SRAT: PXM 0 -> APIC 0x07 -> Node 0
0.000713 SRAT: PXM 0 -> APIC 0x08 -> Node 0
0.000714 SRAT: PXM 0 -> APIC 0x09 -> Node 0
0.000716 SRAT: PXM 0 -> APIC 0x0a -> Node 0
0.000717 SRAT: PXM 0 -> APIC 0x0b -> Node 0
0.000719 SRAT: PXM 0 -> APIC 0x0c -> Node 0
0.000720 SRAT: PXM 0 -> APIC 0x0d -> Node 0
0.000722 SRAT: PXM 0 -> APIC 0x0e -> Node 0
0.000723 SRAT: PXM 0 -> APIC 0x0f -> Node 0
0.000727 ACPI: SRAT: Node 0 PXM 0 [mem 0x00000000-0xf7ffffff] hotplug
0.000732 ACPI: SRAT: Node 0 PXM 0 [mem 0x10000000-0x1f4dffff] hotplug
0.000735 ACPI: SRAT: Node 0 PXM 0 [mem 0x1f4e0000-0x9ffdf000] hotplug
0.000737 ACPI: SRAT: Node 0 PXM 0 [mem 0x100000000-0xf0000000] hotplug
0.000740 ACPI: SRAT: Node 0 PXM 0 [mem 0x1000000000-0x1fffffffffff] hotplug
0.000743 ACPI: SRAT: Node 0 PXM 0 [mem 0x2000000000-0x3fffffffffff] hotplug
0.000746 ACPI: SRAT: Node 0 PXM 0 [mem 0x4000000000-0x7fffffffffff] hotplug
0.000748 ACPI: SRAT: Node 0 PXM 0 [mem 0x8000000000-0xfffffffffff] hotplug
0.000751 ACPI: SRAT: Node 0 PXM 0 [mem 0x10000000000-0x1fffffffffff] hotplug
0.000754 ACPI: SRAT: Node 0 PXM 0 [mem 0x20000000000-0x3fffffffffff] hotplug
0.000757 ACPI: SRAT: Node 0 PXM 0 [mem 0x40000000000-0x7fffffffffff] hotplug
0.000760 ACPI: SRAT: Node 0 PXM 0 [mem 0x80000000000-0xfffffffffff] hotplug
0.000762 ACPI: SRAT: Node 0 PXM 0 [mem 0x100000000000-0x1fffffffffff] hotplug
0.000765 ACPI: SRAT: Node 0 PXM 0 [mem 0x200000000000-0x3fffffffffff] hotplug
0.000768 ACPI: SRAT: Node 0 PXM 0 [mem 0x400000000000-0x7fffffffffff] hotplug
0.000771 ACPI: SRAT: Node 0 PXM 0 [mem 0x800000000000-0xfffffffffff] hotplug
0.000777 NUMA: Node 0 [mem 0x00000000-0xf7ffffff] + [mem 0x100000000-0x1f4dffff] -> [mem 0x00000000-0x1f4dffff]
0.000780 NODE_DATA(0) allocated [mem 0x1f4dde000-0x1f4dffff]
0.000935 Zone ranges:
0.000936 DMA32 [mem 0x0000000000001000-0x0000000000009ffff]
0.000939 Normal [mem 0x0000000010000000-0x000000001f4dffff]
0.000941 Device empty
0.000943 Movable zone start for each node
0.000945 Early memory node ranges
0.000946 node 0: [mem 0x0000000000000000-0x0000000000009ffff]
0.000948 node 0: [mem 0x0000000000020000-0x000000000000ffff]
0.000949 node 0: [mem 0x0000000010000000-0x000000001f4dffff]
0.000951 Initmem setup node 0 [mem 0x0000000000001000-0x000000001f4dffff]
0.000992 On node 0, zone DMA32: 1 pages in unavailable ranges
0.000994 On node 0, zone DMA32: 352 pages in unavailable ranges
0.000996 On node 0, zone Normal: 12800 pages in unavailable ranges
0.000999 ACPI: PM-Timer IO Port: 0x408
0.000951 ACPI: LAPIC_NMI (acpi_id[0x01] dfl dfl lint[0x1])
0.000955 IOAPIC[0]: apic_id 16, version 17, address 0xfec00000, GSI 0-23
0.000958 ACPI: INT_SRC_OVR (bus 0 bus_irq 9 global_irq 9 high level)
0.000959 ACPI: Using ACPI (MADT) for SMP configuration information
0.000957 smpboot: Allowing 16 CPUs, 0 hotplug CPUs
0.000961 PM: hibernation: Registered nosave memory: [mem 0x00000000-0x000000fff]
0.000962 PM: hibernation: Registered nosave memory: [mem 0x000a0000-0x000a00fff]
0.000963 PM: hibernation: Registered nosave memory: [mem 0x000e0000-0x000e00fff]
0.000964 PM: hibernation: Registered nosave memory: [mem 0x000f0000-0x000f00fff]
0.000965 PM: hibernation: Registered nosave memory: [mem 0x00100000-0x001000fff]
0.000966 PM: hibernation: Registered nosave memory: [mem 0xf8000000-0xffffffff]
0.000967 [mem 0xf8000000-0xffffffff] available for PCI devices
```

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agilinggara@RagilAnggara: ~$ cat /proc/kmsg
0.048631] [mem 0xf8000000-0xffffffff] available for PCI devices
0.048634] Booting paravirtualized kernel on Hyper-V
0.048637] clocksource: refined-jiffies: mask: 0xffffffff max_cycles: 0xffffffff, max_idle_ns: 7645519600211568 ns
0.048671] setup_percpu: NR_CPUS:64 nr_cpumask_bits:16 nr_cpu_ids:16 nr_node_ids:1
0.052391] percpu: Embedded 63 pages/cpu <221184 0192 028672 0262144>
0.052402] pcpu-alloc: <221184 0192 028672 0262144> alloc=1*2097152
0.052407] pcpu-alloc: [0] 00 01 02 03 04 05 06 07 [0] 08 09 10 11 12 13 14 15
0.052453] Hyper-V: PV spinlocks enabled
0.052456] PV qspinlock hash table entries: 256 (order: 0, 4096 bytes, linear)
0.052461] Kernel command line: initrd=initrd.img WSL_ROOT_INIT=1 panic=1 nr_cpus=16 hv_utils.timesync_implicit=1 console=hvc0 debug pty.legacy_count=0 WSL_ENABLE_CRASH_DUMP=1
0.052469] Unknown kernel command line parameters "WSL_ROOT_INIT=1 WSL_ENABLE_CRASH_DUMP=1", will be passed to user space.
0.052723] random: crng init done
0.059584] Dentry cache hash table entries: 1048576 (order: 11, 8388608 bytes, linear)
0.062940] Inode-cache hash table entries: 524288 (order: 10, 4194304 bytes, linear)
0.063233] fallback order for Node 0: 0
0.063244] Built 1 zonelists, mobility grouping on. Total pages: 1986248
0.063247] Policy zone: Normal
0.063261] mem auto-init: stack:off, heap alloc:on, heap free:off
0.063270] software IO TLB: area num 16
0.135821] Memory: 4107404K/8073852K available (20480K kernel code, 3457K rdata, 13496K rodata, 4492K init, 6208K bss, 262424K reserved, 0K cma-reserved)
0.136185] SLUB: Hwalign=64, Order=0-3, MinObjects=0, CPUs=16, Nodes=1
0.136266] ftrace: allocating 55808 entries in 218 pages
0.136344] ftrace: allocated 218 pages with 5 groups
0.136347] Dynamic Preempt: none
0.155515] rcu: Preemptible hierarchical RCU implementation.
0.155517] rcu: RCU restricting CPUs from NR_CPUS=8192 to nr_cpu_ids=16.
0.155519] Trampoline variant of Tasks RCU enabled.
0.155520] Rude variant of Tasks RCU enabled.
0.155521] Tracing variant of Tasks RCU enabled.
0.155522] rcu: RCU calculated value of scheduler-enlistment delay is 25 jiffies.
0.155523] rcu: Adjusting geometry for rcu_fanout_leaf=16, nr_cpu_ids=16
0.167327] Using NULL legacy PIC
0.167330] NR_IRQS: 524544, nr_irqs: 552, preallocated irqs: 0
0.167432] rcu: srcu_init: Setting srcu_struct sizes based on contention.
0.168735] Console: colour dummy device 80x25
0.168782] ACPI: Core revision 20230628
0.169170] Failed to register legacy timer interrupt
0.169173] APIC: Switch to symmetric I/O mode setup
0.169189] APIC: Switched APIC routing to: physical flat
0.169194] Hyper-V: Host Build 10.0.26100.6725-6-0
0.169197] Hyper-V: enabling crash_kexec_post_notifiers
0.169304] Hyper-V: Using IPI hypercalls
0.169308] APIC: send_IPI() replaced with hv_send_ipi()
0.169315] APIC: send_IPI_mask() replaced with hv_send_ipi_mask()
0.169318] APIC: send_IPI_allbutself() replaced with hv_send_ipi_mask_allbutself()
0.169320] APIC: send_IPI_allbutself() replaced with hv_send_ipi_allbutself()
0.169323] APIC: send_IPI_all() replaced with hv_send_ipi_all()
0.169325] APIC: send_IPI_self() replaced with hv_send_ipi_self()
0.169327] Hyper-V: Using enlightened APIC (xapic mode)
0.169339] APIC: eoi() replaced with hv_apic_eoi_write()
0.169375] APIC: read() replaced with hv_apic_read()
0.169404] APIC: write() replaced with hv_apic_write()
0.169406] APIC: icr_write() replaced with hv_apic_icr_write()

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agilinggara@RagilAnggara: ~$ cat /proc/kmsg
0.169409] APIC: icr_read() replaced with hv_apic_icr_read()
0.169421] clocksource: tsc-early: mask: 0xffffffffffffffff max_cycles: 0x2f7a2765acc, max_idle_ns: 440795282716 ns
0.169430] Calibrating delay loop (skipped), value calculated using timer frequency.. 6587.45 BogoMIPS (lpj=13174912)
0.169765] x86/cpu: User Mode Instruction Prevention (UMIP) activated
0.169795] Last level iTLB entries: 4KB 512, 2MB 512, 4MB 256
0.169796] Last level dTLB entries: 4KB 2048, 2MB 2048, 4MB 1024, 1GB 0
0.169801] Spectre V1 : Mitigation: usercopy/swapgs barriers and __user pointer sanitization
0.169804] Spectre V2 : Mitigation: Retpolines
0.169805] Spectre V2 : Spectre V2 / SpectreRSB mitigation: Filling RSB on context switch
0.169805] Spectre V2 : Spectre V2 / SpectreRSB : Filling RSB on VMEXIT
0.169806] Spectre V2 : Enabling Restricted Speculation for firmware calls
0.169808] Spectre V2 : mitigation: Enabling conditional Indirect Branch Prediction Barrier
0.169809] Spectre V2 : User space: Mitigation: STIBP always-on protection
0.169811] Speculative Store Bypass: Mitigation: Speculative Store Bypass disabled via prctl
0.169812] Speculative Return Stack Overflow: IBPB-extending microcode not applied!
0.169813] Speculative Return Stack Overflow: WARNING: See https://kernel.org/doc/html/latest/admin-guide/hw-vuln/srso.html for mitigation options.
0.169814] Speculative Return Stack Overflow: Vulnerable: Safe RET, no microcode
0.169839] x86/fpu: Supporting XSAVE feature 0x001: 'x87 floating point registers'
0.169841] x86/fpu: Supporting XSAVE feature 0x002: 'SSE registers'
0.169842] x86/fpu: Supporting XSAVE feature 0x004: 'AVX registers'
0.169843] x86/fpu: Supporting XSAVE feature 0x800: 'Control-flow User registers'
0.169844] x86/fpu: xstate_offset[2]: 576, xstate_sizes[2]: 256
0.169846] x86/fpu: xstate_offset[11]: 892, xstate_sizes[11]: 16
0.169848] x86/fpu: Enabled xstate features 0x807, context size is 848 bytes, using 'compacted' format.
0.173626] Freeing SMP alternatives memory: 44K
0.173626] pid_max: default: 32768 minimum: 301
0.173626] LSM: initializing lsm=capability,landlock,yama,safesetid,selinux,integrity
0.173626] landlock: Up and running.
0.173626] Yama: becoming mindful.
0.173626] SELinux: Initializing.
0.173626] Mount-cache hash table entries: 16384 (order: 5, 131072 bytes, linear)
0.173626] Mountpoint-cache hash table entries: 16384 (order: 5, 131072 bytes, linear)
0.173626] smpboot: CPU0: AMD Ryzen 9 5900HX with Radeon Graphics (family: 0x19, model: 0x50, stepping: 0x0)
0.173626] RCU Tasks: Setting shift to 4 and lim to 1 rcu_task_cb_adjust=1 rcu_task_cpu_ids=16.
0.173626] RCU Tasks Rude: Setting shift to 4 and lim to 1 rcu_task_cb_adjust=1 rcu_task_cpu_ids=16.
0.173626] RCU Tasks Trace: Setting shift to 4 and lim to 1 rcu_task_cb_adjust=1 rcu_task_cpu_ids=16.
0.173626] Performance Events: Fam17h+ core perfctr, AMD PMU driver.
0.173626] ... version: 0
0.173626] ... bit width: 48
0.173626] ... generic registers: 6
0.173626] ... value mask: 0000000000000000
0.173626] ... max period: 0000000000000000
0.173626] ... fixed-purpose events: 0
0.173626] ... event mask: 0000000000000000
0.173626] signal: max sigframe size: 1776
0.173626] rcu: Hierarchical SRCU implementation.
0.173626] rcu: Max phase no-delay instances is 1000.
0.179485] smp: Bringing up secondary CPUs ...
0.179485] smpboot: x86: Booting SMP configuration:
0.179485] ... node #0, CPUs: #2 #4 #6 #8 #10 #12 #14 #1 #3 #5 #7 #9 #11 #13 #15
0.205641] Spectre V2 : Update user space SMT mitigation: STIBP always-on
0.210140] smp: Brought up 1 node, 16 CPUs
0.210140] smpboot: Max logical packages: 1

```

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agilinggara@RagilAnggara: ~/  +  -
0.210151] smpboot: Total of 16 processors activated (105399.29 BogoMIPS)
0.238961] node 0 deferred pages initialised in 12ms
0.241952] devtmpfs: initialized
0.241952] x86/x86_64: Memory block size: 128MB
0.245955] clocksource: jiffies: mask: 0xffffffff max_cycles: 0xffffffff, max_idle_ns: 7645041785180000 ns
0.245982] futex hash table entries: 4096 (order: 6, 262144 bytes, linear)
0.246146] pinctrl core: initialized pinctrl subsystem
0.248407] NET: Registered PF_NETLINK/PF_ROUTE protocol family
0.249859] DMA: preallocated 1024 KiB GFP_KERNEL pool for atomic allocations
0.250470] DMA: preallocated 1024 KiB GFP_KERNEL pool for atomic allocations
0.250533] audit: initializing netlink subsys (disabled)
0.250575] audit: type=2000 audit(1760427505.080:1): state=initialized audit_enabled=0 res=1
0.250575] thermal_sys: Registered thermal governor 'step-wise'
0.250575] thermal_sys: Registered thermal governor 'user-space'
0.250575] cpuidle: using governor ladder
0.250575] cpuidle: using governor menu
0.250575] acpihp: ACPI Hot Plug PCI Controller Driver version: 0.5
0.250575] dca service started, version 1.12.1
0.250575] PCI: Fatal: No config space access function found
0.254692] kprobes: kprobe jump-optimization is enabled. All kprobes are optimized if possible.
0.254730] HugeTLB: registered 1.00 GiB page size, pre-allocated 0 pages
0.254730] HugeTLB: 16380 KiB vmemmap can be freed for a 1.00 GiB page
0.254730] HugeTLB: registered 2.00 MiB page size, pre-allocated 0 pages
0.254730] HugeTLB: 28 KiB vmemmap can be freed for a 2.00 MiB page
0.255060] cryptd: max_cpu_glen set to 1000
0.258057] ACPI: Added _OSI(Module Device)
0.258062] ACPI: Added _OSI(Processor Device)
0.258065] ACPI: Added _OSI(1.0 _SCP Extensions)
0.258068] ACPI: Added _OSI(Processor Aggregator Device)
0.290612] ACPI: 1 ACPI AML tables successfully acquired and loaded
0.292343] ACPI: _OSC evaluation for CPUs failed, trying _PDC
0.292943] ACPI: Interpreter enabled
0.292959] ACPI: PM: (supports S0 S5)
0.292962] ACPI: Using IOAPIC for interrupt routing
0.292990] PCI: Using host bridge windows from ACPI; if necessary, use "pci=nocrs" and report a bug
0.293001] PCI: Using E820 reservations for host bridge windows
0.293546] ACPI: Enabled 2 GPEs in block 00 to 0F
0.299448] iommu: Default domain type: Translated
0.299448] iommu: DMA domain TLB invalidation policy: lazy mode
0.299448] SCSI subsystem initialized
0.299448] libata version 3.00 loaded.
0.299448] pps_core: LinuxPPS API ver. 1 registered
0.299448] pps_core: Software ver. 5.3.6 - Copyright 2005-2007 Rodolfo Giometti <giometti@linux.it>
0.299448] PTP clock support registered
0.299448] EDAC MC: Ver: 3.0.0
0.313637] hv_vmbus: Vmbus version:5.3
0.318697] hv_vmbus: Unknown GUID: 6e382d18-3336-4f4b-acc4-2b7703d4d4f4a
0.318782] hv_vmbus: Unknown GUID: dde9c0c0-5060-4436-9448-ea1254a5d177
0.318789] hv_vmbus: Unknown GUID: 6e382d18-3336-4f4b-acc4-2b7703d4d4f4a
0.318770] NetLabel: Initializing
0.318770] NetLabel: domain hash size = 128
0.318770] NetLabel: protocols = UNLABELED CIPSOv4 CALIPSO
0.318770] NetLabel: unlabeled traffic allowed by default
```

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agilinggara@RagilAnggara: ~/  +  -
0.318770] NetLabel: unlabeled traffic allowed by default
0.318770] PCI: Using ACPI for IRQ routing
0.318770] PCI: System does not support PCI
0.318770] vgaarb: loaded
0.328434] clocksource: Switched to clocksource tsc-early
0.334076] VFS: Disk quotas dquot_6.6.0
0.334117] VFS: Dquot-cache hash table entries: 512 (order 0, 4096 bytes)
0.334216] FS-Cache: Loaded
0.334334] pnp: PnP ACPI init
0.334933] pnp: PnP ACPI: found 1 devices
0.350773] clocksource: acpi_pm: mask: 0xffffffff max_cycles: 0xffffffff, max_idle_ns: 2085701024 ns
0.350927] NET: Registered PF_INET protocol family
0.351559] IP ident hash table entries: 131072 (order: 8, 1048576 bytes, linear)
0.354589] tcp_listen_portaddr_hash hash table entries: 4096 (order: 4, 65536 bytes, linear)
0.354771] Table-perturb hash table entries: 65536 (order: 6, 262144 bytes, linear)
0.355106] TCP established hash table entries: 65536 (order: 7, 524288 bytes, linear)
0.356356] TCP bind hash table entries: 65536 (order: 9, 2097152 bytes, linear)
0.356552] TCP: Hash tables configured (established 65536 bind 65536)
0.356817] UDP hash table entries: 4096 (order: 5, 131072 bytes, linear)
0.356906] UDP-Lite hash table entries: 4096 (order: 5, 131072 bytes, linear)
0.357070] NET: Registered PF_UNIX/PF_LOCAL protocol family
0.357084] NET: Registered PF_XDP protocol family
0.357092] PCI: CLS 0 bytes, default 64
0.357179] PCI-DMA: Using software bounce buffering for IO (SWIOTLB)
0.357182] software IO TLB: mapped [mem 0x00000000f4000000-0x00000000f8000000] (64MB)
0.357438] Trying to unpack rootfs image as initramfs...
0.363030] Freeing initrd memory: 2672K
0.364246] clocksource: tsc: mask: 0xffffffffffffffff max_cycles: 0x2f7a2765acc, max_idle_ns: 440795282716 ns
0.377224] clocksource: Switched to clocksource tsc
0.379134] Initialise system trusted keyrings
0.379147] Key type blacklist registered
0.379382] workingset: timestamp_bits=36 max_order=21 bucket_order=0
0.380211] squashfs: version 4.0 (2009/01/31) Phillip Lougher
0.380252] fuse: init (API version 7.39)
0.380679] SGI XFS with ACLs, security attributes, realtime, verbose warnings, quota, no debug enabled
0.381870] 9p: Installing v9fs 9p2000 file system support
0.402251] Key type asymmetric registered
0.402256] Asymmetric key parser 'x509' registered
0.402296] Block layer SCSI generic (bsg) driver version 0.4 loaded (major 245)
0.402579] io scheduler mq-deadline registered
0.402586] io scheduler kyber registered
0.409965] hv_vmbus: registering driver hv_pci
0.410797] hv_pci c4b741f5-5582-4c98-8f8b-2e082933c396: PCI VMBus probing: Using version 0x10004
0.412453] hv_pci c4b741f5-5582-4c98-8f8b-2e082933c396: PCI host bridge to bus 5582:00
0.412458] pci_bus 5582:00: root bus resource [mem 0x9ffe00000-0x9ffe02fff window]
0.412463] pci_bus 5582:00: No busn resource found for root bus, will use [bus 00-ff]
0.413830] pci 5582:00:00.0: [1af4:1043] type 00 class 0x010000
0.414882] pci 5582:00:00.0: reg 0x10: [mem 0x9ffe00000-0x9ffe00fff 64bit]
0.415511] pci 5582:00:00.0: reg 0x18: [mem 0x9ffe01000-0x9ffe01fff 64bit]
0.416139] pci 5582:00:00.0: reg 0x20: [mem 0x9ffe02000-0x9ffe02fff 64bit]
0.420950] pci_bus 5582:00: busn_res: [bus 00-ff] end is updated to 00
0.420971] pci 5582:00:00.0: BAR 0: assigned [mem 0x9ffe00000-0x9ffe00fff 64bit]
0.421580] pci 5582:00:00.0: BAR 2: assigned [mem 0x9ffe01000-0x9ffe01fff 64bit]
```

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agilinggara@RagilAnggara: ~/  +  -
0.422073] pci 5582:00:00.0: BAR 4: assigned [mem 0x9ffe02000-0x9ffe02fff 64bit]
0.423583] hv_pci b88269c7-de0e-46b7-b243-b65ff865f444: PCI VMBus probing: Using version 0x10004
0.424835] hv_pci b88269c7-de0e-46b7-b243-b65ff865f444: PCI host bridge to bus de0e:00
0.424941] pci bus de0e:00: No busn resource found for root bus, will use [bus 00-ff]
0.425794] pci de0e:00:00.0: [1144:008e] type 00 class 0x030200
0.433381] pci_bus de0e:00: busn_res: [bus 00-ff] end is updated to 00
0.434072] hv_pci a56e290c-a836-4826-910c-4d20421db135: PCI VMBus probing: Using version 0x10004
0.436861] hv_pci a56e290c-a836-4826-910c-4d20421db135: PCI host bridge to bus a836:00
0.436866] pci_bus a836:00: No busn resource found for root bus, will use [bus 00-ff]
0.437602] pci a836:00:00.0: [1144:008e] type 00 class 0x030200
0.440970] pci_bus a836:00: busn_res: [bus 00-ff] end is updated to 00
0.452383] iohatdma: Intel(R) QuickData Technology Driver 5.00
0.452760] virtio-pci 5582:00:00.0: enabling device (0000 -> 0002)
0.459482] Serial: 8250/16550 driver, 8 ports, IRQ sharing enabled
0.531069] Linux agpgart interface v0.103
0.531134] AMD-Vi: AMD IOMMUv2 functionality not available on this system - This is not a bug.
0.531150] ACPI: bus type drv_connector registered
0.534685] printk: console [hvc0] enabled
0.547098] brd: module loaded
0.556580] loop: module loaded
0.557336] Loading iSCSI transport class v2.0-870.
0.558585] rdac: device handler registered
0.559540] Microchip SmartIO Driver (v2.1.24-046)
0.560683] VMware PVSCSI driver - version 1.0.7.0-k
0.561465] hv_vmbus: registering driver hv_storvsc
0.565069] PPP generic driver version 2.4.2
0.565273] scsi host0: storvsc_host.t
0.565935] VMware vmxnet3 virtual NIC driver - version 1.7.0.0-k-NAPI
0.566060] hv_vmbus: registering driver hv_netvsc
0.567517] Fusion MPT base driver: 3.04.20
0.567908] Copyright (C) 1999-2008 LSI Corporation
0.568487] Fusion MPT SPI Host driver 3.04.20
0.569009] Fusion MPT SAS Host driver 3.04.20
0.569670] Fusion MPT misc device (ioctl) driver 3.04.20
0.570520] mptctl: Registered with Fusion MPT base driver
0.571280] mptctl: /dev/mptctl 0 (major,minor=10,220)
0.571833] i8042: PNP: No PS/2 controller found.
0.572376] hv_vmbus: registering driver hyperv_keyboard
0.572965] rtc_cmos 00:00: RTC can wake from S4
0.577214] rtc_cmos 00:00: registered as rtc0
0.578352] rtc_cmos 00:00: setting system clock to 2025-10-14T07:30:26 UTC (1760427506)
0.579357] rtc_cmos 00:00: alarms up to one month, 114 bytes nvram
0.580118] device-mapper: core: CONFIG_IMA_DISABLE_HTABLE is disabled. Duplicate IMA measurements will not be recorded in the IMA log.
0.581436] device-mapper: uevent: version 1.0.3
0.582217] device-mapper: ioctl: 4.48.0-ioctl (2023-03-01) initialised: dm-devel@redhat.com
0.584213] hv_utils: Registering HyperV Utility Driver
0.585364] hv_vmbus: registering driver hv_utils
0.586408] hv_vmbus: registering driver hv_balloon
0.587552] hv_vmbus: registering driver dxgkrnl
0.587575] hv_utils: TimeSync IC version 4.0
0.588009] hv_balloon: Using Dynamic Memory protocol version 2.0
0.591148] Free page reporting enabled
0.591698] hv_balloon: Cold memory discard hint enabled with order 0
```

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agilinggara@RagilAnggara: ~/  +  -
0.598988] drop_monitor: Initializing network drop monitor service
0.630479] NET: Registered PF_INET6 protocol family
0.633532] Segment Routing with IPv6
0.634220] In-situ OAM (IOAM) with IPv6
0.634072] NET: Registered PF_PACKET protocol family
0.636071] 9pnet: Installing 9P2000 support
0.637801] NET: Registered PF_VSOCK protocol family
0.638817] hv_vmbus: registering driver hv_sock
0.644555] IPI shorthand broadcast: enabled
0.645529] AVX2 version of gcm_enc/dec engaged.
0.647996] AES CTR bytestream optimization enabled
0.657545] sched_clock: Marking stable (656089167, 1131992)->(742173656, -85032497)
0.659727] registered taskstats version 1
0.660695] Loading compiled-in X.509 certificates
0.673928] ima: No TPM chip found, activating TPM-bypass!
0.675996] ima: Allocated hash algorithm: sha256
0.678016] ima: No architecture policies found
0.680479] RAS: Correctable Errors collector initialized.
0.684197] clk: Disabling unused clocks
0.685743] Freeing unused decrypted memory: 2028K
0.688593] Freeing unused kernel image (initmem) memory: 4492K
0.689938] Write protecting the kernel read-only data: 34816k
0.692529] Freeing unused kernel image (rodata/data gap) memory: 840K
0.693805] Run /init as init process
0.694522] with arguments:
0.695094] /init
0.695455] with environment:
0.696031] HOME=/
0.696391] TERM=linux
0.696601] WSL_ROOT_INIT=1
0.697097] WSL_ENABLE_CRASH_DUMP=1
0.698199] scsi 0:0:0:0: Direct-Access Msft Virtual Disk 1.0 PQ: 0 ANSI: 5
0.697672] scsi 0:0:0:1: Direct-Access Msft Virtual Disk 1.0 PQ: 0 ANSI: 5
0.902192] sd 0:0:0:0: Attached scsi generic sg0 type 0
0.902666] sd 0:0:0:0: [sda] 795536 512-byte logical blocks: (407 MB/388 MiB)
0.903675] sd 0:0:0:1: Attached scsi generic sg1 type 0
0.904204] sd 0:0:0:1: [sdb] 381016 512-byte logical blocks: (195 MB/186 MiB)
0.904420] sd 0:0:0:1: [sdb] Write Protect is on
0.904498] sd 0:0:0:0: [sda] Write Protect is on
0.904505] sd 0:0:0:0: [sda] Mode Sense: 0f 00 80 00
0.905072] sd 0:0:0:0: [sda] Write cache: disabled, read cache: enabled, doesn't support DPO or FUA
0.906435] sd 0:0:0:1: [sdb] Mode Sense: 0f 00 80 00
0.906569] scsi 0:0:0:2: Direct-Access Msft Virtual Disk 1.0 PQ: 0 ANSI: 5
0.911985] sd 0:0:0:0: [sda] Attached SCSI disk
0.912747] sd 0:0:0:1: [sdb] Write cache: disabled, read cache: enabled, doesn't support DPO or FUA
0.918982] sd 0:0:0:1: [sdb] Attached SCSI disk
0.922708] sd 0:0:0:2: Attached scsi generic sg2 type 0
0.923041] sd 0:0:0:2: [sdc] 4190312 512-byte logical blocks: (2.15 GB/2.00 GiB)
0.924786] sd 0:0:0:2: [sdc] 4096-byte physical blocks
0.925756] sd 0:0:0:2: [sdc] Write Protect is off
0.926633] sd 0:0:0:2: [sdc] Mode Sense: 0f 00 00 00
0.927729] sd 0:0:0:2: [sdc] Write cache: enabled, read cache: enabled, doesn't support DPO or FUA
0.932698] sd 0:0:0:2: [sdc] Attached SCSI disk
```

```
aglinggara@RagilAnggara: ~$  
0.971334] EXT4-fs (sda): mounted filesystem 00000000-0000-0000-0000-000000000000 ro without journal. Quota mode: none.  
0.978091] EXT4-fs (sdb): mounted filesystem 00000000-0000-0000-0000-000000000000 ro without journal. Quota mode: none.  
1.018395] tun: Universal TUN/TAP device driver, 1.6  
1.071837] Adding 2048 swap on /dev/sdc. Priority:-2 extents:1 across:2097152k  
1.071949] bridge: filtering via arp/ip/ip6tables is no longer available by default. Update your scripts to load br_netfilter if you need this.  
1.078618] Bridge firewalling registered  
1.526598] hv_pci bb2572d8-13ed-4be8-8714-15aed3fb0e4b: PCI VMBus probing: Using version 0x10004  
1.530233] hv_pci bb2572d8-13ed-4be8-8714-15aed3fb0e4b: PCI host bridge to bus 13ed:00  
1.531633] pci_bus 13ed:00: root bus resource [mem 0xc0000000-0xe0001fff window]  
1.533176] pci_bus 13ed:00: No busn resource found for root bus, will use [bus 00-ff]  
1.537007] pci 13ed:00:00: [class 05a] type 00 class 0x0000  
1.540288] pci 13ed:00:00: reg 0x18: [mem 0xe0000000-0xe0000fff 64bit]  
1.542534] pci 13ed:00:00: reg 0x18: [mem 0xe0001000-0xe0001fff 64bit]  
1.544684] pci 13ed:00:00: reg 0x20: [mem 0xc0000000-0xdfffffff 64bit]  
1.552929] pci_bus 13ed:00: busn_res: [bus 00-ff] end is updated to 00  
1.554126] pci 13ed:00:00: BAR 4: assigned [mem 0xc0000000-0xdfffffff 64bit]  
1.556104] pci 13ed:00:00: BAR 0: assigned [mem 0xe0000000-0xe0000fff 64bit]  
1.558030] pci 13ed:00:00: BAR 2: assigned [mem 0xe0001000-0xe0001fff 64bit]  
1.560164] virtio-pci 13ed:00:00: enabling device (0000 -> 0002)  
1.597590] virtiofs virtio1: Cache len: 0x20000000 0 0xc0000000  
1.800207] scsi 0:0:0:3: Direct-Access Msft Virtual Disk 1.0 PQ: 0 ANSI: 5  
1.814846] sd 0:0:0:3: Attached scsi generic sg3 type 0  
1.815244] sd 0:0:0:3: [sdd] 2147483648 512-byte logical blocks: (1.10 TB/1.00 TiB)  
1.816817] sd 0:0:0:3: [sdd] 4096-byte physical blocks  
1.817607] sd 0:0:0:3: [sdd] Write Protect is off  
1.818294] sd 0:0:0:3: [sdd] Mode Sense: 0f 00 00 00  
1.819223] sd 0:0:0:3: [sdd] Write cache: enabled, read cache: enabled, doesn't support DPO or FUA  
1.823985] sd 0:0:0:3: [sdd] Attached SCSI disk  
1.857972] EXT4-fs (sdd): mounted filesystem 77bd2e57-3ee6-49e5-99a7-b08c1b47a2f r/w with ordered data mode. Quota mode: none.  
2.870883] pulseaudio[301]: memfd_create() called without MFD_EXEC or MFD_NOEXEC_SEAL set  
3.066842] misc dxg: dxgk: dxgkio_is_feature_enabled: Ioctl failed: -22  
3.076583] misc dxg: dxgk: dxgkio_query_adapter_info: Ioctl failed: -22  
3.077799] misc dxg: dxgk: dxgkio_query_adapter_info: Ioctl failed: -22  
3.079765] misc dxg: dxgk: dxgkio_query_adapter_info: Ioctl failed: -22  
3.081666] misc dxg: dxgk: dxgkio_query_adapter_info: Ioctl failed: -2  
3.089688] misc dxg: dxgk: dxgkio_query_adapter_info: Ioctl failed: -22  
3.091421] misc dxg: dxgk: dxgkio_query_adapter_info: Ioctl failed: -22  
3.092730] misc dxg: dxgk: dxgkio_query_adapter_info: Ioctl failed: -22  
3.094347] misc dxg: dxgk: dxgkio_query_adapter_info: Ioctl failed: -2  
3.196261] Failed to connect to bus: No such file or directory  
3.426510] systemd-journald[44]: Collecting audit messages is disabled.  
3.596486] systemd-journald[44]: Received client request to flush runtime journal.  
3.599217] systemd-journald[44]: File /var/log/journal/5e0b2178f2a64652ac34e2eddec893ce/system.journal corrupted or uncleanly shut down, renaming and replacing.  
3.903585] ACPI: battery: Slot [BAT1] (battery present)  
3.910138] ACPI: AC: Adapter [AC1] (off-line)  
4.275554] kvm_amd: TSC scaling supported  
4.275565] kvm_amd: Nested Virtualization enabled  
4.275568] kvm_amd: Nested Paging enabled  
4.275574] kvm_amd: Hyper-V enlightened NPT TLB flush enabled  
4.275577] kvm_amd: Hyper-V Direct TLB Flush enabled  
4.275594] kvm_amd: Virtual VMLOAD VMSAVE supported  
4.7327784] systemd-journald[44]: /var/log/journal/5e0b2178f2a64652ac34e2eddec893ce/user-1000.journal: Journal file uses a different sequence number ID, rotating.  
11.549481] WSL (266) ERROR: CheckConnection: getaddrinfo() failed: -3  
  
16.556006] WSL (266) ERROR: CheckConnection: getaddrinfo() failed: -3  
48.696382] hv_balloon: Max. dynamic memory size: 7886 MB  
681.169123] winl_init (260): drop caches: 1  
688.761995] TCP: eth0: Driver has suspect GRO implementation, TCP performance may be compromised.  
aglinggara@RagilAnggara:~/praktikum1/buggy$
```

Follow Kernel Log

```
aglinggara@RagilAnggara: ~$  
aglinggara@RagilAnggara:~/praktikum1/buggy$ dmesg -w  
0.000000] Linux version 6.6.87.2-microsoft-standard-WSL2 (root@439a258ad544) (gcc (GCC) 11.2.0, GNU ld (GNU Binutils) 2.37) #1 SMP PREEMPT_DYNAMIC Thu Jun 5 18:30:46 UTC 2025  
0.000000] Command line: initrd=initrd.img WSL_ROOT_INIT=1 panic=-1 nr_cpus=16 hv_utils.timesync_implicit=1 console=hvc0 debug ptty.legacy_count=0 WSL_ENABLE_CRASH_DUMP=1  
0.000000] KERNEL supported cpus:  
0.000000] Intel GenuineIntel  
0.000000] AMD AuthenticAMD  
0.000000] BIOS-provided physical RAM map:  
0.000000] BIOS-e820: [mem 0x0000000000000000-0x000000000009ffff] usable  
0.000000] BIOS-e820: [mem 0x000000000009ffff-0x00000000000cffff] reserved  
0.000000] BIOS-e820: [mem 0x00000000000cffff-0x00000000001fffff] ACPI data  
0.000000] BIOS-e820: [mem 0x0000000000200000-0x00000000007fffff] usable  
0.000000] BIOS-e820: [mem 0x0000000010000000-0x000000001f4dffff] usable  
0.000000] NX (Execute Disable) protection: active  
0.000000] APIC: Static calls initialized  
0.000000] DMI not present or invalid.  
0.000000] Hypervisor detected: Microsoft Hyper-V  
0.000000] Hyper-V: privilege flags low 0xae7f, high 0x3b030, hints 0x9082c, misc 0xe0bed7b6  
0.000000] Hyper-V: Nested features: 0x4a0000  
0.000000] Hyper-V: LAPIC Timer Frequency: 0xc3500  
0.000000] Hyper-V: Using hypercall for remote TLB flush  
0.000000] clocksource: hyperv_clocksource_tsc_page: mask: 0xffffffffffffff max_cycles: 0x24e6a1710, max_idle_ns: 440795202120 ns  
0.000000] clocksource: hyperv_clocksource_msr: mask: 0xffffffffffffff max_cycles: 0x24e6a1710, max_idle_ns: 440795202120 ns  
0.000000] tsc: Detected 3293.728 MHz processor  
0.000114] e820: update [mem 0x00000000-0x0000ffff] usable ==> reserved  
0.000121] e820: remove [mem 0x00000000-0x0000ffff] usable  
0.000128] last_pfn = 0x1f4e00 max_arch_pfn = 0x400000000  
0.000193] MTRR map: 5 entries (4 fixed + 1 variable; max 20), built from 8 variable MTRRs  
0.000200] x86/PAT: Configuration [0-7]: WB WC UC- UC WB WP UC- WT  
0.000310] last_pfn = 0xf8000 max_arch_pfn = 0x400000000  
0.000333] Using 6B pages for direct mapping  
0.000567] RAMDISK: [mem 0x04be4000-0x04e7ffff]  
0.000572] ACPI: Early table checksum verification disabled  
0.000577] ACPI: RSDP 0x000000000000E000 000024 (v02 VIRTUAL)  
0.000585] ACPI: XSDT 0x0000000000100000 000044 (v01 VIRTUAL MICROSOFT 00000001 MSFT 00000001)  
0.000594] ACPI: FACP 0x0000000000101000 000114 (v06 VIRTUAL MICROSOFT 00000001 MSFT 00000001)  
0.000604] ACPI: DSDT 0x0000000000101100 01E11C (v02 MSFTVM DSDT01 00000001 INTL 20230626)  
0.000610] ACPI: FACS 0x0000000000101114 000040  
0.000616] ACPI: OEM0 0x0000000000101154 000064 (v01 VIRTUAL MICROSOFT 00000001 MSFT 00000001)  
0.000622] ACPI: SRAT 0x000000000011F2D4 0003B0 (v02 VIRTUAL MICROSOFT 00000001 MSFT 00000001)  
0.000628] ACPI: APIC 0x000000000011F684 0000C8 (v04 VIRTUAL MICROSOFT 00000001 MSFT 00000001)  
0.000634] ACPI: Reserving FACP table memory at [mem 0x101000-0x101113]  
0.000636] ACPI: Reserving DSDT table memory at [mem 0x101108-0x11f2d3]  
0.000638] ACPI: Reserving FACS table memory at [mem 0x101114-0x101153]  
0.000640] ACPI: Reserving OEM0 table memory at [mem 0x101154-0x1011b7]  
0.000642] ACPI: Reserving SRAT table memory at [mem 0x11f2d4-0x11f683]  
0.000644] ACPI: Reserving APIC table memory at [mem 0x11f684-0x11f74b]  
0.000699] SRAT: PXM 0 -> APIC 0x00 -> Node 0  
0.000782] SRAT: PXM 0 -> APIC 0x01 -> Node 0  
0.000784] SRAT: PXM 0 -> APIC 0x02 -> Node 0  
0.000785] SRAT: PXM 0 -> APIC 0x03 -> Node 0  
0.000787] SRAT: PXM 0 -> APIC 0x04 -> Node 0  
0.000788] SRAT: PXM 0 -> APIC 0x05 -> Node 0  
0.000710] SRAT: PXM 0 -> APIC 0x06 -> Node 0
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agilinggara@RagilAnggara: ~/ +
0.000711] SRAT: PXM 0 -> APIC 0x07 -> Node 0
0.000713] SRAT: PXM 0 -> APIC 0x08 -> Node 0
0.000714] SRAT: PXM 0 -> APIC 0x09 -> Node 0
0.000716] SRAT: PXM 0 -> APIC 0x0a -> Node 0
0.000717] SRAT: PXM 0 -> APIC 0x0b -> Node 0
0.000719] SRAT: PXM 0 -> APIC 0x0c -> Node 0
0.000720] SRAT: PXM 0 -> APIC 0x0d -> Node 0
0.000722] SRAT: PXM 0 -> APIC 0x0e -> Node 0
0.000723] SRAT: PXM 0 -> APIC 0x0f -> Node 0
0.000727] ACPI: SRAT: Node 0 PXM 0 [mem 0x00000000-0xf7ffffff] hotplug
0.000732] ACPI: SRAT: Node 0 PXM 0 [mem 0x10000000-0x1f4dffff] hotplug
0.000735] ACPI: SRAT: Node 0 PXM 0 [mem 0x1f4e0000-0x9fddffff] hotplug
0.000737] ACPI: SRAT: Node 0 PXM 0 [mem 0x100000000-0xfcffffff] hotplug
0.000740] ACPI: SRAT: Node 0 PXM 0 [mem 0x100000000-0x1ffffffff] hotplug
0.000743] ACPI: SRAT: Node 0 PXM 0 [mem 0x200000000-0x3ffffffff] hotplug
0.000746] ACPI: SRAT: Node 0 PXM 0 [mem 0x400000000-0x7ffffffff] hotplug
0.000748] ACPI: SRAT: Node 0 PXM 0 [mem 0x800000000-0xffffffff] hotplug
0.000751] ACPI: SRAT: Node 0 PXM 0 [mem 0x1000000000-0x1ffffffff] hotplug
0.000754] ACPI: SRAT: Node 0 PXM 0 [mem 0x2000000000-0x3ffffffff] hotplug
0.000757] ACPI: SRAT: Node 0 PXM 0 [mem 0x4000000000-0x7ffffffff] hotplug
0.000760] ACPI: SRAT: Node 0 PXM 0 [mem 0x8000000000-0xffffffff] hotplug
0.000762] ACPI: SRAT: Node 0 PXM 0 [mem 0x10000000000-0x1ffffffff] hotplug
0.000765] ACPI: SRAT: Node 0 PXM 0 [mem 0x20000000000-0x3ffffffff] hotplug
0.000768] ACPI: SRAT: Node 0 PXM 0 [mem 0x40000000000-0x7ffffffff] hotplug
0.000771] ACPI: SRAT: Node 0 PXM 0 [mem 0x80000000000-0xffffffff] hotplug
0.000777] NUMA: Node 0 [mem 0x00000000-0xf7ffffff] + [mem 0x10000000-0x1f4dffff] -> [mem 0x00000000-0x1f4dffff]
0.000786] NODE_DATA(0) allocated [mem 0x1f4dde000-0x1f4dffff]
0.000935] Zone ranges:
0.000936] DMA32 [mem 0x000000000000-0x000000000000-0x000000000000]
0.000939] Normal [mem 0x0000000010000000-0x000000001f4dffff]
0.000941] Device empty
0.000943] Movable zone start for each node
0.000945] Early memory node ranges
0.000946] node 0: [mem 0x000000000000-0x000000000000-0x000000000000]
0.000948] node 0: [mem 0x000000000000-0x000000000000-0x000000000000]
0.000949] node 0: [mem 0x0000000010000000-0x000000001f4dffff]
0.000951] Initmem setup node 0 [mem 0x000000000000-0x000000001f4dffff]
0.000992] On node 0, zone DMA32: 1 pages in unavailable ranges
0.045548] On node 0, zone DMA32: 352 pages in unavailable ranges
0.047336] On node 0, zone Normal: 1268 pages in unavailable ranges
0.047439] ACPI: PM-Timer IO Port: 0x008
0.047451] ACPI: LAPIC_NMI (acpi_id[0x01] dfl dfl lint[0x1])
0.048575] IOAPIC[0]: apic_id 16, version 17, address 0xfec00000, GSI 0-23
0.048585] ACPI: INT_SRC_OVR (bus 0 bus_irq 9 global_irq 9 high level)
0.048594] ACPI: Using ACPI (MADT) for SMP configuration information
0.048597] smpboot: Allowing 16 CPUs, 0 hotplug CPUs
0.048616] PM: hibernation: Registered nosave memory: [mem 0x00000000-0x000000ff]
0.048620] PM: hibernation: Registered nosave memory: [mem 0x00000000-0x000000ff]
0.048622] PM: hibernation: Registered nosave memory: [mem 0x00000000-0x000000ff]
0.048623] PM: hibernation: Registered nosave memory: [mem 0x00001000-0x0000ffff]
0.048625] PM: hibernation: Registered nosave memory: [mem 0x00100000-0x001fffff]
0.048628] PM: hibernation: Registered nosave memory: [mem 0xf8000000-0xffffffff]
0.048631] [mem 0xf8000000-0xffffffff] available for PCI devices

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agilinggara@RagilAnggara: ~/ +
0.048631] [mem 0xf8000000-0xffffffff] available for PCI devices
0.048634] Booting paravirtualized kernel on Hyper-V
0.048637] clocksource: refined-jiffies: mask: 0xffffffff max_cycles: 0xffffffff, max_idle_ns: 7645519680211568 ns
0.048713] setup_percpu: NR_CPUS=8192 nr_cpumask_bits=16 nr_cpu_ids=16 nr_node_ids=1
0.052391] percpu: Embedded 63 pages/cpu s221184 r8192 d28672 u262144
0.052402] percpu-alloc: s221184 r8192 d28672 u262144 alloc=1*2097152
0.052407] percpu-alloc: [0] 00 01 02 03 04 05 06 07 [0] 08 09 10 11 12 13 14 15
0.052453] Hyper-V: PV spinlocks enabled
0.052456] PV qspinlock hash table entries: 256 (order: 0, 4096 bytes, linear)
0.052461] Kernel command line: initrd=initrd.img WSL_ROOT_INIT=1 panic=1 nr_cpus=16 hv_utils.timesync_implicit=1 console=hvc0 debug ptty.legacy_count=0 WSL_ENABLE_CRASH_DUMP=1
0.052695] Unknown kernel command line parameters "WSL_ROOT_INIT=1 WSL_ENABLE_CRASH_DUMP=1", will be passed to user space.
0.052723] random: crng init done
0.059584] Dentry cache hash table entries: 1048576 (order: 11, 8388608 bytes, linear)
0.062940] Inode-cache hash table entries: 524288 (order: 10, 4194304 bytes, linear)
0.063233] fallback order for Node 0: 0
0.063240] Built 1 zonelists, mobility grouping on. Total pages: 1986248
0.063247] Policy zone: Normal
0.063261] mem auto-init: stack:off, heap alloc:on, heap free:off
0.063270] software IO TLB: area num 16
0.135821] Memory: 4107404K/8073852K available (20480K kernel code, 3457K rwdma, 13496K rodata, 4492K init, 6208K bss, 262424K reserved, 0K cma-reserved)
0.136185] SLUB: Hwalign=64, Order=0-3, MinObjects=0, CPUs=16, Nodes=1
0.136266] ftrace: allocating 55808 entries in 218 pages
0.153434] ftrace: allocated 218 pages with 5 groups
0.155342] Dynamic Preempt: none
0.155515] rcu: Preemptible hierarchical RCU implementation.
0.155517] rcu: RCU restricting CPUs from NR_CPUS=8192 to nr_cpu_ids=16.
0.155519] Trampoline variant of Tasks RCU enabled.
0.155520] Rude variant of Tasks RCU enabled.
0.155521] Tracing variant of Tasks RCU enabled.
0.155522] rcu: RCU calculated value of scheduler-enlistment delay is 25 jiffies.
0.155523] rcu: Adjusting geometry for rcu_fanout_leaf=16, nr_cpu_ids=16
0.167327] Using NULL legacy PIC
0.167330] NR_IRQS: 524544, nr_irqs: 552, preallocated irqs: 0
0.167432] rcu: rcu_init: Setting srcu_struct sizes based on contention.
0.168735] Console: colour dummy device 80x25
0.168782] ACPI: Core revision 20230628
0.169170] Failed to register legacy timer interrupt
0.169173] APIC: Switch to symmetric I/O mode setup
0.169189] APIC: Switched APIC routing to: physical flat
0.169194] Hyper-V: Host Build 10.0.26100.6725-6-0
0.169197] Hyper-V: enabling crash_hexco_post_notifiers
0.169304] Hyper-V: Using IPI hypercalls
0.169308] APIC: send_IPI() replaced with hv_send_ipi()
0.169315] APIC: send_IPI_mask() replaced with hv_send_ipi_mask()
0.169318] APIC: send_IPI_mask_allbutself() replaced with hv_send_ipi_mask_allbutself()
0.169320] APIC: send_IPI_allbutself() replaced with hv_send_ipi_allbutself()
0.169323] APIC: send_IPI_all() replaced with hv_send_ipi_all()
0.169325] APIC: send_IPI_self() replaced with hv_send_ipi_self()
0.169327] Hyper-V: Using enlightened APIC (xapic mode)
0.169339] APIC: eoi() replaced with hv_apic_eoi_write()
0.169375] APIC: read() replaced with hv_apic_read()
0.169404] APIC: write() replaced with hv_apic_write()
0.169406] APIC: icr_write() replaced with hv_apic_icr_write()

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agilinggara@RagilAnggara: ~/  X + v
0.169486] APIC: icr_write() replaced with hv_apic_icr_write()
0.169489] APIC: icr_read() replaced with hv_apic_icr_read()
0.169621] clocksource: tsc-early: mask: 0xffffffffffffff max_cycles: 0x2f7a2765acc, max_idle_ns: 440795282716 ns
0.169636] Calibrating delay loop (Skipped), value calculated using timer frequency.. 6587.45 BogoMIPS (lpj=13174912)
0.169765] x86/cpu: User Mode Instruction Prevention (UMIP) activated
0.169795] Last level iTLB entries: 4KB 512, 2MB 512, 4MB 256
0.169796] Last level dTLB entries: 4KB 2048, 2MB 2048, 4MB 1024, 1GB 0
0.169801] Spectre V1 : Mitigation: usercopy/swapgs barriers and __user pointer sanitization
0.169804] Spectre V2 : Mitigation: Retpolines
0.169805] Spectre V2 : Spectre v2 / SpectreRSB mitigation: Filling RSB on context switch
0.169806] Spectre V2 : SpectreRSB : Filling RSB on VMEXIT
0.169806] Spectre V2 : Enabling Restricted Speculation for firmware calls
0.169808] Spectre V2 : mitigation: Enabling conditional Indirect Branch Prediction Barrier
0.169809] Spectre V2 : User space: Mitigation: STIBP always-on protection
0.169811] Speculative Store Bypass: Mitigation: Speculative Store Bypass disabled via prctl
0.169812] Speculative Return Stack Overflow: IBPB-extending microcode not applied!
0.169813] Speculative Return Stack Overflow: WARNING: See https://kernel.org/doc/html/latest/admin-guide/hw-vuln/srso.html for mitigation options.
0.169814] Speculative Return Stack Overflow: Vulnerable: Safe RET, no microcode
0.169839] x86/fpu: Supporting XSAVE feature 0x001: 'x87 floating point registers'
0.169841] x86/fpu: Supporting XSAVE feature 0x002: 'SSE registers'
0.169842] x86/fpu: Supporting XSAVE feature 0x004: 'AVX registers'
0.169843] x86/fpu: Supporting XSAVE feature 0x000: 'Control-flow User registers'
0.169844] x86/fpu: xstate_offset[2]: 576, xstate_sizes[2]: 256
0.169846] x86/fpu: xstate_offset[11]: 832, xstate_sizes[11]: 16
0.169846] x86/fpu: Enabled xstate features 0x007, context size is 848 bytes, using 'compacted' format.
0.173626] Freeing SMP alternatives memory: 44K
0.173626] pid_max: default: 32768 minimum: 301
0.173626] LSM: initializing lsm=capability, landlock, yama, safesetid, selinux, integrity
0.173626] landlock: Up and running.
0.173626] Yama: becoming mindful.
0.173626] SELinux: Initializing.
0.173626] Mount-cache hash table entries: 16384 (order: 5, 131072 bytes, linear)
0.173626] Mountpoint-cache hash table entries: 16384 (order: 5, 131072 bytes, linear)
0.173626] smpboot: CPU0: AMD Ryzen 9 5900HX with Radeon Graphics (family: 0x19, model: 0x50, stepping: 0x0)
0.173626] RCU Tasks: Setting shift to 4 and lim to 1 rcu_task_cb_adjust=1 rcu_task_cpu_ids=16.
0.173626] RCU Tasks Rude: Setting shift to 4 and lim to 1 rcu_task_cb_adjust=1 rcu_task_cpu_ids=16.
0.173626] RCU Tasks Trace: Setting shift to 4 and lim to 1 rcu_task_cb_adjust=1 rcu_task_cpu_ids=16.
0.173626] Performance Events: Fam17h+ core perfctr, AMD PMU driver.
0.173626] ... version: 0
0.173626] ... bit width: 48
0.173626] ... generic registers: 6
0.173626] ... value mask: 0000ffffffffffff
0.173626] ... max period: 00007fffffffffff
0.173626] ... fixed-purpose events: 0
0.173626] ... event mask: 000000000000003f
0.173626] signal: max sigframe size: 1776
0.173626] rcu: Hierarchical SRCU implementation.
0.173626] rcu: Max phase no-delay instances is 1000.
0.179435] smp: Bringing up secondary CPUs ...
0.179855] smpboot: x86: Booting SMP configuration:
0.179858] .... node #0 CPUs: #2 #4 #6 #8 #10 #12 #14 #1 #3 #5 #7 #9 #11 #13 #15
0.206641] Spectre V2 : Update user space SMT mitigation: STIBP always-on
0.210140] smp: Brought up 1 node, 16 CPUs

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agilinggara@RagilAnggara: ~/  X + v
0.210148] smpboot: Max logical packages: 1
0.210151] smpboot: Total of 16 processors activated (105399.29 BogoMIPS)
0.238961] node 0 deferred pages initialised in 12ms
0.241952] devtmpfs: initialized
0.241952] x86/mm: Memory block size: 128MB
0.245955] clocksource: jiffies: mask: 0xffffffff max_cycles: 0xffffffff, max_idle_ns: 7645041785100000 ns
0.245982] futex hash table entries: 4096 (order: 6, 262144 bytes, linear)
0.246146] pinctrl core: initialized pinctrl subsystem
0.246407] NET: Registered PF_INETLL/PP_ROUTE protocol family
0.246559] DMA: preallocated 1024 KiB GFP_KERNEL pool for atomic allocations
0.246478] DMA: preallocated 1024 KiB GFP_KERNEL pool for atomic allocations
0.250533] audit: initializing netlink subsys (disabled)
0.250575] audit: type=2000 audit(1760427505.080:1): state=initialized audit_enabled=0 res=1
0.250575] thermal_sys: Registered thermal governor 'step_wise'
0.250575] thermal_sys: Registered thermal governor 'user_space'
0.250575] cpuidle: using governor ladder
0.250575] cpuidle: using governor menu
0.250575] acpihp: ACPI Hot Plug PCI Controller Driver version: 0.5
0.250575] dca service started, version 1.12.1
0.250575] PCI: Fatal: No config space access function found
0.254692] kprobes: kprobe jump-optimization is enabled. All kprobes are optimized if possible.
0.254730] HugeTLB: registered 1.00 GiB page size, pre-allocated 0 pages
0.254730] HugeTLB: 16380 KiB vmemmap can be freed for a 1.00 GiB page
0.254730] HugeTLB: registered 2.00 MiB page size, pre-allocated 0 pages
0.254730] HugeTLB: 28 KiB vmemmap can be freed for a 2.00 MiB page
0.255060] cryptd: max_cpu_qlen set to 1000
0.258057] ACPI: Added _OSI(Module Device)
0.258062] ACPI: Added _OSI(Processor Device)
0.258065] ACPI: Added _OSI(3.0 _SCP Extensions)
0.258068] ACPI: Added _OSI(Processor Aggregator Device)
0.290612] ACPI: 1 ACPI AML tables successfully acquired and loaded
0.292343] ACPI: _OSC evaluation for CPUs failed, trying _PDC
0.292943] ACPI: Interpreter enabled
0.292959] ACPI: PM: (supports S8 S5)
0.292962] ACPI: Using IOAPIC for interrupt routing
0.292999] PCI: Using host bridge windows from ACPI; if necessary, use "pci=noocrs" and report a bug
0.293081] PCI: Using E820 reservations for host bridge windows
0.293546] ACPI: Enabled 2 GPEs in block 00 to 0F
0.299448] iommu: Default domain type: Translated
0.299448] iommu: DMA domain TLB invalidation policy: lazy mode
0.299448] SCSI subsystem initialized
0.299448] libata version 3.00 loaded.
0.299448] pps_core: LinuxPPS API ver. 1 registered
0.299448] pps_core: Software ver. 5.3.6 - Copyright 2005-2007 Rodolfo Giometti <giometti@linux.it>
0.299448] PTP clock support registered
0.299448] EDAC MC: Ver: 3.0.0
0.316597] hv_vmbus: Vmbus version:5.3
0.316597] hv_vmbus: Unknown GUID: 6e382d18-3336-4f4b-acc4-2b7703d4d4f4a
0.318782] hv_vmbus: Unknown GUID: dde9cb08-5060-4436-9448-ea1254a5d177
0.318789] hv_vmbus: Unknown GUID: 6e382d18-3336-4f4b-acc4-2b7703d4d4f4a
0.318770] NetLabel: Initializing
0.318770] NetLabel: domain hash size = 128
0.318770] NetLabel: protocols = UNLABELED CIPSOv4 CALIPSO

```

```
agilinggara@RagilAnggara: ~/ X + v
0.318770 NetLabel: unlabeled traffic allowed by default
0.318770 PCI: Using ACPI for IRQ routing
0.318770 PCI: System does not support PCI
0.318770 vgaarb: loaded
0.328434 clocksource: Switched to clocksource tsc-early
0.334876 VFS: Disk quotas dquot_6.6.0
0.334117 VFS: Dquot-cache hash table entries: 512 (order 0, 4096 bytes)
0.334216 FS-Cache: Loaded
0.334334 pnp: PnP ACPI init
0.334933 pnp: PnP ACPI: found 1 devices
0.358773 clocksource: acpi_pm: mask: 0xffffffff max_cycles: 0xffffffff, max_idle_ns: 2085701024 ns
0.358927 NET: Registered PF_INET protocol family
0.351559 IP ident hash table entries: 131072 (order: 8, 1048576 bytes, linear)
0.354589 tcp_listen_portaddr_hash hash table entries: 4096 (order: 4, 65536 bytes, linear)
0.354771 Table-perturb hash table entries: 65536 (order: 6, 262144 bytes, linear)
0.355186 TCP established hash table entries: 65536 (order: 7, 524288 bytes, linear)
0.355356 TCP bind hash table entries: 65536 (order: 9, 2097152 bytes, linear)
0.356552 TCP: Hash tables configured (established 65536 bind 65536)
0.356817 UDP hash table entries: 4096 (order: 5, 131072 bytes, linear)
0.356986 UDP-Lite hash table entries: 4096 (order: 5, 131072 bytes, linear)
0.357070 NET: Registered PF_UNIX/PF_LOCAL protocol family
0.357084 NET: Registered PF_XDP protocol family
0.357092 PCI: CLS 0 bytes, default 64
0.357179 PCI-DMA: Using software bounce buffering for IO (SWIOTLB)
0.357182 software IO TLB: mapped [mem 0x00000000f4000000-0x00000000f8000000] (64MB)
0.357438 Trying to unpack rootfs image as initramfs...
0.363030 Freeing initrd memory: 2672K
0.364246 clocksource: tsc: mask: 0xffffffffffffffff max_cycles: 0x2f7a2765acc, max_idle_ns: 440795282716 ns
0.377224 clocksource: Switched to clocksource tsc
0.379134 Initialise system trusted keyrings
0.379147 Key type blacklist registered
0.379382 workingset: timestamp_bits=36 max_order=21 bucket_order=0
0.380211 squashfs: version 4.0 (2009/01/31) Phillip Lougher
0.380252 fuse: init (API version 7.39)
0.380679 SGI XFS with ACLs, security attributes, realtime, verbose warnings, quota, no debug enabled
0.381878 9p: Installing v9fs 9p2000 file system support
0.402251 Key type asymmetric registered
0.402256 Asymmetric key parser 'x509' registered
0.402296 Block layer SCSI generic (bsg) driver version 0.4 loaded (major 245)
0.402579 io scheduler mq-deadline registered
0.402586 io scheduler kyber registered
0.409965 hv_vmbus: registering driver hv_pci
0.410797 hv_pci c4b741f5-5582-4c98-8f8b-2e082933c396: PCI VMBus probing: Using version 0x10004
0.412453 hv_pci c4b741f5-5582-4c98-8f8b-2e082933c396: PCI host bridge to bus 5582:00
0.412458 pci_bus 5582:00: root bus resource [mem 0x9ffe0000-0x9ffe02fff window]
0.412463 pci_bus 5582:00: No busn resource found for root bus, will use [bus 00-ff]
0.413830 pci 5582:00:00.0: [1af4:1043] type 00 class 0x010000
0.414082 pci 5582:00:00.0: reg 0x10: [mem 0x9ffe0000-0x9ffe00fff 64bit]
0.414511 pci 5582:00:00.0: reg 0x18: [mem 0x9ffe01000-0x9ffe01fff 64bit]
0.416139 pci 5582:00:00.0: reg 0x20: [mem 0x9ffe02000-0x9ffe02fff 64bit]
0.420958 pci_bus 5582:00: busn_res: [bus 00-ff] end is updated to 00
0.420971 pci 5582:00:00.0: BAR 0: assigned [mem 0x9ffe0000-0x9ffe00fff 64bit]
0.421580 pci 5582:00:00.0: BAR 2: assigned [mem 0x9ffe01000-0x9ffe01fff 64bit]
```

```
agilinggara@RagilAnggara: ~/ X + v
0.422073 pci 5582:00:00.0: BAR 4: assigned [mem 0x9ffe02000-0x9ffe02fff 64bit]
0.423583 hv_pci bb8269c7-de0e-46b7-b243-b65f865f444: PCI VMBus probing: Using version 0x10004
0.424835 hv_pci bb8269c7-de0e-46b7-b243-b65f865f444: PCI host bridge to bus de0e:00
0.424841 pci_bus de0e:00: No busn resource found for root bus, will use [bus 00-ff]
0.425704 pci de0e:00:00.0: [1414:008e] type 00 class 0x030200
0.433381 pci_bus de0e:00: busn_res: [bus 00-ff] end is updated to 00
0.434072 hv_pci a56290c-a836-4826-910c-4d28421db135: PCI VMBus probing: Using version 0x10004
0.436861 hv_pci a56290c-a836-4826-910c-4d28421db135: PCI host bridge to bus a836:00
0.436866 pci_bus a836:00: No busn resource found for root bus, will use [bus 00-ff]
0.437682 pci a836:00:00.0: [1414:008e] type 00 class 0x030200
0.449070 pci_bus a836:00: busn_res: [bus 00-ff] end is updated to 00
0.452383 iohd: Intel(R) QuickData Technology Driver 5.00
0.452760 virtio-pci 5582:00:00.0: enabling device (0000 -> 0002)
0.459482 Serial: 8250/16550 driver, 8 ports, IRQ sharing enabled
0.510609 Linux agpgart interface v0.103
0.531134 AMD-Vi: AMD IOMMUv2 functionality not available on this system - This is not a bug.
0.531150 ACPI: bus type drm_connector registered
0.534685 printk: console [hvc0] enabled
0.547098 brd: module loaded
0.556580 loop: module loaded
0.557336 Loading iSCSI transport class v2.0-870.
0.558585 rdac: device handler registered
0.559540 Microchip SmartPQI Driver (v2.1.24-046)
0.560603 VMware PVSCSI driver - version 1.0.7.0-k
0.561465 hv_vmbus: registering driver hv_storvsc
0.565069 PPP generic driver version 2.4.2
0.566273 scsi host0: storvsc-host-1
0.565935 VMware vmxnet3 virtual NIC driver - version 1.7.0.0-k-NAPI
0.566960 hv_vmbus: registering driver hv_netvsc
0.567517 Fusion MPT base driver 3.04.20
0.567908 Copyright (c) 1999-2008 LSI Corporation
0.568487 Fusion MPT SPI Host driver 3.04.20
0.569069 Fusion MPT SAS Host driver 3.04.20
0.569670 Fusion MPT misc device (ioctl) driver 3.04.20
0.570520 mptctl: Registered with Fusion MPT base driver
0.571200 mptctl: /dev/mptctl @ (major,minor=10,220)
0.571833 i8042: PNP: No PS/2 controller found.
0.572376 hv_vmbus: registering driver hyperv_keyboard
0.572965 rtc_cmos 00:00: RTC can wake from S4
0.577214 rtc_cmos 00:00: registered as rtc0
0.578352 rtc_cmos 00:00: setting system clock to 2025-10-14T07:38:26 UTC (1760427506)
0.579357 rtc_cmos 00:00: alarms up to one month, 114 bytes nvram
0.580118 device-mapper: core: CONFIG_IMA_DISABLE_HTABLE is disabled. Duplicate IMA measurements will not be recorded in the IMA log.
0.581436 device-mapper: uevent: version 1.0.3
0.582217 device-mapper: ioctl: 4.48.0-ioctl (2023-03-01) initialised: dm-devel@redhat.com
0.584213 hv_utils: Registering HyperV Utility Driver
0.585364 hv_vmbus: registering driver hv_utils
0.586400 hv_vmbus: registering driver hv_balloon
0.587552 hv_vmbus: registering driver dxgkrnl
0.587575 hv_utils: TimeSync IC version 4.0
0.588009 hv_balloon: Using Dynamic Memory protocol version 2.0
0.591148 Free page reporting enabled
0.591698 hv_balloon: Cold memory discard hint enabled with order 9
```

```
aglinggara@RagilAnggara: ~/ X + v
0.598988] drop_monitor: Initializing network drop monitor service
0.630479] NET: Registered PF_INET6 protocol family
0.633532] Segment Routing with IPv6
0.634226] In-situ OAM (IOAM) with IPv6
0.634872] NET: Registered PF_PACKET protocol family
0.636071] 9pnet: Installing 9P2000 support
0.637881] NET: Registered PF_VSOCK protocol family
0.638817] hv_vmbus: registering driver hv_sock
0.644555] IPI shorthand broadcast: enabled
0.645529] AVX2 version of gcm_enc/dec engaged.
0.647996] AES CTR mode by8 optimization enabled
0.657545] sched_clock: Marking stable (656809167, 1131992)-->(742173656, -85832497)
0.659727] registered taskstats version 1
0.660695] Loading compiled-in X.509 certificates
0.673928] ima: No TPM chip found, activating TPM-bypass!
0.675996] ima: Allocated hash algorithm: sha256
0.678816] ima: No architecture policies found
0.680479] RAS: Correctable Errors collector initialized.
0.841997] clk: Disabling unused clocks
0.845743] Freeing unused decrypted memory: 2028K
0.848593] Freeing unused kernel image (initmem) memory: 4492K
0.849938] Write protecting the kernel read-only data: 34816k
0.852529] Freeing unused kernel image (rodata/data gap) memory: 840K
0.853005] Run /init as init process
0.854522] with arguments:
0.855094] /init
0.855455] with environment:
0.856031] HOME=/
0.856391] TERM=linux
0.856691] WSL_ROOT_INIT=1
0.857097] WSL_ENABLE_CRASH_DUMP=1
0.868199] scsi 0:0:0:0: Direct-Access Msft Virtual Disk 1.0 PQ: 0 ANSI: 5
0.876752] scsi 0:0:0:1: Direct-Access Msft Virtual Disk 1.0 PQ: 0 ANSI: 5
0.902192] sd 0:0:0:0: Attached scsi generic sg0 type 0
0.902666] sd 0:0:0:0: [sda] 795536 512-byte logical blocks: (407 MB/388 MiB)
0.903675] sd 0:0:0:1: Attached scsi generic sg1 type 0
0.904204] sd 0:0:0:1: [sdb] 381016 512-byte logical blocks: (195 MB/186 MiB)
0.904420] sd 0:0:0:1: [sdb] Write Protect is on
0.904498] sd 0:0:0:0: [sda] Write Protect is on
0.904585] sd 0:0:0:0: [sda] Mode Sense: 0f 00 80 00
0.905072] sd 0:0:0:0: [sda] Write cache: disabled, read cache: enabled, doesn't support DPO or FUA
0.906435] sd 0:0:0:1: [sdb] Mode Sense: 0f 00 80 00
0.906569] scsi 0:0:0:2: Direct-Access Msft Virtual Disk 1.0 PQ: 0 ANSI: 5
0.911985] sd 0:0:0:1: [sdb] Attached SCSI disk
0.912747] sd 0:0:0:1: [sdb] Write cache: disabled, read cache: enabled, doesn't support DPO or FUA
0.918982] sd 0:0:0:1: [sdb] Attached SCSI disk
0.922708] sd 0:0:0:2: Attached scsi generic sg2 type 0
0.923041] sd 0:0:0:2: [sdc] 4194312 512-byte logical blocks: (2.15 GB/2.00 GiB)
0.924786] sd 0:0:0:2: [sdc] 4096-byte physical blocks
0.925750] sd 0:0:0:2: [sdc] Write Protect is off
0.926633] sd 0:0:0:2: [sdc] Mode Sense: 0f 00 80 00
0.927729] sd 0:0:0:2: [sdc] Write cache: enabled, read cache: enabled, doesn't support DPO or FUA
0.932698] sd 0:0:0:2: [sdc] Attached SCSI disk
```

```
aglinggara@RagilAnggara: ~/ X + v
0.971334] EXT4-fs (sda): mounted filesystem 00000000-0000-0000-0000-000000000000 ro without journal. Quota mode: none.
0.972001] EXT4-fs (sdb): mounted filesystem 00000000-0000-0000-0000-000000000000 ro without journal. Quota mode: none.
1.018395] tun: Universal TUN/TAP device driver, 1.6
1.071837] Adding 2097152k swap on /dev/sdc. Priority: -2 extents:1 across:2097152k
1.073049] bridge: filtering via arp/ip/ip6tables is no longer available by default. Update your scripts to load br_netfilter if you need this.
1.078618] Bridge firewalling registered
1.526590] hv_pci bb2572d8-13ed-4be8-8714-15aed3fb0e4b: PCI VMBus probing: Using version 0x10004
1.538233] hv_pci bb2572d8-13ed-4be8-8714-15aed3fb0e4b: PCI host bridge to bus 13ed:00
1.531633] pci bus 13ed:00:00: root bus resource [mem 0xc0000000-0xc0000fff 64bit]
1.533176] pci bus 13ed:00: No bus resource found for root bus, will use [bus 00-ff]
1.537007] pci 13ed:00:00:0: [laf4:105a] type 00 class 0x088000
1.540288] pci 13ed:00:00:0: reg 0x10: [mem 0xc0000000-0xc0000fff 64bit]
1.542534] pci 13ed:00:00:0: reg 0x18: [mem 0xc0000100-0xc00001fff 64bit]
1.544684] pci 13ed:00:00:0: reg 0x20: [mem 0xc0000000-0xc0000fff 64bit]
1.552929] pci bus 13ed:00: busn_res: [bus 00-ff] end is updated to 00
1.551263] pci 13ed:00:00:0: BAR 4: assigned [mem 0xc0000000-0xc0000fff 64bit]
1.556104] pci 13ed:00:00:0: BAR 0: assigned [mem 0xc0000000-0xc0000fff 64bit]
1.558030] pci 13ed:00:00:0: BAR 2: assigned [mem 0xc0000100-0xc00001fff 64bit]
1.560164] virtio-pci 13ed:00:00:0: enabling device (0000 -> 0002)
1.597590] virtiofs virtio1: Cache len: 0x200000000 0 0xc0000000
1.800207] scsi 0:0:0:3: Direct-Access Msft Virtual Disk 1.0 PQ: 0 ANSI: 5
1.814046] sd 0:0:0:3: Attached scsi generic sg3 type 0
1.815244] sd 0:0:0:3: [sdd] 2147483648 512-byte logical blocks: (1.10 TB/1.00 TiB)
1.816817] sd 0:0:0:3: [sdd] 4096-byte physical blocks
1.817667] sd 0:0:0:3: [sdd] Write Protect is off
1.818294] sd 0:0:0:3: [sdd] Mode Sense: 0f 00 80 00
1.819233] sd 0:0:0:3: [sdd] Write cache: enabled, read cache: enabled, doesn't support DPO or FUA
1.823985] sd 0:0:0:3: [sdd] Attached SCSI disk
1.857972] EXT4-fs (sdd): mounted filesystem 77bd2e57-3ae6-49e5-99a7-b08c01b07a2f r/w with ordered data mode. Quota mode: none.
2.070803] pulseaudio[11]: memfd_create() called without MFD_EXEC or MFD_NOEXEC_SEAL set
3.066042] misc dxgk: dxgk: dxgkio_is_feature_enabled: Ioctl failed: -22
3.075683] misc dxgk: dxgk: dxgkio_query_adapter_info: Ioctl failed: -22
3.077799] misc dxgk: dxgk: dxgkio_query_adapter_info: Ioctl failed: -22
3.079676] misc dxgk: dxgk: dxgkio_query_adapter_info: Ioctl failed: -22
3.081666] misc dxgk: dxgk: dxgkio_query_adapter_info: Ioctl failed: -2
3.089680] misc dxgk: dxgk: dxgkio_query_adapter_info: Ioctl failed: -22
3.091421] misc dxgk: dxgk: dxgkio_query_adapter_info: Ioctl failed: -22
3.092730] misc dxgk: dxgk: dxgkio_query_adapter_info: Ioctl failed: -22
3.094347] misc dxgk: dxgk: dxgkio_query_adapter_info: Ioctl failed: -2
3.196261] Failed to connect to bus: No such file or directory
3.426510] systemd-journald[44]: Collecting audit messages is disabled.
3.596486] systemd-journald[44]: Received client request to flush runtime journal.
3.599217] systemd-journald[44]: File /var/log/journal/5e0b2178f2a64652ac34e2eddec893ce/system.journal corrupted or uncleanly shut down, renaming and replacing.
3.903505] ACPI: Battery Slot [BAT1] (battery present)
3.910130] ACPI: AC: AC Adapter [AC1] (off-line)
4.275554] kvm_amd: TSC scaling supported
4.275565] kvm_amd: Nested Virtualization enabled
4.275568] kvm_amd: Nested Paging enabled
4.275574] kvm_amd: kvm_amd: Hyper-V enlightened NPT TLB flush enabled
4.275577] kvm_amd: kvm_amd: Hyper-V Direct TLB Flush enabled
4.275594] kvm_amd: Virtual VMX00D VMSAVE supported
7.327784] systemd-journald[44]: /var/log/journal/5e0b2178f2a64652ac34e2eddec893ce/user-1000.journal: Journal file uses a different sequence number ID, rotating.
11.549401] WSL (266) ERROR: CheckConnection: getaddrinfo() failed: -3
```

```
16.850006] WSL (266) ERROR: CheckConnection: getaddrinfo() failed: -3
40.696382] hv_balloon: Max. dynamic memory size: 7886 MB
601.169123] mini_init (250): drop_caches: 1
688.761995] TCP: eth0: Driver has suspect GRO implementation, TCP performance may be compromised.
```

Filter By Level

```

agilnggara@RagilAnggara:~$ dmesg -l err
[ 0.258575] PCI: Fatal: No config space access function found
[ 3.066842] misc dxg: dxgk: dxgkio_is_feature_enabled: Ioctl failed: -22
[ 3.075683] misc dxg: dxgk: dxgkio_query_adapter_info: Ioctl failed: -22
[ 3.077799] misc dxg: dxgk: dxgkio_query_adapter_info: Ioctl failed: -22
[ 3.079676] misc dxg: dxgk: dxgkio_query_adapter_info: Ioctl failed: -22
[ 3.081666] misc dxg: dxgk: dxgkio_query_adapter_info: Ioctl failed: -2
[ 3.089688] misc dxg: dxgk: dxgkio_query_adapter_info: Ioctl failed: -22
[ 3.091421] misc dxg: dxgk: dxgkio_query_adapter_info: Ioctl failed: -22
[ 3.092730] misc dxg: dxgk: dxgkio_query_adapter_info: Ioctl failed: -22
[ 3.094347] misc dxg: dxgk: dxgkio_query_adapter_info: Ioctl failed: -2
[ 11.549401] WSL (266) ERROR: CheckConnection: getaddrinfo() failed: -3
[ 16.556066] WSL (266) ERROR: CheckConnection: getaddrinfo() failed: -3

```

```

agilnggara@RagilAnggara:~/praktikum1/buggy$ dmesg -l warn
[ 0.169812] Speculative Return Stack Overflow: IBPB-extending microcode not applied!
[ 0.169813] Speculative Return Stack Overflow: WARNING: See https://kernel.org/doc/html/latest/admin-guide/hw-vuln/srso.html for mitigation options.
[ 0.318770] PCI: System does not support PCI
[ 0.580118] device-mapper: core: CONFIG_IMA_DISABLE_HTABLE is disabled. Duplicate IMA measurements will not be recorded in the IMA log.
[ 2.870893] pulseaudio[301]: memfd_create() called without MFD_EXEC or MFD_NOEXEC_SEAL set
[ 3.196261] Failed to connect to bus: No such file or directory
[ 3.599217] systemd-journald[44]: File /var/log/journal/5e0b2178f2a64652ac34e2eddec893ce/system.journal corrupted or uncleanly shut down, renaming and replacing.
[ 688.761995] TCP: eth0: Driver has suspect GRO implementation, TCP performance may be compromised.

```

10.5 Memory Leak Detection dengan Valgrind

Install valgrind

```

agilnggara@RagilAnggara:~$ sudo apt install valgrind
[sudo] password for agilnggara:
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Suggested packages:
  valgrind-dbg valgrind-mpi kcachegrind alleyoop valkyrie
The following NEW packages will be installed:
  valgrind
0 upgraded, 1 newly installed, 0 to remove and 28 not upgraded.
Need to get 14.9 MB of archives.
After this operation, 78.8 MB of additional disk space will be used.
Get:1 http://archive.ubuntu.com/ubuntu noble/main amd64 valgrind amd64 1:3.22.0-0ubuntu3 [14.9 MB]
Fetched 14.9 MB in 1min 5s (230 kB/s)
Selecting previously unselected package valgrind.
(Reading database ... 49803 files and directories currently installed.)
Preparing to unpack .../valgrind_1%3a3.22.0-0ubuntu3_amd64.deb ...
Unpacking valgrind (1:3.22.0-0ubuntu3) ...
Setting up valgrind (1:3.22.0-0ubuntu3) ...
Processing triggers for man-db (2.12.0-4build2) ...

```

```

agilnggara@RagilAnggara:~$ cat > memleak.c << 'EOF'
> #include <stdlib.h>
int main() {
    int *ptr = (int*)malloc(100 * sizeof(int));
    // Forgot to free(ptr)
return 0;
}
EOF
agilnggara@RagilAnggara:~$ gcc -g -o memleak memleak.c
agilnggara@RagilAnggara:~$ valgrind --leak-check=full ./memleak
==484== Memcheck, a memory error detector
==484== Copyright (C) 2002-2022, and GNU GPL'd, by Julian Seward et al.
==484== Using Valgrind-3.22.0 and LibVEX; rerun with -h for copyright info
==484== Command: ./memleak
==484==
==484== HEAD SUMMARY:
==484==    in use at exit: 400 bytes in 1 blocks
==484==    total heap usage: 1 allocs, 0 frees, 400 bytes allocated
==484==
==484== 400 bytes in 1 blocks are definitely lost in loss record 1 of 1
==484==    at 0x4846828: malloc (in /usr/libexec/valgrind/vgpreload_memcheck-amd64-linux.so)
==484==    by 0x10915E: main (memleak.c:3)
==484==
==484== LEAK SUMMARY:
==484==    definitely lost: 400 bytes in 1 blocks
==484==    indirectly lost: 0 bytes in 0 blocks
==484==    possibly lost: 0 bytes in 0 blocks
==484==    still reachable: 0 bytes in 0 blocks
==484==    suppressed: 0 bytes in 0 blocks
==484==
==484== For lists of detected and suppressed errors, rerun with: -s
==484== ERROR SUMMARY: 1 errors from 1 contexts (suppressed: 0 from 0)
agilnggara@RagilAnggara:~$

```

10.6 Performance Analysis

CPU profiling dengan perf

```
agilnggara@RagilAnggara:~$ sudo apt install linux-tools-common linux-tools-generic
[sudo] password for agilnggara:
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  hwdetect libnumal libtraceevent1 linux-tools-6.8.0-85 linux-tools-6.8.0-85-generic
The following NEW packages will be installed:
  hwdetect libnumal libtraceevent1 linux-tools-6.8.0-85 linux-tools-6.8.0-85-generic linux-tools-common linux-tools-generic
0 upgraded, 7 newly installed, 0 to remove and 28 not upgraded.
Need to get 6303 kB of archives.
After this operation, 28.9 MB of additional disk space will be used.
Get:1 http://archive.ubuntu.com/ubuntu noble/main amd64 libnumal amd64 2.0.18-1build1 [23.3 kB]
Get:2 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 libtraceevent1 amd64 1:1.8.2-1ubuntu2.1 [55.3 kB]
Get:3 http://archive.ubuntu.com/ubuntu noble/main amd64 hwdetect all 0.379-1 [29.1 kB]
Get:4 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 linux-tools-common all 6.8.0-85.85 [762 kB]
Get:5 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 linux-tools-6.8.0-85 amd64 6.8.0-85.85 [9421 kB]
Get:6 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 linux-tools-6.8.0-85-generic amd64 6.8.0-85.85 [1804 B]
Get:7 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 linux-tools-generic amd64 6.8.0-85.85 [19.8 kB]
Fetched 6303 kB in 6s (1042 kB/s)
Selecting previously unselected package libnumal:amd64.
(Reading database ... 58175 files and directories currently installed.)
Preparing to unpack .../8-libnumal_2.0.18-1build1_amd64.deb ...
Unpacking libnumal:amd64 (2.0.18-1build1) ...
Selecting previously unselected package libtraceevent1:amd64.
Preparing to unpack .../1-libtraceevent1_1.8.2-1ubuntu2.1_amd64.deb ...
Unpacking libtraceevent1:amd64 (1:1.8.2-1ubuntu2.1) ...
Selecting previously unselected package hwdetect.
Preparing to unpack .../2-hwdetect_0.379-1_all.deb ...
Unpacking hwdetect (0.379-1) ...
Selecting previously unselected package linux-tools-common.
Preparing to unpack .../3-linux-tools-common_6.8.0-85.85_all.deb ...
Unpacking linux-tools-common (6.8.0-85.85) ...
Selecting previously unselected package linux-tools-6.8.0-85.
Preparing to unpack .../4-linux-tools-6.8.0-85_6.8.0-85.85_amd64.deb ...
Unpacking linux-tools-6.8.0-85 (6.8.0-85.85) ...
Selecting previously unselected package linux-tools-6.8.0-85-generic.
Preparing to unpack .../5-linux-tools-6.8.0-85-generic_6.8.0-85.85_amd64.deb ...
Unpacking linux-tools-6.8.0-85-generic (6.8.0-85.85) ...
Selecting previously unselected package linux-tools-generic.
Preparing to unpack .../6-linux-tools-generic_6.8.0-85.85_amd64.deb ...
Unpacking linux-tools-generic (6.8.0-85.85) ...
Setting up hwdetect (0.379-1) ...
Setting up libnumal:amd64 (2.0.18-1build1) ...
Setting up libtraceevent1:amd64 (1:1.8.2-1ubuntu2.1) ...
Setting up linux-tools-common (6.8.0-85.85) ...
Setting up linux-tools-6.8.0-85 (6.8.0-85.85) ...
Setting up linux-tools-6.8.0-85-generic (6.8.0-85.85) ...
Setting up linux-tools-generic (6.8.0-85.85) ...
Processing triggers for man-db (2.12.0-4ubuntu2) ...
Processing triggers for libc-bin (2.39-0ubuntu8.6) ...
agilnggara@RagilAnggara:~$
```

```
agilnggara@RagilAnggara:~$ sudo perf record -a sleep 10
WARNING: perf not found for kernel 6.6.87.2-microsoft

You may need to install the following packages for this specific kernel:
  linux-tools-6.6.87.2-microsoft-standard-WSL2
  linux-cloud-tools-6.6.87.2-microsoft-standard-WSL2

You may also want to install one of the following packages to keep up to date:
  linux-tools-standard-WSL2
  linux-cloud-tools-standard-WSL2
agilnggara@RagilAnggara:~$ sudo perf trace ls
WARNING: perf not found for kernel 6.6.87.2-microsoft

You may need to install the following packages for this specific kernel:
  linux-tools-6.6.87.2-microsoft-standard-WSL2
  linux-cloud-tools-6.6.87.2-microsoft-standard-WSL2

You may also want to install one of the following packages to keep up to date:
  linux-tools-standard-WSL2
  linux-cloud-tools-standard-WSL2
agilnggara@RagilAnggara:~$ sudo perf top
WARNING: perf not found for kernel 6.6.87.2-microsoft

You may need to install the following packages for this specific kernel:
  linux-tools-6.6.87.2-microsoft-standard-WSL2
  linux-cloud-tools-6.6.87.2-microsoft-standard-WSL2

You may also want to install one of the following packages to keep up to date:
  linux-tools-standard-WSL2
  linux-cloud-tools-standard-WSL2
```